

QUANTUM Series

Semester - 4 Electronics & Allied Branches

Signal System



Session
2024-25
Even Semester

- Topic-wise coverage of entire syllabus in Question-Answer form.
- Short Questions (2 Marks)

Includes solution of following AKTU Question Papers:
2017-18 • 2018-19 • 2021-22 • 2022-23 • 2023-24

सजाग सूचना

छात्रों, डीलरों, दुकानदारों, कॉपियर्स और जिनसे यह संबंधित हो सकता है क्वांटम पुस्तकों के कॉपीराइट उल्लंघन के संबंध में दिल्ली के माननीय उच्च न्यायालय का आदेश

यह छात्रों, डीलरों, दुकानदारों, कॉपियर्स और आम जनता को सूचित करता है कि क्वांटम पेज प्राइवेट लिमिटेड पुस्तकों की क्वांटम श्रृंखला में कॉपीराइट का मालिक है।

क्वांटम पुस्तकों (चाहे सॉफ्ट कॉपी या हार्ड कॉपी में) की कोई भी अनधिकृत प्रतिलिपि, स्कैनिंग, पुनरुत्पादन, वितरण या संचलन, क्वांटम पेज प्राइवेट लिमिटेड के कॉपीराइट और ट्रेडमार्क अधिकारों के उल्लंघन के बराबर है, जो दिवानी अपराध के साथ-साथ दंडिक अपराध है, जिनकी सजा कारावास तक है।

क्वांटम पेज प्राइवेट लिमिटेड ने दिल्ली के माननीय उच्च न्यायालय के समक्ष क्वांटम पुस्तकों के उल्लंघनकर्ताओं के खिलाफ क्वांटम पेज प्राइवेट लिमिटेड बनाम देलीग्राम एफजेड एलएलसी और अन्य, सीएस (कॉम) १२१/२०२२ नामक एक मुकदमा शुरू किया है। 23 दिसंबर, 2022 के अपने आदेश में, माननीय उच्च न्यायालय ने माना है कि क्वांटम पुस्तकों का अनधिकृत पुनरुत्पादन कॉपीराइट उल्लंघन के बराबर है और देलीग्राम चैनलों सहित विभिन्न स्रोतों से क्वांटम पुस्तकों की उल्लंघनकारी प्रतियों को हटाने का निर्देश दिया है।

छात्रों, डीलरों, दुकानदारों, कॉपीयरों और आम जनता को एतद्वारा आगाह किया जाता है कि वे क्वांटम पुस्तकों (चाहे सॉफ्ट कॉपी या हार्ड कॉपी में) की कोई भी अनधिकृत नकल, स्कैनिंग, पुनरुत्पादन, वितरण और वितरण न करें।

क्वांटम पुस्तकों के ऐसे किसी भी अनधिकृत उपयोग से उल्लंघनकर्ताओं पर क्वांटम पेज प्राइवेट लिमिटेड द्वारा दीवानी और आपराधिक कार्यवाही शुरू की जाएगी।

QUANTUM SERIES

For

B.Tech Students of Second Year
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Signal System

By

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Signal System (EC : Sem-4)

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BEC403 : Signal System

UNIT-1 : INTRODUCTION TO SIGNALS & SYSTEMS

(1-1 J to 1-21 J)

Signals and systems as seen in everyday life, and in various branches of engineering and science, energy and power signals, continuous and discrete time signals, continuous and discrete amplitude signals, system properties: linearity, additivity and homogeneity, shift-invariance, causality, stability, realizability.

UNIT-2 : LINEAR SHIFT-INVARIANT SYSTEM

(2-1 J to 2-17 J)

Linear shift-invariant (LSI) systems, impulse response and step response, convolution, input-output behaviour with aperiodic convergent inputs, characterization of causality and stability of linear shift invariant systems, system representation through differential equations and difference equations, Periodic and semi-periodic inputs to an LSI system, the notion of a frequency response and its relation to the impulse response.

UNIT-3 : TRANSFORMS

(3-1 J to 3-40 J)

Fourier series representation, Fourier transform, convolution/multiplication and their effect in the frequency domain, magnitude and phase response, Fourier domain duality, Discrete-Time Fourier Transform (DTFT) and the Discrete Fourier Transform (DFT), Parseval's Theorem, the idea of signal space and orthogonal bases, the Laplace transform, notion of Eigen functions of LSI systems, a basis of Eigen functions, region of convergence, poles and zeros of system, Laplace domain analysis, solution to differential equations and system behaviour.

UNIT-4 : Z-TRANSFORM**(4-1 J to 4-20 J)**

The z-Transform for discrete time signals and systems- Eigen functions, region of convergence, z-domain analysis.

UNIT-5 : SAMPLING OF TIME SIGNALS (5-1 J to 5-22 J)

The sampling theorem and its implications- spectra of sampled signals, reconstruction: ideal interpolator, zero-order hold, first-order hold, and so on, aliasing and its effects, relation between continuous and discrete time systems.

SHORT QUESTIONS**(SQ-1 J to SQ-22 J)****SOLVED PAPERS (2017-18 TO 2023-24) (SP-1 J to SP-26 J)**

1
UNIT

Introduction to Signals and Systems

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Part-4 : Shift-invariance, Causality, 1-12J to 1-20J Stability, Realizability.	

PART-1

Signals and Systems as Seen in Everyday Life, and in Various Branches of Engineering and Science.

Que 1.1. Explain signal and also mention some signals seen in our daily life.

Answer

1. Signal can be defined as a function of one or more independent variables which contains some information.
2. It may be a function of time, temperature, position, pressure, distance etc.
3. Some signals in our daily life are music, speech, picture and video signals.
4. In electrical sense the signal may be voltage or current.
5. In daily life many electrical signals such as TV signals, radio signals, computer signals, telephone signals etc.

Que 1.2. Discuss the application of signals and systems in various branches of engineering and science.

Answer

Application of signals and systems :

Control applications :

1. Industrial control and automation (Control the velocity or position of an object)

2. Examples : Controlling the position of a valve or shaft of a motor

Communication applications :

1. Transmission of information (signal) over a channel.

2. The channel may be free space, coaxial cable, fiber optic cable.

Signal processing applications :

1. Application of algorithms to modify signals in a way to make them more useful.

2. Efficient and reliable transmission, storage and display of information.

3. Information extraction and enhancement.

Multimedia applications :

1. Compression : Fast, efficient, reliable transmission and storage of data.

2. Applied on audio, image and video data for transmission over the internet, storage.

Examples : CDs, DVDs, MP3, MPEG4, JPEG

Biological signal analysis :

Examples :

1. Brain signals (EEG)
2. Cardiac signals (ECG)
3. Medical images (X-Ray, PET, MRT)

Audio signal processing :

1. Active noise cancellation : Adaptive filtering
2. Add special music effects such as delay, echo, reverb
3. Separate speech from interference

PART-2

Energy and Power Signals, Continuous and Discrete Time Signals, Continuous and Discrete Amplitude Signals.

Que 1.3.

Give the classification of signals.

OR

Differentiate between the following:

- i. Continuous-time signals and discrete-time signals.
- ii. Periodic and aperiodic signals.
- iii. Deterministic and random signals.

OR

What is signal ? Give brief classification of signals.

AKTU 2017-18, Marks 3.5

OR

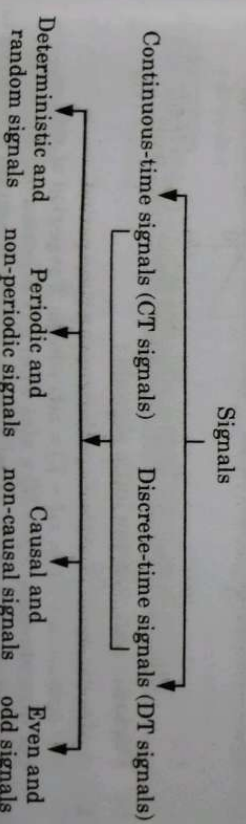
Classify signals according to signal characteristics.

AKTU 2018-19, Marks 07

Answer

Signal : It is a physical quantity that varies with time, space or any other independent variable or variables. It gives the information about behaviour or nature of the phenomenon.

Classification of signals : The signals are broadly classified as :



- i. **Continuous-time and discrete-time signals :** A signal $x(t)$ is a continuous-time signal if time t , is a continuous variable. If time t , is a discrete variable, then signal is called as discrete-time signal and denoted by $x[n]$.

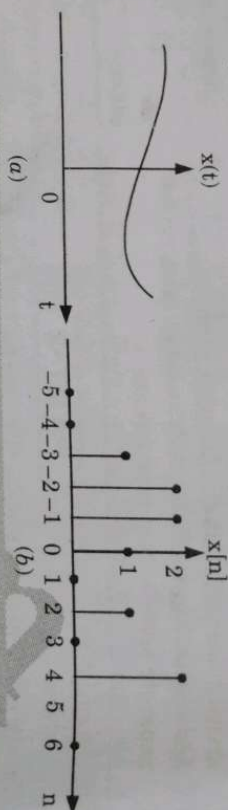


Fig. 1.3.1. (a) Continuous-time, (b) Discrete-time signals.

The sequence shown in Fig. 1.3.1(b) can be written as

$$x[n] = \{1, 2, 2, 1, 0, 1, 0, 2\}$$

ii. **Deterministic and random signals :**

1. **Deterministic signals** can be completely specified in time. The pattern of this signal is regular and can be characterized mathematically. So, its future value at any instant can be determined easily.
For example : $y(t) = b \sin(\pi t)$.
2. **Random signals** take random values at any given time. The pattern of this signal is irregular and it cannot be defined mathematically. So, future value of these signals at any instant cannot be predicted.
For example : noise.
- iii. **Even and odd signals :** A signal $x(t)$ or $x[n]$ is referred to as symmetric or even signal if,

$$x(-t) = x(t) \text{ or } x[-n] = x[n]$$

and referred to as non-symmetric or odd signal if,

$$x(-t) = -x(t) \text{ or } x[-n] = -x[n]$$

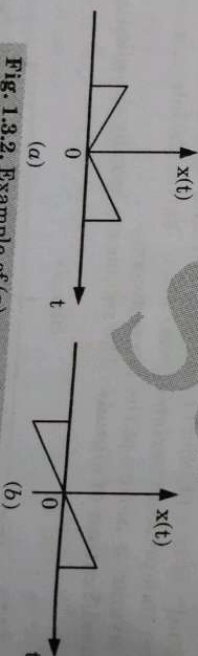


Fig. 1.3.2. Example of (a) even signal and (b) odd signal.

iv. **Periodic and non-periodic signals :**

1. **Periodic signal** repeats its pattern over a time period. For continuous-time, $x(t+T) = x(t)$ where, T is period of $x(t)$.
2. **Discrete-time signal** $x[n+N] = x[n]$ where, N is period of $x[n]$. A signal which does not satisfy the periodicity condition is referred to as non-periodic signal.

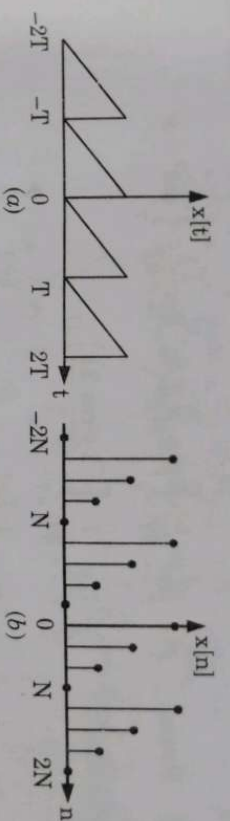


Fig. 1.3.3. Examples of periodic signals.

v. Energy and power signals :

Energy signal : The energy signal is one that has a finite energy and zero average power i.e., $0 < E < \infty$ and $P = 0$.

For continuous-time signal, $x(t)$; $E = \int_{-\infty}^{\infty} |x(t)|^2 dt$

For discrete-time signal, $x[n]$; $E = \sum_{n=-\infty}^{\infty} |x[n]|^2$

Power signal : A power signal is defined as a signal having finite average power and infinite energy i.e., $0 < P < \infty$ and $E = \infty$.

For continuous-time signal, $x(t)$; $P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$

For discrete-time signal, $x[n]$; $P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$

vi. One-dimensional and multi-dimensional signals :

One-dimensional signal : If the signal is a function of single independent variable, it is called one-dimensional signal.

For example : $x(t) = a \cos \omega_0 t$.

Multi-dimensional signal :

1. If a signal is a function of more than one independent variable then it is called multi-dimensional signal.

2. If a signal is having n independent variable, it is called n -dimensional signal. For example : A picture is an example of a two dimensional signal since the intensity $I(x, y)$ at each point is a function of two independent variables.

Que 1.4. Find energy and power of the signal.

i. $x(t) = \cos t$

ii. $x(t) = Ae^{-\alpha t} u(t)$, $\alpha > 0$

Answer

- i. 1. Given, $x(t) = \cos t$
2. Since the signal is periodic, it is necessarily a power signal and its energy is infinite.

$$\begin{aligned}
 3. \text{ Power, } P &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \cos^2 t dt \\
 &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \left[\frac{1 + \cos 2t}{2} \right] dt \\
 &= \lim_{T \rightarrow \infty} \frac{1}{4T} \left[\int_{-T}^T dt + \int_{-T}^T \cos 2t dt \right] \\
 &\quad \left(\because \lim_{T \rightarrow \infty} \frac{1}{4T} \int_{-T}^T \cos 2t dt = 0 \right)
 \end{aligned}$$

$$P = \lim_{T \rightarrow \infty} \frac{1}{4T} [t]_{-T}^T = \frac{1}{2}$$

ii. 1. Given,

$$x(t) = Ae^{-\alpha t} u(t), \alpha > 0$$

$$\begin{aligned}
 2. \text{ The energy of } x(t), E &= \lim_{T \rightarrow \infty} \int_{-T}^T x^2(t) dt \\
 &= \int_0^{\infty} A^2 e^{-2\alpha t} dt = A^2 \int_0^{\infty} e^{-2\alpha t} dt \\
 &= A^2 \left[\frac{e^{-2\alpha t}}{-2\alpha} \right]_0^{\infty} = \frac{A^2}{2\alpha}
 \end{aligned}$$

3. Power,

$$\begin{aligned}
 P &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x^2(t) dt \\
 &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^{\infty} A^2 e^{-2\alpha t} dt = \lim_{T \rightarrow \infty} \frac{1}{2T} \left[\frac{A^2}{-2\alpha} \right] = 0
 \end{aligned}$$

Que 1.5. Consider $x(t) = \cos(2\pi f_0 t)$. Determine it is a power signal or energy signal.

AKTU 2021-22, Marks 10

Answer

- Given, $x(t) = \cos(2\pi f_0 t)$
- Since the signal is periodic, it is necessarily a power signal its energy is infinite.
- We know, $2\pi f_0 = \omega_0$
- So $x(t) = \cos \omega_0 t$

$$\begin{aligned}
 5. \text{ Power, } P &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt \\
 &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \cos^2 \omega_0 t dt \\
 &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \frac{1 + \cos 2\omega_0 t}{2} dt
 \end{aligned}$$

$$\begin{aligned}
 &= \lim_{T \rightarrow \infty} \frac{1}{4T} \left[\int_{-T}^T dt + \int_{-T}^T \cos 2\omega_0 t dt \right] \\
 &\quad \left(\because \lim_{T \rightarrow \infty} \frac{1}{4T} \int_{-T}^T \cos 2\omega_0 t dt = 0 \right)
 \end{aligned}$$

$$P = \lim_{T \rightarrow \infty} \frac{1}{4T} [T]_{-T}^T = \frac{1}{2} W$$

Que 1.6. Prove that power of energy signal is zero over infinite time.

AKTU 2017-18, Marks 3.5

Answer

- Let $x(t)$ be a finite energy signal i.e.,

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

- Power of the signal,

$$\begin{aligned}
 P &= \lim_{T \rightarrow \infty} \frac{1}{2T} \left[\int_{-T}^T |x(t)|^2 dt \right] \\
 &= \lim_{T \rightarrow \infty} \frac{1}{2T} \left[\int_{-\infty}^{\infty} |x(t)|^2 dt \right] \\
 &= \lim_{T \rightarrow \infty} \frac{1}{2T} [E] \quad \left[\text{since } E = \int_{-\infty}^{\infty} |x(t)|^2 dt \right] \\
 &= 0 \times E = 0 \quad \left[\text{since } \lim_{T \rightarrow \infty} \frac{1}{2T} = 0 \right]
 \end{aligned}$$

Thus, the power of the energy signal is zero over infinite time.

Que 1.7. Determine whether the following signal is energy or power signal.

$$x[n] = u[n] - u[n-6]$$

Answer

- Given, $x[n] = u[n] - u[n-6]$
- The function $x[n]$ exist only for the range $n = 0$ to $n = 6$, i.e., finite duration signal and it is not periodic signal. Thus, this is an energy signal.
- Energy of the signal can be calculated as

$$\begin{aligned}
 E &= \sum_{n=-\infty}^{\infty} |x(n)|^2 \\
 &= \sum_{n=0}^6 |x(n)|^2 = 1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 1^2 = 7
 \end{aligned}$$

4. Power of the signal, $P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2 = 0$
5. As the energy of the signal is finite and average power is zero, hence it is energy signal.

Que 1.8. Determine whether or not signal is periodic. If periodic find its fundamental period.

- i. $x(t) = \sin 15\pi t$ ii. $x(n) = \cos\left(\frac{\pi n}{5}\right) \sin\left(\frac{\pi n}{5}\right)$.

Answer

- i. Given, $x(t) = \sin 15\pi t$

on comparing with $\sin \omega t$

$$\omega = 15\pi$$

$$\therefore \frac{2\pi}{T} = 15\pi$$

$$T = \frac{2}{15}$$

which is a rational number, hence it is a periodic signal with fundamental period $2/15$.

- ii. Given, $x[n] = \cos\left[\frac{\pi n}{5}\right] \sin\left[\frac{\pi n}{5}\right] = \frac{1}{2} \sin\left[\frac{2\pi n}{5}\right]$

on comparing with $A \sin[\Omega n]$

$$\Omega = \frac{2\pi}{5}$$

$$\therefore \frac{2\pi}{N} = \frac{2\pi}{5}$$

so, $x[n]$ is periodic signal with fundamental period, $N = 5$.

Que 1.9. Plot $x(t) = u(t) - r(t-1) + 2r(t-2) - r(t-3) + u(t-4) - 2u(t-5)$. Find the even and odd parts of this signal.

AKTU 2017-18, Marks 07

Answer

Given,

$$x(t) = u(t) - r(t-1) + 2r(t-2) - r(t-3) + u(t-4) - 2u(t-5)$$

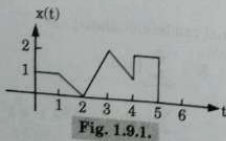


Fig. 1.9.1.

$$x(-t) = u(-t) - r(-t-1) + 2r(-t-2) - r(-t-3) + u(-t-4) - 2u(-t-5)$$

Even Part: $x_e(t) = \frac{1}{2} [x(t) + x(-t)]$

$$= \frac{1}{2} [u(t) + u(-t)] - [r(t-1) + r(-t-1)]$$

$$+ 2[r(t-2) + r(-t-2)] - [r(t-3) + r(-t-3)]$$

$$+ [u(t-4) + u(-t-4)] - 2[u(t-5) + u(-t-5)]$$

Odd Part: $x_o(t) = \frac{1}{2} [x(t) - x(-t)]$

$$= \frac{1}{2} [u(t) - u(-t)] - [r(t-1) - r(-t-1)]$$

$$+ 2[r(t-2) - r(-t-2)] - [r(t-3) - r(-t-3)]$$

$$+ [u(t-4) - u(-t-4)] - 2[u(t-5) - u(-t-5)]$$

Que 1.10. State whether the following signals $x(t)$ is periodic or not, giving reasons. If it is periodic, find the corresponding period.
 $x(t) = 2 \cos 100\pi t + 5 \sin 50t$.

AKTU 2018-19, Marks 07

Answer

1. Given, $x(t) = 2 \cos 100\pi t + 5 \sin 50t$

2. Let $x(t) = x_1(t) + x_2(t)$

where, $x_1(t) = 2 \cos 100\pi t$

3. Comparing with $A \sin \omega_1 t$, $\omega_1 = 100\pi$ rad/sec

$$T_1 = \frac{2\pi}{\omega_1} = \frac{2\pi}{100\pi} = \frac{1}{50} \text{ sec}$$

and $x_2(t) = 5 \sin 50t$

4. Comparing with $A \cos \omega_2 t$, $\omega_2 = 50$ rad/sec

$$T_2 = \frac{2\pi}{\omega_2} = \frac{2\pi}{50} = \frac{\pi}{25} \text{ sec}$$

$$\frac{T_1}{T_2} = \frac{1/50}{\pi/25} = \frac{1}{2\pi}$$

which is irrational number, hence it is not a periodic function.

Que 1.11. Sketch the following signal

- i. $2u(t+2) - 2u(t-3)$

- ii. $r(t) - r(t-1) - r(t-3) + r(t-4)$

AKTU 2022-23, Marks 10

Answer

i. $2u(t+2) - 2u(t-3)$

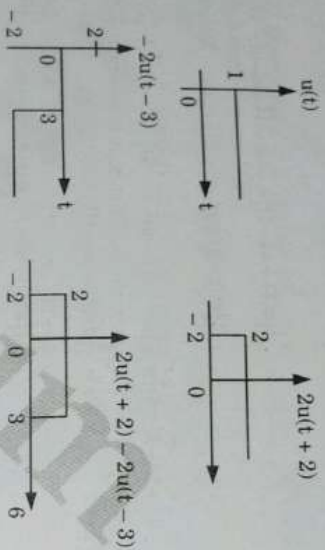


Fig. 1.11.1.

The signal $-2u(t-3)$ is obtained by shifting $u(t)$ to the right by 3 units and multiplying by 2 as shown.

$$2u(t+2) = 2 \text{ for } t \geq -2$$

$$\text{and } -2u(t-3) = -2 \text{ for } t \geq 3$$

$$\therefore 2u(t+2) - 2u(t-3) = \begin{cases} 2 & \text{For } -2 \leq t \leq 3 \\ 0 & \text{Otherwise} \end{cases}$$

ii. $r(t) - r(t-1) - r(t-3) + r(t-4)$

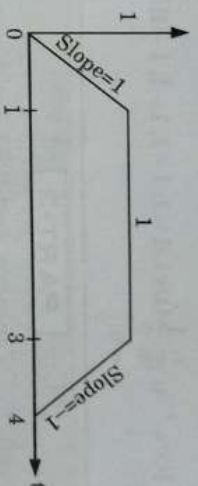
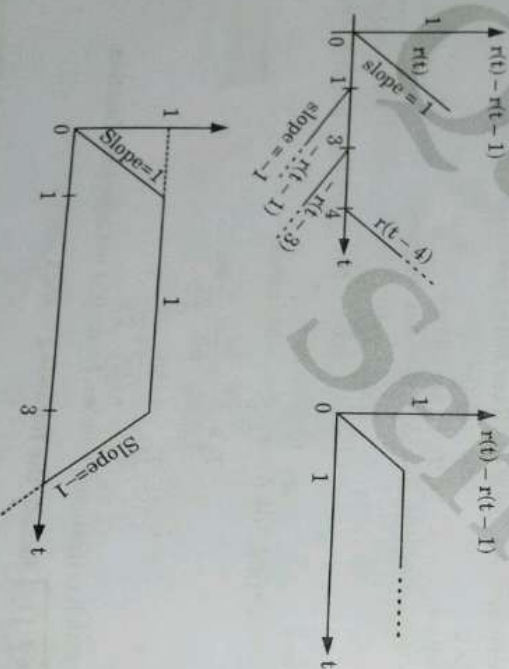


Fig. 1.11.2.

The sum $r(t) - r(t-1) - r(t-3) + r(t-4)$ is a trapezoidal signal as shown.

Que 1.12. Determine the even and odd components of the following signals :

i. $x(t) = \cos(t) + \sin(t) + \cos(t) \cdot \sin(t)$

ii. $x(n) = \{1, 1, 1, 1, 1\}$

Answer

i.

$$\begin{aligned} x(t) &= \cos(t) + \sin(t) + \sin(t) \cos(t) \\ x(-t) &= \cos(-t) + \sin(-t) + \sin(-t) \cos(-t) \\ &= \cos(t) - \sin(t) - \sin(t) \cos(t) \end{aligned}$$

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)]$$

$$\begin{aligned} &= \frac{1}{2} [\cos(t) + \sin(t) + \sin(t) \cos(t) + \cos(t) - \sin(t) - \sin(t) \cos(t)] \\ &= \cos(t) \end{aligned}$$

$$x_o(t) = \frac{1}{2} [x(t) - x(-t)]$$

$$\begin{aligned} &= \frac{1}{2} [\cos(t) + \sin(t) + \sin(t) \cos(t) - \cos(t) + \sin(t) + \sin(t) \cos(t)] \\ &= \sin(t) + \sin(t) \cos(t) \end{aligned}$$

ii. Given, $x(n) = \{1, 1, 1, 1, 1\}$

$$x(-n) = \{1, 1, 1, 1, 1\}$$

$$x_e(n) = \frac{1}{2} [x(n) + x(-n)]$$

$$= \frac{1}{2} [1 + 1, 1 + 1, 1 + 1, 1 + 1, 1 + 1] = \{1, 1, 1, 1, 1\}$$

$$x_o(n) = \frac{1}{2} [x(n) - x(-n)]$$

$$= \frac{1}{2} [1-1, 1-1, 1-1, 1-1, 1-1] \\ = 0$$

PART-3*System Properties : Linearity, Additivity and Homogeneity.***Que 1.13.** What are the linear and non-linear systems ?

OR

Explain the principle of linearity of DT system.

AKTU 2018-19, Marks 07**Answer**

1. A system which obeys the principle of superposition and principle of homogeneity is called a linear system, and a system which does not obey the principle of superposition and homogeneity is called a non-linear system.
2. Homogeneity property means a system which produces an output $y(t)$ for an input $x(t)$ must produce an output $ay(t)$ for an input $ax(t)$.
3. If arbitrary inputs $x_1(t)$ and $x_2(t)$ produce the outputs $y_1(t)$ and $y_2(t)$ respectively
 $y_1(t) = T[x_1(t)]$ and $y_2(t) = T[x_2(t)]$
 Let weighted sum of inputs is
 $x(t) = x_1(t) + x_2(t)$
 $y(t) = T[x(t)] = T[x_1(t) + x_2(t)]$
 If
 $y(t) = ay_1(t) + by_2(t)$
 $T[ax_1(t) + bx_2(t)] = T[ax_1(t)] + T[bx_2(t)] = aT[x_1(t)] + bT[x_2(t)]$
 $\dots(1.13.1)$
 The eq. (1.13.1) gives the superposition principle.
 For discrete-time linear system,
 $T[ax_1(n) + bx_2(n)] = aT[x_1(n)] + bT[x_2(n)]$
 4. Simply we can say that a system is linear if the output due to weighted sum of inputs is equal to the weighted sum of outputs.
 5. Few examples of linear systems are filters, communication channels, etc.

PART-4*Shift-invariance, Causality, Stability, Realizability.***Que 1.14.**

Determine whether the system is linear, time-invariant, causal and with memory.

i. $y[n] = x^2[n]$

ii. $y(t) = \frac{d}{dt} [e^{-t} x(t)]$

Answer

i. Given, $y[n] = x^2[n]$

Test for linearity :

1. For input $x_1[n] \Rightarrow y_1[n] = x_1^2[n]$ $\dots(1.14.1)$

For input $x_2[n] \Rightarrow y_2[n] = x_2^2[n]$ $\dots(1.14.2)$

2. For input $[ax_1[n] + bx_2[n]] \Rightarrow$

$ay_1[n] + by_2[n] = [ax_1[n] + bx_2[n]]^2$ $\dots(1.14.3)$

3. Multiplying eq.(1.14.1) by a and eq. (1.14.2) by b and adding we get, $\dots(1.14.4)$

$ax_1^2[n] + bx_2^2[n] = ax_1^2[n] + bx_2^2[n]$ $\dots(1.14.4)$

Since eq. (1.14.3) and (1.14.4) are not same, so this is non-linear system.

Test for time-variant or invariant :

1. The output due to input delayed by k units is :

$y[n, k] = x^2[n - k]$

2. The output due to input delayed by k units is :

$y[n - k] = x^2[n - k]$

As $y[n, k] = y[n - k]$, so system is time-invariant.**Test for causality and memory :**

Here system is only based on present input so system is causal and without memory or memoryless.

ii. Given, $y(t) = \frac{d}{dt} [e^{-t} x(t)]$

Test for linearity :

1. For input $x_1(t) \Rightarrow y_1(t) = \frac{d}{dt} [e^{-t} x_1(t)]$ $\dots(1.14.5)$

For input $x_2(t) \Rightarrow y_2(t) = \frac{d}{dt} [e^{-t} x_2(t)]$ $\dots(1.14.6)$

2. For input $[ax_1(t) + bx_2(t)] \Rightarrow$

$ay_1(t) + by_2(t) = \frac{d}{dt} [e^{-t}(ax_1(t) + bx_2(t))]$ $\dots(1.14.7)$

3. Multiplying eq. (1.14.5) and (1.14.6) by a and b respectively and adding them, we get

$ay_1(t) + by_2(t) = \frac{d}{dt} [e^{-t}(ax_1(t) + bx_2(t))]$ $\dots(1.14.8)$

Since eq. (1.14.7) and (1.14.8) are same, so this is a linear system.

Test for time-variant or invariant :

1. Now the output due to input delayed by T sec is

$y(t, T) = \frac{d}{dt} [e^{-t} x(t - T)]$

2. The output due to output delayed by T sec is :

$$y(t-T) = \frac{d}{dt} [e^{-(t-T)} x(t-T)]$$

as
So, the system is time-variant.

Test for causality and memory :

The system is described by a differential equation and output is only dependent on present value of input so system is causal and with memory.

Que 1.15. Check whether the system is static/dynamic and causal/non-causal and why ?

$$y(n) = \log_{10} |x(n)|$$

Answer

- Given,
For $n = 0$, $y[0] = \log_{10} |x[0]|$
For $n = 1$, $y[1] = \log_{10} |x[1]|$
For $n = -1$, $y[-1] = \log_{10} |x[-1]|$
- As the output at any instant depends on the input at that instant. Hence, the given system is static.
- As the output depends on present input only. Hence, the given system is causal.

Que 1.16. Check whether the system is :

$$y(t) = E_0 [x(t)]$$

- Static or dynamic.
- Linear or non-linear.
- Causal or non-causal.
- Time-variant or invariant.

Answer

- Since $x(t)$ requires memory, the system is dynamic.

$$\text{ii. Given, } y(t) = E_0 [x(t)] = \frac{1}{2} [x(t) + x(-t)]$$

$$y_1(t) = \frac{1}{2} [x_1(t) + x_1(-t)]$$

$$y_2(t) = \frac{1}{2} [x_2(t) + x_2(-t)]$$

$$\text{and, } y_3(t) = a_1 y_1(t) + a_2 y_2(t)$$

$$= \frac{1}{2} [a_1 x_1(t) + a_1 x_1(-t) + a_2 x_2(t) + a_2 x_2(-t)]$$

$$y_4(t) = f[a_1 x_1(t) + a_2 x_2(t)]$$

$$= \frac{1}{2} [a_1 \{x_1(t) + x_1(-t)\} + a_2 \{x_2(t) + x_2(-t)\}]$$

Since,

$$y_3(t) = y_4(t)$$

So, the system is linear.

$$\text{iii. Given, } y(t) = E_0 [x(t)] = \frac{1}{2} [x(t) + x(-t)]$$

$$y(-1) = \frac{1}{2} [x(-1) + x(1)]$$

The output depends on the present value of $x(-1)$ and also the future value of $x(1)$. Hence, the system is non-causal.

$$\text{iv. Given, } y(t) = E_0 [x(t)] = \frac{1}{2} [x(t) + x(-t)]$$

The output due to the delayed input is,

$$y'(t, t_0) = \frac{1}{2} [x(t - t_0) + x(-t - t_0)]$$

The delayed output is obtained by putting $t = (t - t_0)$

$$y'(t - t_0) = \frac{1}{2} [x(t - t_0) + x(-t - t_0)]$$

$$y'(t, t_0) \neq y'(t - t_0)$$

Hence, the system is time-variant.

Que 1.17. Determine whether the system is BIBO stable or not.

$$y[n] = \max \{x[n+1], x[n], x[n-1]\}$$

Answer

- Given,
- If $x[n] = \delta[n]$, then $y[n] = h[n]$
 $h[n] = \max \{\delta[n+1], \delta[n], \delta[n-1]\}$
 $h[0] = \max \{\delta[1], \delta[0], \delta[-1]\} = 1$
 $h[1] = \max \{\delta[2], \delta[1], \delta[0]\} = 1$
 $h[2] = \max \{\delta[3], \delta[2], \delta[1]\} = 0$

$$\sum_{n=-\infty}^{\infty} h[n] = |h[0]| + |h[1]| + |h[2]| + \dots = 1 + 1 + 0 + 0 + \dots = 2$$

So, the given system is BIBO stable.

Que 1.18. Check whether the system with impulse response is :

$$y[n] = \sum_{k=-\infty}^{n+5} x[k]$$

- Causal/non-causal

ii. Stable/unstable

Answer

- Causal/non-causal :

$$\text{Given, } y[n] = \sum_{k=-\infty}^{n+5} x[k]$$

Putting $n = 0$, $y[0] = \sum_{k=-\infty}^{\infty} x[k]$

$$= x[-\infty] + x[-\infty + 1] + \dots + x[-1] + x[0] + x[1] + x[2] + x[3] + x[4] + x[5]$$

$$x[-1] = \text{past input}$$

$$x[0] = \text{present input}$$

$$x[1], x[2], x[3], x[4], x[5] \text{ are future input}$$

Since, the output depends on present, past and future input, hence The system is non-causal.

ii. Stable/unstable :

Given, $y[n] = \sum_{k=-\infty}^{n+5} x[k]$

here as $n \rightarrow \infty$, $y[n] \rightarrow \infty$ and the output is unbounded. For bounded input, n should be a finite number.

In that case $y[n]$ is bounded and the system is stable.

The system is stable

; for $n = \text{finite}$

; for $n = \infty$

The system is unstable

Que 1.19. A discrete-time system is given as $y[n] = y^2[n-1] + x[n]$.

A bounded input of $x[n] = 2\delta[n]$ is applied to the system. Assume that the system is initially relaxed. Check whether the system is stable or unstable.

Answer

1. The input $x[n] = 2\delta[n]$ is applied. We know that,

$$\delta[n] = \begin{cases} 1 & \text{for } n = 0 \\ 0 & \text{for } n \neq 0 \end{cases}$$

$$x[n] = \begin{cases} 2 & \text{for } n = 0 \\ 0 & \text{for } n \neq 0 \end{cases}$$

2. $y[0] = y^2[0-1] + x[0]$

Since system is initially relaxed, $y^2[0-1] = y^2[-1] = 0$

$$x[0] = 2$$

$$y[0] = 2$$

3. So,

$$y[1] = y^2[1-1] + x[1] = y^2[0] + x[1]$$

Now, $x[n] = 0$ when $n \neq 0$, eq. (1.25.1) becomes,

$$y[1] = y^2[0] = 2^2$$

...(1.19.1)

4. Similarly

$$y[2] = y^2[2-1] + x[2] = y^2[1] = (2^2)^2 = 2^4$$

$$y[3] = y^2[3-1] + x[3] = y^2[2] = (2^4)^2 = 2^8$$

$$y[n] = 2^{2^n}$$

5. Here as $n \rightarrow \infty$, $y[n] \rightarrow \infty$. Thus the input $x[n] = 2\delta[n]$ is bounded but output $y[n]$ is not bounded for all n . Hence, the system is unstable.

Que 1.20.

Determine whether the following continuous time system $y(t) = x(t) \cos(100\pi t)$ is

- Static or dynamic
- Linear or non-linear
- Shift variant or shift invariant
- Causal or non-causal
- Stable or unstable.

AKTU 2017-18, Marks 07

Answer

Given, $y(t) = x(t) \cos(100\pi t)$

i. Static or dynamic :

For $t = 1$, $y(1) = x(1) \cos(100\pi)$

For $t = 2$, $y(2) = x(2) \cos(200\pi)$

For $t = -1$, $y(-1) = x(-1) \cos(-100\pi)$

Since, output depends upon present input only, hence, this system is static (memoryless) system.

ii.

Linear or non-linear : The response of the system with two inputs

For input $x_1(t)$, $y_1(t) = f[x_1(t)] = x_1(t) \cos(100\pi t)$

For input $x_2(t)$, $y_2(t) = f[x_2(t)] = x_2(t) \cos(100\pi t)$

Hence, linear combination of the output becomes,

$$y_3(t) = a_1 y_1(t) + a_2 y_2(t)$$

$$y_3(t) = a_1 x_1(t) \cos(100\pi t) + a_2 x_2(t) \cos(100\pi t)$$

$$y_3(t) = [a_1 x_1(t) + a_2 x_2(t)] \cos(100\pi t) \quad \dots(1.26.1)$$

The response of linear combination of two inputs becomes

$$y_4(t) = f[a_1 x_1(t) + a_2 x_2(t)]$$

$$y_4(t) = [a_1 x_1(t) + a_2 x_2(t)] \cos(100\pi t) \quad \dots(1.26.2)$$

Since eq. (1.26.1) and (1.26.2) are equal, hence this system is a linear system.

iii. Shift variant or shift invariant : The output due to delayed input becomes

$$y(t, t_1) = f[x(t - t_1)]$$

$$y(t, t_1) = x[(t - t_1)] \cos(100\pi t) \quad \dots(1.26.3)$$

$$\text{and the delayed output becomes (by replacing } t \text{ by } (t - t_1) \text{ in eq. (1.26.3))}$$

$$y(t - t_1) = x(t - t_1) \cos[100\pi(t - t_1)] \quad \dots(1.26.4)$$

$$y(t, t_1) \neq y(t - t_1)$$

Here, Hence, system is shift variant.

iv. Causal or non-causal : Since the system depends upon present input only, hence this system is causal.

v. Stable or unstable :

$$y(t) = x(t) \cos(100\pi t)$$

$$-1 < \cos(100\pi t) < 1$$

$$-x(t) < y(t) < x(t)$$

So,

Here, input is multiplied by cosine function and maximum and minimum value is 1 and -1 respectively. Hence, as long as $x(t)$ is bounded, $y(t)$ is bounded. Hence, the system is stable.

Que 1.21.

Define invertible system and state whether the following systems are invertible or not.

- $y[n] = x[n]$
- $y[n] = x^2[n] + 1$

Answer

Invertible system :

- A system is said to be invertible, if distinct input leads to distinct outputs.
- If a system is invertible, then an inverse system exists that, when cascaded with the original system, yield an output $w[n]$ equal to the input $x[n]$ of the first system.



Numerical :

- Given, $y[n] = x[n]$

The inverse system is

$$w[n] = y[n] = x[n]$$

Since, after reverting the system, the output is not affected. So the given system is an invertible system.

- Given, $y[n] = x^2[n] + 1$

The inverse system is

$$w[n] = \pm [y[n] - 1]^{1/2}$$

In this case, we cannot determine the sign of input from knowledge of the output. Hence, this system is non-invertible.

Que 1.22. Check whether the given system is static, linear, causal and time invariant system or not $y(t) = at^2 x(t) + bt x(t - 4)$.

AKTU 2022-23, Marks 10

Answer

Given : $y(t) = at^2 x(t) + bt x(t - 4)$

- The output depends on past input so it is dynamic system.
- For an input $x_1(t)$

$$y_1(t) = at^2 x_1(t) + bt x_1(t - 4)$$

For an input $x_2(t)$

$$y_2(t) = at^2 x_2(t) + bt x_2(t - 4)$$

The weighted sum of output is

$$Py_1(t) + qy_2(t)$$

$$= [P at^2 x_1(t) + P bt x_1(t - 4) + q at^2 x_2(t) + q bt x_2(t - 4)]$$

$$= at^2 [P x_1(t) + q x_2(t)] + bt [P x_1(t - 4) + q x_2(t - 4)]$$

The output due to weighted sum of input is

$$y_3(t) = T[P x_1(t) + q x_2(t)]$$

$$= at^2 [P x_1(t) + q x_2(t)] + bt [P x_1(t - 4) + q x_2(t - 4)]$$

$$y_3(t) = ay_1(t) + by_2(t)$$

$$y_3(t) = ay_1(t) + by_2(t)$$

3.

Superposition principle is satisfied. Hence system is linear.

The output depends only on the present and past inputs. It does not depend on future inputs. Hence the system is causal.

- Given

$$y(t) = T[x(t)] = at^2 x(t) + bt x(t - 4)$$

The output delayed due to input delayed by T sec is

$$y(t, T) = T[x(t - T)] = y(t) \big|_{t(t) = x(t - T)} = at^2 x(t - T) + bt x(t - 4 - T)$$

The output delayed by T sec

$$y(t - T) = y(t) \big|_{t = t - T} = at(t - T)^2 x(t - T) + b(t - T) x(t - T - 4)$$

$$y(t, T) \neq y(t - T)$$

Hence system is time variant

So, the given system is dynamic, linear, causal and time variant.

Que 1.23. Check whether the following digital systems are BIBO stable or not

- $y(n) = ax(n) + b$
- $y(n) = e^{-x(n)}$

AKTU 2022-23, Marks 10

Answer

- $y(n) = ax(n) + b$

$$\text{If } x(n) = \delta(n)$$

$$\text{Then } y(n) = h(n)$$

Hence the impulse response is

$$h(n) = a \delta(n) + b$$

$$\text{When } n = 0, h(0) = a \delta(0) + b = a + b$$

$$\text{When } n = 1, h(1) = a \delta(1) + b = b$$

$$\text{Here } h(1) = h(2) = \dots h(n) = b$$

$$\therefore h(n) = \begin{cases} a + b & \text{When } n = 0 \\ b & \text{When } n \neq 0 \end{cases}$$

The necessary condition for BIBO stability is

$$= \sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

$$\therefore \sum_{n=-\infty}^{\infty} |h(n)| = |h(0)| + |h(1)| + |h(2)| + \dots + |h(n)| + \dots$$

$$= |a + b| + |b| + |b| + \dots + |b| + \dots$$

This series never converges since the ratio between the successive term is one. Hence the given system is BIBO unstable.

-

$$y(n) = e^{-x(n)}$$

If

$$x(n) = \delta(n)$$

Then

$$y(n) = h(n)$$

Hence impulse response is

$$h(n) = e^{-\delta(n)}$$

$$\text{When } n = 0, h(0) = e^{-\delta(0)} = e^{-1}$$

$$\text{When } n = 1, h(1) = e^{-\delta(1)} = e^0 = 1$$

$$\text{In general } h(n) = \begin{cases} e^{-1} & \text{When } n = 0 \\ 1 & \text{When } n \neq 0 \end{cases}$$

The necessary condition for BIBO stability is

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

$$\therefore \sum_{n=-\infty}^{\infty} |h(n)| = |h(0)| + |h(1)| + |h(2)| + \dots + |h(n)|$$

$$= e^{-1} + 1 + 1 + 1 + \dots + 1 + \dots$$

The given sequence never converges, it is BIBO unstable.

Que 1.24. Determine whether the system is BIBO stable or not.

Answer

1. Given,
2. If

$$y[n] = \max \{x[n+1], x[n], x[n-1]\}$$

$$x[n] = \delta[n], \text{ then } y[n] = h[n]$$

$$h[n] = \max \{\delta[n+1], \delta[n], \delta[n-1]\}$$

$$h[0] = \max \{\delta[1], \delta[0], \delta[-1]\} = 1$$

$$h[1] = \max \{\delta[2], \delta[1], \delta[0]\} = 1$$

$$h[2] = \max \{\delta[3], \delta[2], \delta[1]\} = 0$$

$$\sum_{n=-\infty}^{\infty} h[n] = |h[0]| + |h[1]| + |h[2]| + \dots$$

$$= 1 + 1 + 0 + 0 + \dots = 2$$

So, the given system is BIBO stable.

Que 1.25. If a system is represented by the following difference

equation : $y(n) = 3y^2(n-1) - nx(n) + 4x(n-1) - x(n+1)$; $n \geq 0$

- a. Is the system linear ? Explain
- b. Is the system shift-invariant ? Explain
- c. Is the system causal ? Why or why not ?

Answer

- Given :** $y(n) = 3y^2(n-1) - nx(n) + 4x(n-1) - x(n+1)$; $n \geq 0$
- a. No, the system is non linear because there is a square term of delayed output in the difference equation.
 - b. No, the system is shift variant because the coefficient of $x(n)$ is not a constant. It is a function of time.
 - c. No, the system is non causal because the output depends on future inputs.

VERY IMPORTANT QUESTIONS

Following questions are very important. These questions may be asked in your SESSIONALS as well as UNIVERSITY EXAMINATION.

Q. 1. Classify signals according to signal characteristics.
Ans. Refer Q. 1.3.

Q. 2. Consider $x(t) = \cos(2\pi f_0 t)$. Determine it is a power signal or energy signal.
Ans. Refer Q. 1.5.

Q. 3. Prove that power of energy signal is zero over infinite time.
Ans. Refer Q. 1.6.

Q. 4. Sketch the following signal
i. $2u(t+2) - 2u(t-3)$
ii. $r(t) - r(t-1) - r(t-3) + r(t-4)$
Ans. Refer Q. 1.11.

Q. 5. Determine whether the following continuous time system $y(t) = x(t) \cos(100\pi t)$ is

- i. Static or dynamic
- ii. Linear or non-linear
- iii. Shift variant or shift invariant
- iv. Causal or non-causal
- v. Stable or unstable.

Ans. Refer Q. 1.20.

Q. 6. Check whether the given system is static, linear, causal and time invariant system or not $y(t) = a^2 x(t) + bt x(t-4)$.
Ans. Refer Q. 1.22.

Q. 7. Check whether the following digital systems are BIBO stable or not
i. $y(n) = ax(n) + b$ ii. $y(n) = e^{-x(n)}$
Ans. Refer Q. 1.23.



2 UNIT

Linear Shift-Invariant (LSI) System

CONTENTS

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PART-1

Linear Shift-Invariant (LSI) Systems, Impulse Response and Step Response, Convolution, Input-Output Behaviour With Aperiodic Convergent Inputs.

Que 2.1. What are linear shift-invariant (LSI) systems ?

Answer

- LSI systems are a class of systems used in signals and systems that are both linear and shift-invariant.
- Linear systems are systems whose outputs for a linear combination of inputs are the same as a linear combination of individual responses to those inputs.
 - Shift-invariant systems are systems where the output does not depend on when an input was applied.

Que 2.2. Define the following terms :

- Impulse response function of an LTI system.
- Unit-sample response sequence of an LSI system.

Answer

i. **Impulse response of systems :**

The impulse response, $h(t)$, of a linear time-invariant continuous-time system is defined as the response of the system to a unit impulse given to it as input when the system is in ground state.



Fig. 2.2.1. Impulse response of a system.

ii. **Unit-sample response sequence :**

- The unit sample response sequence, $h[n]$, of a linear shift-invariant (LSI) discrete-time system is defined as the response of the system in ground state, to a unit-sample sequence applied to it as input.
- A discrete-time system is said to be in ground state if all its memory elements contain only zeros.

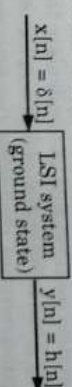


Fig. 2.2.2. Unit-sample response sequence.

Que 2.3. Explain impulse response characterization of LTI systems ?

OR

Write a short note on convolution integral.

Answer

1. An LTI system could be completely characterized by its impulse response function, $h(t)$.
2. We know that

$$x(t) \xrightarrow{T} y(t) = T[x(t)]$$

but,

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

so,

$$\begin{aligned} y(t) &= T \left[\int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau \right] \\ &= \int_{-\infty}^{\infty} x(\tau) T[\delta(t - \tau)] d\tau = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \end{aligned}$$

3. Therefore we have,

$$x(t) \xrightarrow{T} \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = y(t) = h(t) \otimes x(t) \quad \dots (2.3.1)$$

4. From eq. (2.3.1), we find that the output $y(t)$, of an LTI system in ground state to an input $x(t)$ is given by the convolution of the input signal $x(t)$ with the impulse response, $h(t)$, of the system.

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \quad \dots (2.3.2)$$

5. By a change of the variable, $\lambda = (t - \tau)$, we may write eq. (2.3.2) as

$$y(t) = \int_{-\infty}^{\infty} x(t - \tau) h(\tau) d\tau \quad \dots (2.3.3)$$

hence,

$$\begin{aligned} y(t) &= x(t) \otimes h(t) \\ &= \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = \int_{-\infty}^{\infty} x(t - \tau) h(\tau) d\tau \end{aligned}$$

6. Integral on the RHS of eq. (2.3.3) and (2.3.2) are called convolution integrals, or the superposition integrals.
7. From eq. (2.3.1), we find that a knowledge of the $h(t)$, the impulse response of the system would enable us to calculate the output, $y(t)$, of the system for any given input signal, $x(t)$. It is in this sense that we say that $h(t)$, the impulse response, completely characterizes an LTI system.

Que 2.4. What are the properties of convolution ?**Answer**

1. Let us consider two signals $x_1(t)$ and $x_2(t)$. The convolution of two signals $x_1(t)$ and $x_2(t)$ is given by

$$x_1(t) \otimes x_2(t) = \int_{-\infty}^{\infty} x_1(\tau) x_2(t - \tau) d\tau = \int_{-\infty}^{\infty} x_2(\tau) x_1(t - \tau) d\tau$$

2. The properties of convolution are as follows :
i. **Commutative property** : The commutative property of convolution states that

$$x_1(t) \otimes x_2(t) = x_2(t) \otimes x_1(t)$$

- ii. **Distributive property** : The distributive property of convolution states that

$$x_1(t) \otimes [x_2(t) + x_3(t)] = [x_1(t) \otimes x_2(t)] + [x_1(t) \otimes x_3(t)]$$

- iii. **Associative property** : The associative property of convolution states that

$$x_1(t) \otimes [x_2(t) \otimes x_3(t)] = [x_1(t) \otimes x_2(t)] \otimes x_3(t)$$

- iv. **Shift property** : The shift property of convolution states that if

$$x_1(t) \otimes x_2(t) = z(t)$$

$$\text{then } x_1(t) \otimes x_2(t - T) = z(t - T)$$

$$x_1(t - T) \otimes x_2(t) = z(t - T)$$

$$\text{and } x_1(t - T_1) \otimes x_2(t - T_2) = z(t - T_1 - T_2)$$

- v. **Convolution with an impulse** : Convolution of a signal $x(t)$ with a unit impulse is the signal itself, i.e.,

$$x(t) \otimes \delta(t) = x(t)$$

- vi. **Width property** : Let the duration of $x_1(t)$ and $x_2(t)$ be T_1 and T_2 respectively. Then the duration of the signal obtained by convolving $x_1(t)$ and $x_2(t)$ is $T_1 + T_2$.

- vii. **Differentiation property** :

According to the differentiation property,

$$x(t) \otimes h(t) = y(t)$$

then

$$\frac{dx(t)}{dt} \otimes h(t) = x(t) \otimes \frac{dh(t)}{dt} = \frac{dy(t)}{dt}$$

- viii. **Time-scaling property** :

If $x(t) \otimes h(t) = y(t)$,

$$\text{then } x(at) \otimes h(at) = \frac{1}{|a|} y(at)$$

Que 2.5. Obtain the convolution of

$$x(t) = u(t) \text{ and } h(t) = 1 \text{ for } -1 \leq t \leq 1$$

Answer

1. We can also write $h(t) = u(t+1) - u(t-1)$

2. We know that $x(t) \otimes h(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$

$$= \int_{-\infty}^{\infty} [u(\tau+1) - u(\tau-1)] u(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} u(\tau+1) u(t-\tau) d\tau - \int_{-\infty}^{\infty} u(\tau-1) u(t-\tau) d\tau$$



Fig. 2.5.1.

3. In the 1st part, $u(\tau+1) = 1$ for $\tau > -1$ and $u(t-\tau) = 1$ for $\tau < t$, and in 2nd part, $u(\tau-1) = 1$ for $\tau > 1$.

$$4. \quad x(t) \otimes h(t) = \int_{-1}^t 1 d\tau - \int_1^t 1 d\tau = [\tau]_{-1}^t - [\tau]_1^t \\ = (t+1) u(t+1) - (t-1) u(t-1)$$

Que 2.6.

Implement the convolution integral of the signals

$x(t) = e^{-2t} u(t)$ and $h(t) = u(t)$.

AKTU 2021-22, Marks 10

Answer

1. Given,

$$x(t) = e^{-2t} u(t), \quad h(t) = u(t)$$

2. $y(t) = x(t) * h(t)$

$$x(\tau) = e^{-2\tau} u(\tau)$$

$$h(t-\tau) = u(t-\tau)$$



Fig. 2.6.1.

$$= \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} e^{-2\tau} u(\tau) \cdot u(t-\tau) \cdot d\tau$$

$$x(\tau) \cdot h(t-\tau) = \begin{cases} e^{-2\tau} & 0 < \tau < t \\ 0 & \text{elsewhere} \end{cases}$$

So,

$$y(t) = \int_0^t e^{-2\tau} d\tau$$

$$= \frac{1}{-2} [e^{-2\tau}]_0^t = \frac{-1}{2} [e^{-2t} - e^0]$$

$$= \frac{1 - e^{-2t}}{2}$$

Que 2.7.

Find the convolution of the following signal by graphical method

$$x(t) = e^{-3t} u(t); \quad h(t) = u(t+3)$$

AKTU 2022-23, Marks 10

Answer

Given :

$$x(t) = e^{-3t} u(t)$$

$$h(t) = u(t+3)$$

The output

$$y(t) = x(t) * h(t)$$

$$= \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

The two functions $x(\tau)$ and $h(\tau)$ will be

$$x(\tau) = e^{-3\tau} u(\tau) = e^{-3\tau} \text{ for } \tau \geq 0$$

and $h(\tau) = u(\tau+3) = 1 \text{ for } \tau \geq -3$

$$h(-\tau) = u(-\tau+3)$$

$h(\tau)$ can be obtained by folding $h(\tau)$ about $\tau = 0$

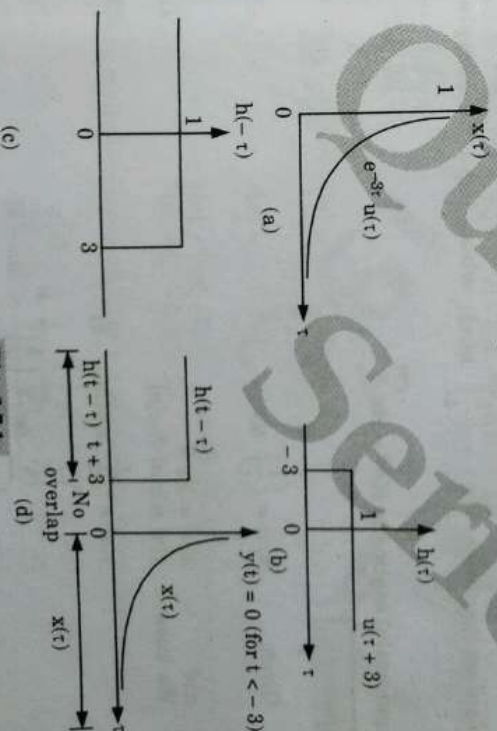


Fig. 2.7.1.

The plot given of $x(t)$ and $h(t-\tau)$ when there is no overlap. Now increase the shift t until the signal $h(t-\tau)$ intersects $x(\tau)$.

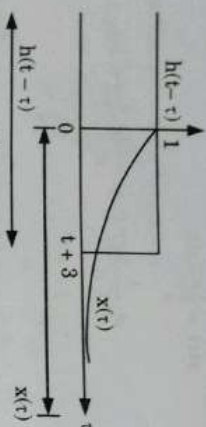


Fig. 2.7.2.

Thus $y(t) = \int_0^{t+3} x(\tau)h(t-\tau)d\tau$

$$= \int_0^{t+3} e^{-3\tau} d\tau = \left[\frac{e^{-3\tau}}{-3} \right]_0^{t+3} = \frac{e^{-3(t+3)} - 1}{-3} = \frac{1 - e^{-3(t+3)}}{3}$$

Que 2.8.

Determine the output sequence of the system with

impulse response $h[n] = \left(\frac{1}{4}\right)^n u[n]$; when input is complexexponential sequence $x[n] = Ae^{\frac{j\pi n}{2}}$.

Answer

1. Given, $h[n] = \left(\frac{1}{4}\right)^n u[n]$

and $x[n] = Ae^{\frac{j\pi n}{2}}$
2. We know, $y[n] = h[n] \otimes x[n]$

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \\ &= \sum_{k=-\infty}^{\infty} Ae^{\frac{j\pi k}{2}} \left(\frac{1}{4}\right)^{n-k} u[n-k] \\ &= \sum_{k=-\infty}^n Ae^{\frac{j\pi k}{2}} \left(\frac{1}{4}\right)^{n-k} = \sum_{k=-n}^{\infty} Ae^{\frac{-j\pi k}{2}} \left(\frac{1}{4}\right)^{n+k} \end{aligned}$$

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$$= \sum_{k=-n}^{\infty} A \left(\frac{1}{4}\right)^n \left(\frac{e^{-j\pi/2}}{4}\right)^k = A \left(\frac{1}{4}\right)^n \sum_{k=-n}^{\infty} \left(\frac{1}{4}e^{-j\pi/2}\right)^k$$

$$= A \left(\frac{1}{4}\right)^n \frac{\left(\frac{1}{4}e^{-j\pi/2}\right)^{-n}}{1 - \left(\frac{1}{4}e^{-j\pi/2}\right)} = \frac{A \left(\frac{1}{4}\right)^n \left(\frac{1}{4}\right)^{-n} e^{\frac{j\pi n}{2}}}{1 - \frac{1}{4}e^{-j\pi/2}}$$

so,

$$y[n] = \frac{Ae^{\frac{j\pi n}{2}}}{1 - \frac{1}{4}e^{-j\pi/2}}$$

Que 2.9.

Calculate the convolution for given sequence

$$x[n] = \begin{cases} 1 & ; \text{for } n = -2, 0, 1 \\ 2 & ; \text{for } n = -1 \\ 0 & ; \text{else} \end{cases}$$

$$h[n] = \delta[n] - \delta[n-1] + \delta[n-2] - \delta[n-3]$$

OR

Find the convolution for given sequence

$$x[n] = 1 \text{ for } n = -2, 0, 1$$

$$2 \text{ for } n = -1$$

$$0 \text{ else}$$

and

$$h[n] = \delta[n] - \delta[n-1] + \delta[n-2] - \delta[n-3]$$

AKTU 2017-18, Marks 07

Answer

1. Given, $x[n] = \begin{cases} 1 & ; \text{for } n = -2, 0, 1 \\ 2 & ; \text{for } n = -1 \\ 0 & ; \text{else} \end{cases}$

$$x[n] = \{1, 2, 1, 1\}$$

$$h[n] = \delta[n] - \delta[n-1] + \delta[n-2] - \delta[n-3]$$

$$h[n] = \{1, -1, 1, -1\}$$

2. Using multiplication method the convolution can be obtained as,

$x[n] \Rightarrow$	1	2	1	1	
$h[n] \Rightarrow$	1	-1	1	-1	
	-1	-2	-1	-1	-1
	1	2	1	1	$\times$
	-1	-2	-1	-1	$\times$
	1	2	1	1	$\times$
	1	2	1	1	$\times$
	1	1	0	-2	0
	1	1	0	-2	-1

3. The output is,
 $x[n] \otimes h[n] = \{1, 1, 0, 1, -2, 0, -1\}$

Que 2.10. The accumulator is excited by the sequence $x[n] = nu[n]$. Accumulator can be defined by following input and output relationship.

$$y[n] = \sum_{k=-\infty}^n x[k]$$

Determine its output under the condition :

- It is initially relaxed.
- Initially $y[-1] = 1$.

Answer

1. The output of the system is defined as

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^n x[k] = \sum_{k=-\infty}^{-1} x[k] + \sum_{k=0}^n x[k] \\ &= y[-1] + \sum_{k=0}^n x[k] \\ &= y[-1] + \frac{2}{n(n+1)}, \quad n \geq 0 \end{aligned}$$

2. If the system is initially relaxed, $y[-1] = 0$ and hence

$$y[n] = \frac{n(n+1)}{2}, \quad n \geq 0$$

3. On the other hand, if the initial condition is $y[-1] = 1$, then

$$y[n] = 1 + \frac{n(n+1)}{2} = \frac{n^2 + n + 2}{2}, \quad n \geq 0$$

PART-2

Characterization of Causality and Stability of Linear Shift Invariant Systems, Representation Through Differential Equations and Difference Equations.

Que 2.11.

Explain in detail about the causality of LSI systems.

Answer

1. The output of a causal system depends only on the present and past values of the input to the system.

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2. The impulse response of a causal discrete-time LSI system satisfy the condition

$$h[n] = 0 \quad \text{for } n < 0 \quad \dots(2.11.1)$$

3. According to eq. (2.11.1) the impulse response of a causal LSI system must be zero before the impulse occurs, which is consistent with the intuitive concept of causality.

4. Causality for a linear system is equivalent to the condition of initial rest, i.e., if the input to a causal system is 0 upto some point in time, then the output must also be 0 upto that time.

4. The equivalent of causality and the condition of initial rest applies only to linear system.

5. For a causal discrete-time LSI system, the condition in eq. (2.11.1) implies that the convolution sum representation becomes

$$y[n] = \sum_{k=-\infty}^n x[k]h[n-k] \quad \dots(2.11.2)$$

and the alternative equivalent form,

$$y[n] = \sum_{k=0}^{\infty} h[k]x[n-k] \quad \dots(2.11.3)$$

6. Similarly, a continuous-time LSI system is causal if

$$h(t) = 0 \quad \text{for } t < 0 \quad \dots(2.11.4)$$

and in this case, the convolution integral is given by

$$y(t) = \int_{-\infty}^t x(\tau)h(t-\tau)d\tau = \int_0^{\infty} h(\tau)x(t-\tau)d\tau$$

7. Causality of an LSI system is equivalent to its impulse response being a causal signal.

Que 2.12. Discuss about the stability of LSI systems.

Answer

1. A system is stable if every bounded input produced a bounded output. In order to determine conditions under which LSI system are stable, consider an input $x[n]$ that is bounded in magnitude :

$$|x[n]| < B \quad \text{for all } n \quad \dots(2.12.1)$$

2. Suppose that we apply this input to an LSI system with unit impulse response $h[n]$. Then, using the convolution sum, we obtain an expression for the magnitude of the output :

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} h[k]x[n-k] \right| \quad \dots(2.12.2)$$

3. Since the magnitude of the sum of a set of numbers is not larger than the sum of the magnitudes of the numbers, it follows from eq. (2.12.2) that

$$|y[n]| \leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]| \quad \dots(2.12.3)$$

4. From eq. (2.12.1) $|x[n-k]| < B$ for all values of k and n . Together with eq. (2.12.3), this implies that

$$|y[n]| \leq B \sum_{k=-\infty}^{\infty} |h[k]| \quad \text{for all } n \quad \dots(2.12.4)$$

5. From eq. (2.12.4) we can conclude that if the impulse response is absolutely summable, that is, if

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty \quad \dots(2.12.5)$$

the $y[n]$ is bounded in magnitude, and hence, the system is stable. Therefore, eq. (2.12.5) is a sufficient condition to guarantee the stability of a discrete-time LSI system.

6. In continuous time, we obtain an analogous characterization of stability in terms of the impulse response of an LTI system. Specifically, if $|x(t)| < B$ for all t , then in analogy with eq. (2.12.2), (2.12.3) and eq. (2.12.4), it follows that

$$\begin{aligned} |y(t)| &= \left| \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau \right| \\ &\leq \int_{-\infty}^{\infty} |h(\tau)| |x(t-\tau)| d\tau \\ &\leq B \int_{-\infty}^{\infty} |h(\tau)| d\tau \end{aligned}$$

7. Therefore, the system is stable if the impulse response is absolutely integrable i.e., if

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty \quad \dots(2.12.6)$$

Que 2.13. Discuss about the linear constant-coefficient differential equation and linear constant-coefficient difference equations.

Answer

Linear constant-coefficient differential equation :

1. A general N th-order linear constant-coefficient differential equation is given by

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k} \quad \dots(2.13.1)$$

2. The order refers to the highest derivative of the output $y(t)$ appearing in the equation. In the case, when $N = 0$, eq. (2.13.1) reduces to

$$y(t) = \frac{1}{a_0} \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k} \quad \dots(2.13.2)$$

3. In this case, $y(t)$ is an explicit function of the input $x(t)$ and its derivatives. For $N \geq 1$, eq. (2.13.1) specifies the output implicitly in terms of the input. The solution $y(t)$ consists of two parts : a particular solution to eq. (2.13.1) plus a solution to the homogeneous differential equation.

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = 0 \quad \dots(2.13.3)$$

The solutions to this eq. (2.13.3) are referred to as the natural responses of the system.

5. If $x(t) = 0$ for $t \leq t_0$, we assume that $y(t) = 0$ for $t \leq t_0$ and therefore, the response for $t > t_0$ can be calculated from the differential eq. (2.13.1) with the initial conditions,

$$y(t_0) = \frac{dy(t_0)}{dt} = \dots = \frac{d^{N-1}y(t_0)}{dt^{N-1}} = 0 \quad \dots(2.13.4)$$

6. Under the condition of initial rest, the system described by eq. (2.13.1) is causal and LTI.

Linear constant-coefficient difference equations :

1. The discrete-time counterpart of eq. (2.13.1) is the N th-order linear constant-coefficient difference equation

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k] \quad \dots(2.13.5)$$

2. An equation of this type can be solved in a manner exactly analogous to that for differential equations. Specifically, the solution $y[n]$ can be written as the sum of a particular solution to eq. (2.13.5) and a solution to the homogeneous equation

$$\sum_{k=0}^N a_k y[n-k] = 0 \quad \dots(2.13.6)$$

3. The solutions to this homogeneous equation are often referred to as the natural responses of the system described by eq. (2.13.5).

4. If $x[n] = 0$ for $n < n_0$, then $y[n] = 0$ for $n < n_0$ as well. With initial rest, the system described by eq. (2.13.5) is LSI and causal.

Que 2.14. Write a differential equation describing the output to the input of the circuit shown in Fig. 2.14.1.

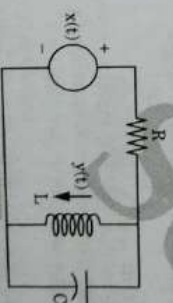


Fig. 2.14.1.

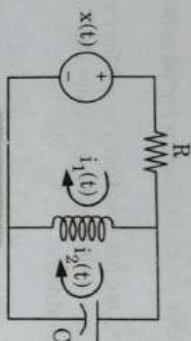
Answer

Fig. 2.14.2.

1. Now taking KVL in loop 1.

$$Ri_1(t) + L \frac{d}{dt} [i_1(t) - i_2(t)] = x(t) \quad \dots(2.14.1)$$

2. Taking KVL in loop 2.

$$L \frac{d}{dt} [i_2(t) - i_1(t)] + \frac{1}{C} \int i_2(t) dt = 0 \quad \dots(2.14.2)$$

$$y(t) = L \frac{d}{dt} [i_1(t) - i_2(t)] \quad \dots(2.14.3)$$

3. Taking integration of eq. (2.14.3)

$$\int y(t) dt = L [i_1(t) - i_2(t)]$$

$$\text{or, } \frac{1}{L} \int y(t) dt = [i_1(t) - i_2(t)] \quad \dots(2.14.4)$$

4. From eq. (2.14.1) and eq. (2.14.3) then we get

$$Ri_1(t) + y(t) = x(t)$$

$$\therefore i_1(t) = \frac{x(t) - y(t)}{R} \quad \dots(2.14.5)$$

5. From eq. (2.14.2) and eq. (2.14.3) then we get

$$-y(t) + \frac{1}{C} \int i_2(t) dt = 0$$

$$\text{or, } -\frac{dy(t)}{dt} + \frac{1}{C} i_2(t) = 0$$

$$\text{or, } i_2(t) = +C \frac{dy(t)}{dt}$$

6. Putting eq. (2.14.5) and (2.14.6) in eq. (2.14.4)

$$\frac{1}{L} \int y(t) dt = \frac{x(t) - y(t)}{R} - \left(+C \frac{dy(t)}{dt} \right) \quad \dots(2.14.6)$$

or

$$\frac{R}{L} \int y(t) dt = x(t) - y(t) - RC \frac{dy(t)}{dt}$$

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7. Now, differentiating both sides

$$R y(t) = \frac{dx(t)}{dt} - \frac{dy(t)}{dt} - RC \frac{d^2 y(t)}{dt^2}$$

$$\text{or, } RC \frac{d^2 y(t)}{dt^2} + \frac{dy(t)}{dt} + \frac{R}{L} y(t) = \frac{dx(t)}{dt}$$

$$\text{or, } \frac{d^2 y(t)}{dt^2} + \frac{1}{RC} \frac{dy(t)}{dt} + \frac{1}{LC} y(t) = \frac{1}{RC} \frac{dx(t)}{dt}$$

PART-3

Periodic and Semi-Periodic Inputs to an LSI System, The Notion of a Frequency Response and Its Relation to Impulse Response.

Que 2.15. Explain the complex exponentials response of LSI system.

Answer**Complex exponentials :**

Complex exponentials are an extremely useful class of functions for representing signals in LSI systems.

1. A very wide class of real world signals can be represented, to virtually any desired level of accuracy, by complex exponentials.

2. The responses of LSI systems to this broad class of signals can be represented and analyzed quite effectively using complex exponentials.

Response of LSI system :

1. The response of an LSI system, to a complex exponential, is that same complex exponential multiplied by another complex variable.



Fig. 2.15.1.

$x(t) = e^{st}$ = Complex exponential input

s = Complex variable = $\sigma + j\omega$

$y(t)$ = Output

$h(t)$ = LTI system impulse response

where the output is obtained by the convolution integral

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau = \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau$$

2. Now, the exponential of the difference can be written as a product, so

$$y(t) = \int_{-\infty}^{\infty} h(\tau) e^{st} e^{-s\tau} d\tau = e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

and we define

$$H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

This integral is the transfer function of the LTI system.

- Substituting back into the equation for the output we obtain a simple expression for the output in terms of the input

$$y(t) = H(s) e^{st}$$

- Such signals, for which the system output is simply a multiplication of the input by another complex variable, are called eigen functions and the multiplicative factor is the called the eigen value. Thus complex exponentials are eigen functions of LSI systems.

Linear combination of response :

- If the input to a system is made up of the sum of two signals, then the total output is the sum of the outputs that result from the system operating on each of the two inputs separately.

- In particular, the system

$$x(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

- Hence if the input signals are complex exponentials, where C_1 and C_2 are constants, Then

$$y(t) = H(s_1) C_1 e^{s_1 t} + H(s_2) C_2 e^{s_2 t}$$

Que 2.16. Find the sinusoidal response of LSI system ? Discuss about frequency response and its relation to impulse response.

Answer

Sinusoidal response of LSI system :

- The signal is a tone (sinusoid) that we designate as $u(t)$, so

$$u(t) = A \sin \omega_0 t$$

- In order to analyze the effects of $u(t)$, we will use a complex exponential representation of $u(t)$. We can write Euler's Equation as

$$e^{j\theta} = \cos \theta + j \sin \theta$$

and if we define

$$s_1 = +j\omega_0 \quad ; \quad s_2 = -j\omega_0$$

then $u(t)$ can be written as the difference of two complex exponentials

$$u(t) = A \left(\frac{e^{s_1 t} - e^{s_2 t}}{2j} \right)$$

- Also, the transmission phenomena can be modeled as a LSI system

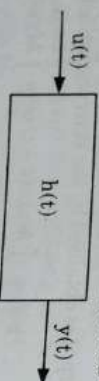


Fig. 2.16.1.

Let the impulse response of the transmission is

$$h(t) = k\delta(t - T)$$

2-16 J (EC-Sem-4)

Linear Shift-Invariant (LSI) System

- To determine the output $y(t)$ we need the system transfer function. From above the system transfer function is

$$H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau = \int_{-\infty}^{\infty} k\delta(\tau - T) e^{-s\tau} d\tau$$

$$= k e^{-sT}$$

- The sinusoidal input is the difference of two complex exponentials,

$$u(t) = \frac{A e^{s_1 t}}{2j} - \frac{A e^{s_2 t}}{2j}$$

- The output $y(t)$,

$$\begin{aligned} y(t) &= \frac{A e^{s_1 t}}{2j} H(s_1) - \frac{A e^{s_2 t}}{2j} H(s_2) \\ &= \frac{A e^{s_1 t}}{2j} k e^{-s_1 T} - \frac{A e^{s_2 t}}{2j} k e^{-s_2 T} \\ &= kA \left[\frac{e^{j\omega_0 t} e^{-j\omega_0 T} - e^{-j\omega_0 t} e^{-j\omega_0 T}}{2j} \right] \\ &= kA \left[\frac{e^{j\omega_0(t-T)} - e^{-j\omega_0(t-T)}}{2j} \right] \\ &= kA \sin \omega_0(t - T) \end{aligned}$$

Frequency response and its relation to impulse response :

- The transfer function of the LSI system given as :

$$H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

- By setting $s = \pm j\omega_0$ in the transfer function we have ignored the real part of the complex variable s .

- In doing so we have eliminated the transient (homogeneous) part of the solution.

- When we do this we are ignoring any startup process. Physically this means that the stable system has been operating long enough with the sinusoidal input so that all effects of the startup process have disappeared.

- In general, when we do this substitution into the system transfer function, we obtain the system frequency response.

$$\begin{aligned} H(s)|_{s=j\omega} &= H(j\omega) = \text{Frequency response} \\ |H(j\omega)| &= M(\omega) \\ \angle H(j\omega) &= \phi(\omega) \end{aligned}$$

VERY IMPORTANT QUESTIONS

Following questions are very important. These questions may be asked in your SESSIONALS as well as UNIVERSITY EXAMINATION.

Q.1. Implement the convolution integral of the signals $x(t) = e^{-2t}u(t)$ and $h(t) = u(t)$.

Ans. Refer Q. 2.6.

Q.2. Find the convolution of the following signal by graphical method

$$x(t) = e^{-3t}u(t); h(t) = u(t+3)$$

Ans. Refer Q. 2.7.

Q.3. The accumulator is excited by the sequence $x[n] = nu[n]$. Accumulator can be defined by following input and output relationship.

$$y[n] = \sum_{k=-\infty}^n x[k]$$

Determine its output under the condition :

- It is initially relaxed.
- Initially $y[-1] = 1$.

Ans. Refer Q. 2.10.

Q.4. Write a differential equation relating the output to the input of the circuit shown in Fig. 1.

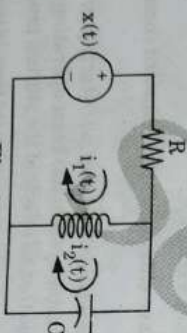


Fig. 1.

Ans. Refer Q. 2.14.

Q.5. Find the sinusoidal response of LSI system ? Discuss about frequency response and its relation to impulse response.

Ans. Refer Q. 2.16.



3 UNIT

Transforms

CONTENTS

Part-1 :	Fourier Series Representation	3-2J to 3-6J
Part-2 :	Fourier Transform, Convolution/ Multiplication and their Effect in Frequency Domain, Magnitude and Phase Response, Fourier Domain Duality	3-6J to 3-17J
Part-3 :	Discrete-Time Fourier Transform (DTFT)	3-17J to 3-23J
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Part-5 :	The Laplace Transform, Notion of Eigen Functions of LSI Systems, a Region of Convergence, Poles and Zeros of System	3-26J to 3-35J
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PART-1

Fourier Series Representation.

Que 3.1.

Briefly explain the three forms of Fourier series.

Answer

Three forms of Fourier series are :

- i. **Trigonometric Fourier series** : The trigonometric Fourier series representation of a periodic signal $x(t)$ with fundamental period T_0 is given by :

$$x(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

where,

$$\omega_0 = \frac{2\pi}{T_0}$$

 a_0, a_n and b_n are the Fourier coefficients given by :

$$a_0 = \frac{2}{T_0} \int_{T_0} x(t) dt$$

$$a_n = \frac{2}{T_0} \int_{T_0} x(t) \cos n\omega_0 t dt$$

$$b_n = \frac{2}{T_0} \int_{T_0} x(t) \sin n\omega_0 t dt$$

- ii. **Harmonic Fourier series** : Another form of Fourier series representation of a real periodic signal $x(t)$ with fundamental period T_0 is :

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos (n\omega_0 t - \theta_n),$$

where,

$$\omega_0 = \frac{2\pi}{T_0}$$

 c_0 = DC component $c_n \cos (n\omega_0 t - \theta_n)$ = n^{th} harmonic component of $x(t)$

$$c_0 = \frac{a_0}{2}, \quad c_n = \sqrt{a_n^2 + b_n^2}, \quad \theta_n = \tan^{-1} \frac{b_n}{a_n}$$

- iii. **Exponential Fourier series** :

1. The exponential Fourier series is the most widely used form of Fourier series.

2. In this, the function $x(t)$ is expressed as a weighted sum of the complex exponential functions.

$$x(t) = C_0 + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} C_n e^{jn\omega_0 t}$$

where,

$$C_n = \frac{1}{T} \int_{T_0} x(t) e^{-jn\omega_0 t} dt$$

$$C_0 = \frac{1}{T} \int_{T_0} x(t) dt$$

Que 3.2.

Give the properties of continuous-time Fourier series.

Answer

1. **Linearity property** :

The linearity property states that, if

$$x_1(t) \xrightarrow{\text{FS}} C_n \text{ and } x_2(t) \xrightarrow{\text{FS}} D_n$$

then $Ax_1(t) + Bx_2(t) \xrightarrow{\text{FS}} AC_n + BD_n$

2. **Time shifting property** :

The time shifting property states that, if

$$x(t) \xrightarrow{\text{FS}} C_n$$

$$x(t - t_0) \xrightarrow{\text{FS}} e^{-jn\omega_0 t_0} C_n$$

3. **Time reversal property** :

The time reversal property states that, if

$$x(t) \xrightarrow{\text{FS}} C_n$$

$$x(-t) \xrightarrow{\text{FS}} C_{-n}$$

4. **Time scaling property** :

The time scaling property states that, if

$$x(t) \xrightarrow{\text{FS}} C_n$$

then $x(\alpha t) \xrightarrow{\text{FS}} C_n$ with $\omega_0 \rightarrow \alpha\omega_0$

5. **Time differentiation property** :

The time differentiation property states that, if

$$x(t) \xrightarrow{\text{FS}} C_n$$

then $\frac{dx(t)}{dt} \xrightarrow{\text{FS}} jn\omega_0 C_n$

6. **Time integration property** :

The time integration property states that, if

$$x(t) \xrightarrow{\text{FS}} C_n$$

then $\int_{-\infty}^{\infty} x(\tau) d\tau \xleftrightarrow{FS} \frac{C_n}{jn\omega_0}$ (if $C_0 = 0$)

Que 3.3. Explain the trigonometric and exponential forms of Fourier series representation of periodic signals. Find the trigonometric Fourier series for the periodic signal shown in Fig. 3.3.1.

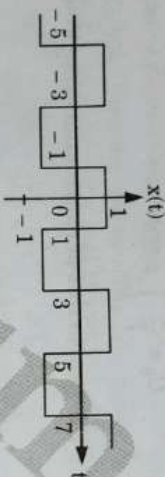


Fig. 3.3.1.

Answer

Trigonometric & exponential forms : Refer Q. 3.1, Page 3-2J, Unit-3.

Numerical :

Assumption : Shift the signal upward by 1 unit.

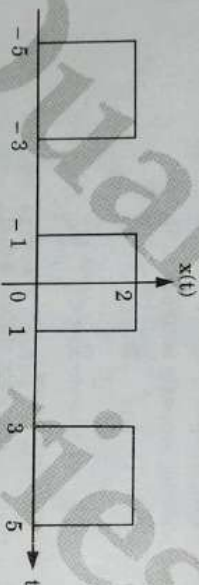


Fig. 3.3.2.

$$x(t) = \begin{cases} 2 & -1 < t < 1 \\ 0 & 1 < t < 3 \end{cases}$$

2. Fundamental time period = 4

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$$

$$a_0 = \frac{4}{T} \int_0^{T/2} x(t) dt = \frac{4}{4} \left(\int_0^1 2 dt + \int_1^2 0 dt \right) = 2$$

$$a_n = \frac{4}{T} \int_0^{T/2} x(t) \cos n\omega_0 t dt$$

$$= \frac{4}{T} \int_0^1 2 \cos n\omega_0 t dt + \int_1^2 0 \cos n\omega_0 t dt$$

$$= 2 \int_0^1 \cos n\omega_0 t dt + 0 = 2 \left[\frac{\sin n\omega_0 t}{n\omega_0} \right]_0^1$$

$$= 2 \left[\frac{\sin n\omega_0 - \sin n\omega_0 \cdot 0}{n\omega_0} \right] = \frac{2 \sin n\omega_0}{n\omega_0}$$

Putting

$$\omega_0 = \frac{\pi}{2}$$

$$a_n = \frac{2 \sin \frac{n\pi}{2}}{n\pi} = \frac{4}{n\pi} \sin \frac{n\pi}{2}$$

$$= \begin{cases} 0 & ; n \text{ even} \\ \frac{4(-1)^n}{n\pi} & ; n \text{ odd} \end{cases}$$

5. Since, given signal is even, $b_n = 0$

$$6. \quad x(t) = 1 + \sum_{n=\text{odd}} \frac{4(-1)^n}{n\pi} \cos \frac{n\pi}{2} t$$

Que 3.4. Determine the exponential Fourier series for the waveform shown

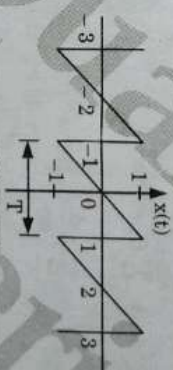


Fig. 3.4.1.

AKTU 2022-23, Marks 10

Answer

The waveform is periodic with period T and amplitude A

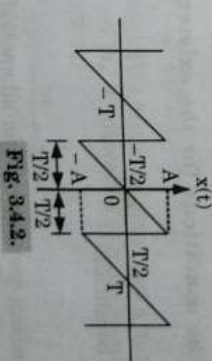


Fig. 3.4.2.

If the interval $-T/2 \leq t \leq T/2$

$$x(t) = \frac{A}{T/2} t \quad \text{and fundamental frequency } \omega_0 = \frac{2\pi}{T}$$

$$\text{Exponential series : } C_0 = \frac{1}{T} \int_0^T x(t) dt = \frac{1}{T} \int_{-T/2}^{T/2} \frac{2A}{T} t dt$$

$$= \frac{2A}{T^2} \left(\frac{t^2}{2} \right)_{-T/2}^{T/2} = \frac{2A}{2T^2} \left[\left(\frac{T}{2} \right)^2 - \left(-\frac{T}{2} \right)^2 \right] = 0$$

$$C_n = \frac{1}{T} \int_0^T x(t) e^{jn\omega_0 t} dt = \frac{1}{T} \int_{-T/2}^{T/2} \frac{2A}{T} t e^{jn\omega_0 t} dt$$

$$= \frac{2A}{T^2} \left[\left(\frac{t e^{jn\omega_0 t}}{jn\omega_0} \right)_{-T/2}^{T/2} - \int_{-T/2}^{T/2} \frac{e^{jn\omega_0 t}}{jn\omega_0} dt \right]$$

$$= \frac{2A}{T^2} \left\{ \frac{T \cos n\pi}{jn(2\pi/T)} - \frac{e^{jn\pi} - e^{-jn\pi}}{jn(2\pi/T)^2} \right\}$$

$$= \frac{A}{jn\pi} (-1)^n = j \frac{A}{n\pi} (-1)^{n+1}$$

Thus exponential series is

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \\ = \sum_{n=-\infty}^{\infty} j \frac{A}{n\pi} (-1)^{n+1} e^{jn(2\pi/T)t}$$

PART-2

Fourier Transform, Convolution / Multiplication and their Effect in the Frequency Domain, Magnitude and Phase Response, Fourier Domain Duality.

Que 3.5. State the conditions for the existence of Fourier transform.

Explain Dirichlet's condition.

OR

Answer

- The Fourier transform does not exist for all aperiodic functions. The conditions for a function $x(t)$ to have Fourier transform, called Dirichlet's conditions which are as follows :
 - $x(t)$ is absolutely integrable over the interval $-\infty$ to ∞ , i.e.,

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

- $x(t)$ has a finite number of discontinuities in every finite time interval. Further, each of these discontinuities must be finite.
- $x(t)$ has a finite number of maxima and minima in every finite time interval.
- Dirichlet's condition is a sufficient condition but not necessary condition for the signal to be Fourier transformable.

Que 3.6. What are the different properties of Fourier transform ?

OR

Explain the following properties of Fourier transform : time scaling, conjugate functions.

AKTU 2018-19, Marks 07

Answer

Properties of Fourier transform : The Fourier transform (FT) has following properties :

- Linearity :**

$$\text{If } x_1(t) \xrightarrow{\text{FT}} X_1(\omega)$$

$$\text{and } x_2(t) \xrightarrow{\text{FT}} X_2(\omega)$$

then according to this property,

$$a_1 x_1(t) + a_2 x_2(t) \xrightarrow{\text{FT}} a_1 X_1(\omega) + a_2 X_2(\omega)$$

- Symmetry or duality property :**

$$\text{If } x(t) \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then, } X(t) \xrightarrow{\text{FT}} 2\pi x(-\omega)$$

- Time-shifting :** This property states that a shift in the time-domain by an amount of a is equivalent to multiplication by $e^{-j\omega a}$ in the frequency-domain.

$$\text{If } x(t) \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then, } x(t-a) \xrightarrow{\text{FT}} e^{-j\omega a} X(\omega)$$

- Frequency-shifting :** This property states that a shift in frequency domain by an amount ' ω_0 ' is equivalent to multiplication by ' $e^{j\omega_0 t}$ ' to time-domain function $x(t)$.

$$\text{If } x(t) \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then, } e^{j\omega_0 t} x(t) \xrightarrow{\text{FT}} X(\omega - \omega_0)$$

5. Differentiation in time-domain :

$$\text{If } x(t) \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then, } \frac{dx(t)}{dt} \xrightarrow{\text{FT}} j\omega \cdot X(\omega)$$

6. Differentiation in frequency-domain :

$$\text{If } x(t) \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then, } t x(t) \xrightarrow{\text{FT}} j \frac{d}{d\omega} X(\omega)$$

7. Time-scaling: This property states that "Time compression of a signal results in its spectral (frequency-domain) expansion, whereas, time expansion of a signal results in its spectral (frequency-domain) compression".

$$\text{If } x(t) \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then, } x(at) \xrightarrow{\text{FT}} \frac{1}{|a|} X(\omega/a)$$

where $a \rightarrow$ real constant.

8. Time-integration :

$$\text{If } x(t) \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then, } \int_{-\infty}^t x(t) dt \xrightarrow{\text{FT}} \frac{1}{j\omega} X(\omega)$$

Assuming all initial conditions equal to zero.

9. Time-reversal :

$$\text{If } x(t) \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then, } x(-t) \xrightarrow{\text{FT}} X(-\omega)$$

10. Conjugation :

$$\text{If } x(t) \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then, } x^*(t) \xrightarrow{\text{FT}} X^*(-\omega)$$

11. Multiplication property (frequency-convolution) :

$$\text{If } x_1(t) \xrightarrow{\text{FT}} X_1(\omega)$$

$$\text{and } x_2(t) \xrightarrow{\text{FT}} X_2(\omega)$$

$$\text{then, } x_1(t) \cdot x_2(t) \xrightarrow{\text{FT}} \frac{1}{2\pi} [X_1(\omega) \otimes X_2(\omega)]$$

12. Convolution (time-convolution) :

$$\text{If } x_1(t) \xrightarrow{\text{FT}} X_1(\omega)$$

Signal System

3-9 J (EC-Sem-4)

$$\text{and } x_2(t) \xrightarrow{\text{FT}} X_2(\omega)$$

$$\text{then, } x_1(t) \otimes x_2(t) \xrightarrow{\text{FT}} X_1(\omega) \cdot X_2(\omega)$$

Que 3.7. Describe the magnitude and phase representation of Fourier transform.

Answer

1. The magnitude and phase representation of the Fourier transform is the tool used to analyse the transformed signal.

2. In general, $X(\omega)$ is a complex valued function of ω .

$$X(\omega) = X_R(\omega) + jX_I(\omega)$$

where, $X_R(\omega)$ is the real part of $X(\omega)$ and $X_I(\omega)$ is the imaginary part of $X(\omega)$.

3. The magnitude of $X(\omega)$ is given by

$$|X(\omega)| = \sqrt{X_R(\omega)^2 + X_I(\omega)^2}$$

and the phase of $X(\omega)$ is given by

$$\angle X(\omega) = \tan^{-1} \frac{X_I(\omega)}{X_R(\omega)}$$

4. The plot of $|X(\omega)|$ versus ω is known as amplitude spectrum, and the plot of $\angle X(\omega)$ versus ω is known as phase spectrum. The amplitude spectrum and phase spectrum together is called frequency spectrum.

Que 3.8. Show that the convolution in time-domain is same as product in frequency-domain.

OR

State and prove the convolution theorem.

Answer

i. **Time-convolution theorem :** It states that the convolution of two signals in time-domain is equivalent to the product of their spectra in frequency-domain, i.e.,

$$\text{if } x_1(t) \xrightarrow{\text{FT}} X_1(\omega) \text{ and } x_2(t) \xrightarrow{\text{FT}} X_2(\omega)$$

$$\text{then, } x_1(t) \otimes x_2(t) \xrightarrow{\text{FT}} X_1(\omega) X_2(\omega)$$

$$\text{Proof : We know, } x_1(t) \otimes x_2(t) = \int_{-\infty}^{\infty} x_1(\tau) x_2(t-\tau) d\tau$$

$$F[x_1(t) \otimes x_2(t)] = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x_1(\tau) x_2(t-\tau) d\tau \right] e^{-j\omega t} dt$$

Interchanging the order of integration, we have

$$F[x_1(t) \otimes x_2(t)] = \int_{-\infty}^{\infty} x_1(\tau) \left[\int_{-\infty}^{\infty} x_2(t-\tau) e^{-j\omega t} dt \right] d\tau$$

Substituting

$$t - \tau = p$$

$$t = \tau + p \text{ and } dt = dp$$

$$\begin{aligned} \therefore F[x_1(t) \otimes x_2(t)] &= \int_{-\infty}^{\infty} x_1(\tau) \left[\int_{-\infty}^{\infty} x_2(p) e^{-j\omega(\tau+p)} dp \right] d\tau \\ &= \int_{-\infty}^{\infty} x_1(\tau) \left[\int_{-\infty}^{\infty} x_2(p) e^{-j\omega p} dp \right] e^{-j\omega\tau} d\tau \\ &= \left[\int_{-\infty}^{\infty} x_1(\tau) e^{-j\omega\tau} d\tau \right] \left[\int_{-\infty}^{\infty} x_2(p) e^{-j\omega p} dp \right] \\ &= X_1(\omega) X_2(\omega) \end{aligned}$$

Hence, $x_1(t) \otimes x_2(t) \xrightarrow{FT} X_1(\omega) X_2(\omega)$

ii. Frequency-convolution theorem or multiplication theorem :

It states that multiplication of two functions in time-domain is equivalent to the convolution of their spectra in the frequency-domain, i.e.,

$$\text{if } x_1(t) \xrightarrow{FT} X_1(\omega) \text{ and } x_2(t) \xrightarrow{FT} X_2(\omega)$$

$$\text{then } x_1(t)x_2(t) \xrightarrow{FT} \frac{1}{2\pi} [X_1(\omega) \otimes X_2(\omega)]$$

Proof : We know, $F[x(t)] = X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$

$$\text{and } F^{-1}[X(\omega)] = x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega$$

$$\text{Now, } F[x_1(t)x_2(t)] = \int_{-\infty}^{\infty} x_1(t)x_2(t)e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\lambda)e^{j\lambda t} d\lambda \right] x_2(t)e^{-j\omega t} dt$$

Interchanging the order of integration, we have

$$\begin{aligned} F[x_1(t)x_2(t)] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\lambda) \left[\int_{-\infty}^{\infty} x_2(t)e^{j\lambda t} e^{-j\omega t} dt \right] d\lambda \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\lambda) \left[\int_{-\infty}^{\infty} x_2(t)e^{-j(\omega-\lambda)t} dt \right] d\lambda \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\lambda) X_2(\omega-\lambda) d\lambda \\ &= \frac{1}{2\pi} [X_1(\omega) \otimes X_2(\omega)] \end{aligned}$$

$$\text{Hence, } x_1(t) \cdot x_2(t) \xrightarrow{FT} \frac{1}{2\pi} [X_1(\omega) \otimes X_2(\omega)]$$

Que 3.9. Find the Fourier transform of the following function using the properties of Fourier transform.

$$y(t) = \frac{d}{dt} te^{-3t} u(t) \otimes e^{-2t} u(t)$$

Answer

1. The function $y(t)$ can be written as,

$$y(t) = \underbrace{\frac{d}{dt} te^{-3t} u(t)}_{x_1(t)} \otimes \underbrace{e^{-2t} u(t)}_{x_2(t)}$$

$$y(t) = x_1(t) \otimes x_2(t)$$

2. By Fourier transform property of convolution,

$$Y(\omega) = X_1(\omega) \cdot X_2(\omega) \quad \dots(3.9.1)$$

3. Now, if

$$x_1(t) = \frac{d}{dt} te^{-3t} u(t)$$

$$F[te^{-3t} u(t)] = \frac{1}{3+j\omega}$$

using the FT property of differentiation in frequency, we get

$$F[te^{-3t} u(t)] = \frac{j\omega}{d\omega} \left[\frac{1}{3+j\omega} \right] = \frac{1}{(3+j\omega)^2} \quad \dots(3.9.2)$$

now using the FT property of differentiation in time, we get

$$F\left[\frac{d}{dt} te^{-3t} u(t)\right] = F[x_1(t)] = j\omega F[te^{-3t} u(t)] = \frac{j\omega}{(3+j\omega)^2} \quad \dots(3.9.3)$$

and,

$$x_2(t) = e^{-2t} u(t)$$

... (3.9.4)

so,

$$X_2(\omega) = \frac{1}{2+j\omega}$$

4. Substituting eq. (3.9.3) and (3.9.4) in eq. (3.9.1), we get

$$Y(\omega) = \frac{j\omega}{(3+j\omega)^2} \cdot \frac{1}{(2+j\omega)} = \frac{j\omega}{(3+j\omega)^2 (2+j\omega)}$$

Que 3.10. Explain Fourier transform of single sided exponential pulse.

AKTU 2018-19, Marks 07

OR

Find the Fourier transform of the signals given below :

i.
$$x(t) = \begin{cases} A, & |t| < T_0 \\ 0, & |t| > T_0 \end{cases}$$

ii.
$$x(t) = e^{-at} u(t)$$

Draw the magnitude and phase response of the transformed signal.

AKTU 2017-18, Marks 07

OR

Find the Fourier transform of the signal $x(t) = e^{-at} u(t)$ and plot its magnitude and phase spectrum.

AKTU 2017-18, Marks 07

AKTU 2021-22, Marks 10

Answer

i. Given,
$$x(t) = \begin{cases} A, & |t| < T_0 \\ 0, & |t| > T_0 \end{cases}$$

Taking Fourier transform

So,
$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= A \int_{-T_0}^{T_0} e^{-j\omega t} dt = A \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-T_0}^{T_0}$$

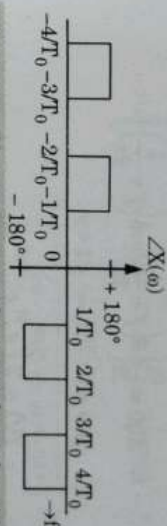
$$= -\frac{A}{j\omega} [e^{-j\omega T_0} - e^{j\omega T_0}] = \frac{A}{j\omega} [e^{j\omega T_0} - e^{-j\omega T_0}]$$

$$X(\omega) = \frac{2A}{\omega} \left[\frac{e^{j\omega T_0} - e^{-j\omega T_0}}{2j} \right] = \frac{2A}{\omega} \sin \omega T_0$$

$$= 2AT_0 \frac{\sin \omega T_0}{\omega T_0} = 2AT_0 \text{ sinc } \omega T_0$$

Magnitude and phase response of $X(\omega)$:

$$|X(\omega)|$$

Fig. 3.10.1. Magnitude spectrum of a rectangular pulse $x(t)$.Fig. 3.10.2. Phase spectrum of rectangular pulse $x(t)$.

ii. Given,

$$x(t) = e^{-at} u(t)$$

$$x(t) = \begin{cases} e^{-at}, & t > 0 \\ 0, & t < 0 \end{cases}$$

The Fourier transform of the given signal is

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$

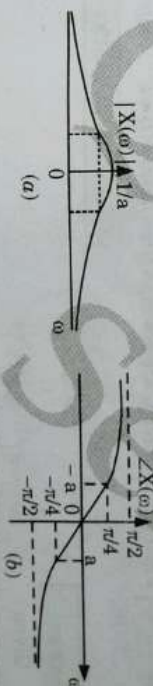
$$= \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty} = \frac{-1}{(a+j\omega)} [0 - 1] = \frac{1}{a+j\omega}$$

Magnitude and phase response :

$$|X(\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}}$$

$$\text{and } \angle X(\omega) = -\tan^{-1} \left(\frac{\omega}{a} \right)$$

Fig. 3.16.3. (a). Magnitude spectrum and (b). Phase spectrum of $x(t)$.Que 3.11. Given $x(t) = 5 \cos t$, $y(t) = 2e^{-|t|}$, find the convolution of $x(t)$ and $y(t)$ using Fourier transform.**Answer**

- Given, $x(t) = 5 \cos t$ and $y(t) = 2e^{-|t|}$
- The convolution of $x(t)$ and $y(t)$ is $z(t) = x(t) * y(t)$ then $Z(f) = X(f) \cdot Y(f)$... (3.11.1)
- Now, $x(t) = 5 \cos t$; $-\infty < t < \infty$

$$\text{so, } X(f) = \frac{5}{2} \left[\delta\left(f - \frac{1}{2\pi}\right) + \delta\left(f + \frac{1}{2\pi}\right) \right] \quad \dots(3.11.2)$$

$$\text{and } y(t) = 2e^{-|t|}; -\infty < t < \infty$$

$$\text{so, } Y(f) = 2 \int_{-\infty}^{\infty} e^{-|t|} e^{-j2\pi ft} dt$$

$$= 2 \left[\int_{-\infty}^0 e^{t} e^{-j2\pi ft} dt + \int_0^{\infty} e^{-t} e^{-j2\pi ft} dt \right] = \frac{4}{1 + \omega^2} \quad \dots(3.11.3)$$

4. Put the values of $X(f)$ and $Y(f)$ in eq. (3.11.1)

$$Z(f) = X(f) \cdot Y(f) = \frac{10}{1 + \omega^2} \left[\delta\left(f - \frac{1}{2\pi}\right) + \delta\left(f + \frac{1}{2\pi}\right) \right] \quad \dots(3.11.4)$$

5. From the sampling property of impulses, we know that

$$f(t)\delta(t - \tau) = f(\tau)\delta(t - \tau)$$

$$Z(f) = \left[\frac{10}{1 + 4\pi^2 f^2} \right]_{f=\frac{1}{2\pi}} \left[\delta\left(f - \frac{1}{2\pi}\right) + \delta\left(f + \frac{1}{2\pi}\right) \right] + \left[\frac{10}{1 + 4\pi^2 f^2} \right]_{f=-\frac{1}{2\pi}} \left[\delta\left(f - \frac{1}{2\pi}\right) + \delta\left(f + \frac{1}{2\pi}\right) \right] \quad \dots(3.11.5)$$

Taking the inverse Fourier transform on both sides,

$$z(t) = x(t) * y(t) = 10 \cos t; -\infty < t < \infty$$

Que 3.12. Find the fourier transform of the signal $x(t) = e^{-a|t|} \operatorname{sgn}(t)$.

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Answer

Given

$$x(t) = e^{-a|t|} \operatorname{sgn}(t)$$

$$x(t) = e^{-a|t|} \operatorname{sgn}(t) = \begin{cases} e^{-a(-t)}(-1) = -e^{-at} & \text{For } t < 0 \\ e^{-a(t)}(1) = e^{-at} & \text{For } t > 0 \end{cases}$$

$$F[x(t)] = X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 -e^{at} e^{-j\omega t} dt + \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$

$$= - \int_{-\infty}^0 e^{(a-j\omega)t} dt + \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= - \int_{-\infty}^0 e^{(a-j\omega)t} dt + \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= \left[- \frac{e^{(a-j\omega)t}}{(a-j\omega)} + \frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

$$= \left[\frac{-e^{\infty} - e^0}{-(a-j\omega)} + \frac{e^{\infty} - e^0}{-(a+j\omega)} \right] = - \frac{1}{(a-j\omega)} + \frac{1}{(a+j\omega)}$$

$$X(\omega) = - \frac{2j\omega}{a^2 + \omega^2}$$

Que 3.13. Using Fourier transform, find the convolution of :

$$x_1(t) = e^{-2t} \cdot u(t)$$

$$x_2(t) = e^{-3t} \cdot u(t)$$

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Answer

1. Given,

$$x_1(t) = e^{-2t} u(t)$$

$$\therefore X_1(\omega) = \frac{1}{j\omega + 2}$$

and

$$x_2(t) = e^{-3t} u(t)$$

$$\therefore X_2(\omega) = \frac{1}{j\omega + 3}$$

2. Using convolution property of Fourier transform

$$x_1(t) \otimes x_2(t) = F^{-1}[X_1(\omega) \cdot X_2(\omega)]$$

$$= F^{-1} \left[\frac{1}{(j\omega + 2)(j\omega + 3)} \right]$$

$$= F^{-1} \left[\frac{A}{j\omega + 2} + \frac{B}{j\omega + 3} \right]$$

$$= F^{-1} \left[\frac{1}{j\omega + 2} - \frac{1}{j\omega + 3} \right]$$

$$x_1(t) \otimes x_2(t) = e^{-2t} u(t) - e^{-3t} u(t)$$

Que 3.14.

Show that if $x_3(t) = ax_1(t) + bx_2(t)$, then

$$X_3(\omega) = aX_1(\omega) + bX_2(\omega)$$

Answer

1. Given,

$$x_3(t) = ax_1(t) + bx_2(t)$$

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2. Taking the Fourier transform of eq. (3.14.1)

$$\begin{aligned}
 X_g(\omega) &= F[ax_1(t)] + F[bx_2(t)] \\
 &= \int_{-\infty}^{\infty} ax_1(t)e^{-j\omega t} dt + \int_{-\infty}^{\infty} bx_2(t)e^{-j\omega t} dt \\
 &= a \int_{-\infty}^{\infty} x_1(t)e^{-j\omega t} dt + b \int_{-\infty}^{\infty} x_2(t)e^{-j\omega t} dt
 \end{aligned}$$

Thus,

$$X_g(\omega) = aX_1(\omega) + bX_2(\omega)$$

Que 3.15. A signal, $x(t)$ has a Fourier transform given by

$$X(\omega) = \frac{1}{(1 + \omega^2)}. \text{ Write down the Fourier transform of } x\left(\frac{3t}{2} - 1\right).$$

Answer

1. Given, $F[x(t)] = X(\omega) = \frac{1}{1 + \omega^2}$... (3.15.1)

We have to find out FT of $x\left(\frac{3t}{2} - 1\right)$, we have to apply two theorems:

- (i) Time shifting and (ii) Time scaling.
2. Now, $F[x(t - 1)] = e^{-j\omega} X(\omega)$ [$\because F[x(t - a)] = e^{-j\omega a} X(\omega)$]

$$\begin{aligned}
 F\left[x\left(\frac{3}{2}t - 1\right)\right] &= \frac{1}{|3/2|} e^{-j\omega(3/2)} X\left(\frac{\omega}{3/2}\right) \\
 &= \left[\because F[x(at)] = \frac{1}{|a|} X\left(\frac{\omega}{a}\right)\right]
 \end{aligned}$$

Thus,
$$F\left[x\left(\frac{3}{2}t - 1\right)\right] = \frac{2}{3} e^{-j\frac{3\omega}{2}} \frac{1}{1 + \left(\frac{2\omega}{3}\right)^2}$$

$$= \frac{2}{3} e^{-j\frac{3\omega}{2}} \frac{9}{9 + 4\omega^2} = \frac{6e^{-j\frac{3\omega}{2}}}{9 + 4\omega^2}$$

Que 3.16. Find the Fourier transform of a signal $x(t) = \frac{1}{a^2 + t^2}$.

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Answer

Given $x(t) = \frac{1}{a^2 + t^2}$

$$F[e^{-a|t|}] = \frac{2a}{a^2 + \omega^2}$$

Applying duality theorem

$$\text{i.e., } X(t) \longleftrightarrow 2\pi X(-\omega)$$

$$\begin{aligned}
 F\left(\frac{2a}{a^2 + t^2}\right) &= 2\pi e^{-a|-\omega|} \\
 &= F\left(\frac{1}{a^2 + t^2}\right) = \frac{1}{2a} 2\pi e^{-a|\omega|} = \frac{\pi}{a} e^{-a|\omega|}
 \end{aligned}$$

PART-3

Discrete-Time Fourier Transform (DTFT).

Que 3.17. What are the different properties of DTFT?

Answer

The DTFT has following properties:

1. **Linearity:** The DTFT is linear in nature.

If $x_1[n] \xrightarrow{\text{FT}} X_1(\omega)$

and $x_2[n] \xrightarrow{\text{FT}} X_2(\omega)$

then $a_1x_1[n] + a_2x_2[n] \xrightarrow{\text{FT}} a_1X_1(\omega) + a_2X_2(\omega)$

2. **Periodicity:** The DTFT is always periodic with period 2π .

If $x[n] \xrightarrow{\text{FT}} X(\omega)$

then $X(\omega + 2\pi k) = X(\omega)$

3. **Time-shifting:** This property states that if a discrete-time signal is shifted in the time-domain by an amount of ' n_0 ', a factor ' $e^{-j\omega n_0}$ ' is multiplied with $X(\omega)$.

If $x[n] \xrightarrow{\text{FT}} X(\omega)$

then $x[n - n_0] \xrightarrow{\text{FT}} e^{-j\omega n_0} X(\omega)$

4. **Frequency-shifting:** This property states that the multiplication by a factor $e^{j\omega_0 n}$ in time-domain sequence $x[n]$ results shift in frequency in $X(\omega)$ by an amount of ω_0 .

$$\text{If } x[n] \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then } e^{j\omega_0 n} \cdot x[n] \xrightarrow{\text{FT}} X(\omega - \omega_0)$$

5. **Time-reversal:** This property states that if a discrete-time signal is folded about the origin in time, its magnitude spectrum remains unchanged while the phase spectrum of this signal undergoes just change in sign.

$$\text{If } x[n] \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then } x[-n] \xrightarrow{\text{FT}} X(-\omega)$$

6. **Scaling:**

$$\text{If } x[n] \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then } y[n] = x[kn] \xrightarrow{\text{FT}} \approx X\left(\frac{\omega}{k}\right) = Y(\omega)$$

7. **Differentiation in frequency-domain:**

$$\text{If } x[n] \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then } nx[n] \xrightarrow{\text{FT}} j \frac{dX(\omega)}{d\omega}$$

8. **Multiplication in time-domain:** This property states that the multiplication of two time-domain sequence is equivalent to the convolution of their FT.

$$\text{If } x_1[n] \xrightarrow{\text{FT}} X_1(\omega)$$

$$\text{and } x_2[n] \xrightarrow{\text{FT}} X_2(\omega)$$

$$\text{then } y[n] = x_1[n] x_2[n] \xrightarrow{\text{FT}} Y(\omega) = [X_1(\omega) \otimes X_2(\omega)]$$

9. **Conjugation and conjugate symmetry:**

$$\text{If } x[n] \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then } x^*[n] \xrightarrow{\text{FT}} X^*(-\omega)$$

10. **Convolution:**

$$\text{If } x_1[n] \xrightarrow{\text{FT}} X_1(\omega)$$

$$\text{and } x_2[n] \xrightarrow{\text{FT}} X_2(\omega)$$

$$\text{then } y[n] = x_1[n] \otimes x_2[n] \xrightarrow{\text{FT}} Y(\omega) = X_1(\omega) \cdot X_2(\omega)$$

- Que 3.18.** Compare CTFT and DTFT.

Answer

S. No.	CTFT	DTFT
1.	The CTFT, $X(f)$ or $X(\omega)$ is not periodic in ω .	The DTFT, $X(\omega)$ is periodic in ω .
2.	The inverse CTFT involves integration over all frequencies from $-\infty$ to $+\infty$.	The inverse DTFT involves integration over a frequency interval of only 2π .
3.	Scaling theorem: $F[x(at)] = \frac{1}{ a } X\left(\frac{\omega}{a}\right)$	Scaling theorem: $F[x(an)] = X\left(\frac{\omega}{a}\right)$
4.	In the case of CTFT, we observed a symmetry or duality between the Fourier transform and inverse Fourier transform expressions.	In the case of DTFT, no corresponding duality exists between the DTFT and IDFT expressions.

- Que 3.19.** Calculate the DTFT of the following using properties of DTFT:

$$x[n] = u[n+3] - u[n-3]$$

Answer

1. Given, $x[n] = u[n+3] - u[n-3]$

2. DTFT, $X(\omega) = F[u[n+3] - u[n-3]]$

$$\begin{aligned}
 &= \sum_{n=-\infty}^{\infty} \{u[n+3] - u[n-3]\} e^{-j\omega n} \\
 &= \sum_{n=-3}^{\infty} (1) e^{-j\omega n} - \sum_{n=3}^{\infty} (1) e^{-j\omega n} \\
 &= e^{j3\omega} + e^{j2\omega} + e^{j\omega} + 1 + e^{-j\omega} + e^{-j2\omega} + \dots - e^{-j3\omega} - e^{-j4\omega} - \dots \\
 &= e^{j3\omega} + e^{j2\omega} + e^{j\omega} + 1 + e^{-j\omega} + e^{-j2\omega}
 \end{aligned}$$

- Que 3.20.** Find DTFT of the signal:

$$x[n] = n \cdot 3^{-n} \cdot u[-n]$$

Answer

1. Given, $x[n] = n \cdot 3^{-n} \cdot u[-n]$
2. Using differentiation in frequency-domain and time reversal properties, we have,

$$\begin{aligned}
 F[n3^{-n}u[-n]] &= j \frac{d}{d\omega} [F[3^{-n}u[-n]]] \\
 &= j \frac{d}{d\omega} [F[3^n u[n]]]_{\omega=-\omega} \\
 &= j \frac{d}{d\omega} \left[\frac{1}{1-3e^{-j\omega}} \right]_{\omega=-\omega} \\
 &= j \frac{d}{d\omega} \left[\frac{1}{1-3e^{j\omega}} \right] = j \frac{-[-3e^{j\omega}(j)]}{[1-3e^{j\omega}]^2} \\
 X(\omega) &= \frac{-3e^{j\omega}}{(1-3e^{j\omega})^2}
 \end{aligned}$$

Que 3.21.An LTI system with impulse response $h_1(n)$

$$h_1(n) = \left(\frac{1}{3}\right)^n u(n)$$

is connected in parallel with another causal LTI system with impulse response $h_2(n)$. The resulting parallel interconnection has the frequency response.

$$H(e^{j\omega}) = \frac{-12 + 5e^{-j\omega}}{12 - 7e^{-j\omega} + e^{-2j\omega}}$$

Find the impulse response $h_2(n)$.**AKTU 2017-18, Marks 07****Answer**

1. When two LTI systems are connected in parallel, the impulse response of overall system is the sum of the impulse response of the individual system. Therefore,

$$h[n] = h_1[n] + h_2[n]$$

So, the Fourier transform of $h[n]$ is given by

$$H(e^{j\omega}) = H_1(e^{j\omega}) + H_2(e^{j\omega})$$

2. Given that $h_1[n] = \left(\frac{1}{3}\right)^n u(n)$

So, the Fourier transform of $h_1[n]$ is

$$H_1(e^{j\omega}) = \frac{1}{1 - \frac{1}{3}e^{-j\omega}} = \frac{3}{3 - e^{-j\omega}}$$

3. Therefore,

$$\begin{aligned}
 H_2(e^{j\omega}) &= H(e^{j\omega}) - H_1(e^{j\omega}) \\
 &= \frac{-12 + 5e^{-j\omega}}{(12 - 7e^{-j\omega} + e^{-2j\omega})} - \frac{3}{(3 - e^{-j\omega})}
 \end{aligned}$$

So,

$$\begin{aligned}
 h_2(n) &= -2 \left(\frac{1}{4}\right)^n u(n) \\
 H_2(e^{j\omega}) &= \frac{-8}{4 - e^{-j\omega}} = -2 \cdot \frac{1}{1 - \left(\frac{1}{4}\right)e^{-j\omega}} \\
 &= \frac{-24 + 8e^{-j\omega}}{(4 - e^{-j\omega})(3 - e^{-j\omega})} = \frac{-8(3 - e^{-j\omega})}{(4 - e^{-j\omega})(3 - e^{-j\omega})} \\
 &= \frac{-12 + 5e^{-j\omega} - 3(4 - e^{-j\omega})}{(4 - e^{-j\omega})(3 - e^{-j\omega})} \\
 &= \frac{-12 + 5e^{-j\omega} - 12 + 3e^{-j\omega}}{(4 - e^{-j\omega})(3 - e^{-j\omega})} \\
 &= \frac{-24 + 8e^{-j\omega}}{(4 - e^{-j\omega})(3 - e^{-j\omega})}
 \end{aligned}$$

Que 3.22.

State and prove frequency shifting theorem of DTFT.

AKTU 2018-19, Marks 07**Answer**

- A. **Statement :** This property states that the multiplication by a factor $e^{j\omega_0 n}$ in time-domain sequence $x[n]$ results shift in frequency in $X(\omega)$ by an amount of ω_0 .

$$\text{If } x[n] \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then, } e^{j\omega_0 n} x[n] \xrightarrow{\text{FT}} X(\omega - \omega_0)$$

B. Prove :

1. Fourier transform of signal $x[n]$

$$F[x[n]] = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

2. Multiplying $e^{j\omega_0 n}$ in the signal $x[n]$ and taking Fourier transform

$$\begin{aligned}
 F[x[n] e^{j\omega_0 n}] &= \sum_{n=-\infty}^{\infty} x[n] e^{j\omega_0 n} e^{-j\omega n} \\
 &= \sum_{n=-\infty}^{\infty} x[n] e^{-j(\omega - \omega_0)n} \\
 &= X(\omega - \omega_0)
 \end{aligned}$$

Que 3.23.

Analyze the Discrete Time Fourier Transform of the following

$$x(n) = 0.5^n u(n) + 2^{-n} u(-n-1)$$

AKTU 2021-22, Marks 10

Answer

1. Given, $x(n) = 0.5u(n) + 2^{-n}u(-n-1)$
 $X(\omega) = F\{x[n]\}$
2. DTFT,

$$\begin{aligned}
 &= \sum_{n=-\infty}^{\infty} [0.5u[n] + 2^{-n}u[-n-1]]e^{-j\omega n} \\
 &= \sum_{n=-\infty}^{\infty} 0.5u[n]e^{-j\omega n} + \sum_{n=-\infty}^{\infty} 2^{-n}u[-n-1]e^{-j\omega n} \\
 &= \sum_{n=0}^{\infty} (0.5e^{-j\omega})^n + \sum_{n=-\infty}^{-1} 2^{-n}e^{-j\omega n} \\
 &= \sum_{n=0}^{\infty} (0.5e^{-j\omega})^n + \sum_{n=1}^{\infty} (2e^{j\omega})^n \\
 &= \frac{1}{1-0.5e^{-j\omega}} + \frac{2e^{j\omega}}{1-2e^{j\omega}}
 \end{aligned}$$

Que 3.24. State and prove the frequency shifting theorem of CTFT.

AKTU 2021-22, Marks 10

Answer

1. **Frequency shifting :** Frequency shifting property states that the multiplication of a time domain signal $x(t)$ by $e^{j\omega_0 t}$ results in the frequency spectrum shifted by ω_0 i.e.,

$$\text{if } x(t) \xrightarrow{\text{FT}} X(\omega)$$

$$\text{then, } e^{j\omega_0 t} x(t) \xrightarrow{\text{FT}} X(\omega - \omega_0)$$

Proof : By definition,

$$\begin{aligned}
 F\{e^{j\omega_0 t} x(t)\} &= \int_{-\infty}^{\infty} e^{j\omega_0 t} x(t) e^{-j\omega t} dt \\
 &= \int_{-\infty}^{\infty} x(t) e^{-j(\omega - \omega_0)t} dt \\
 &= X(\omega - \omega_0) \\
 e^{j\omega_0 t} x(t) &\xrightarrow{\text{FT}} X(\omega - \omega_0) \\
 e^{-j\omega_0 t} x(t) &\xrightarrow{\text{FT}} X(\omega + \omega_0)
 \end{aligned}$$

Que 3.25.

Consider a causal LTI system with frequency response

$$H(\omega) = \frac{1}{4 + j\omega}. \text{ For a particular input } x(t) \text{ the system is observed to}$$

produce the output

$$y(t) = e^{-2t}u(t) - e^{-4t}u(t). \text{ Find the input } x(t).$$

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Answer

$$\begin{aligned}
 y(t) &= e^{-2t}u(t) - e^{-4t}u(t) \\
 Y(j\omega) &= \frac{1}{(j\omega + 2)} - \frac{1}{(j\omega + 4)} \\
 Y(j\omega) &= \frac{2}{(j\omega + 2)(j\omega + 4)} \\
 H(j\omega) &= \frac{Y(j\omega)}{X(j\omega)} \\
 X(j\omega) &= \frac{Y(j\omega)}{H(j\omega)} \\
 &= \frac{2}{(j\omega + 4)} \\
 x(t) &= 2e^{-4t}u(t)
 \end{aligned}$$

PART-4

Discrete Fourier Transform (DFT). Parseval's Theorem, the Idea of Signal Space and Orthogonal Bases.

Que 3.26. Explain the properties of DFT.

Answer

1. **Linearity :**

$$a_1x_1[n] + a_2x_2[n] \leftrightarrow a_1X_1[k] + a_2X_2[k] \quad \dots(3.26.1)$$

2. **Time shifting :**

$$x[n - n_0] \bmod N \leftrightarrow W_N^{kn_0} X[k], \quad W_N = e^{-j2\pi/N} \quad \dots(3.26.2)$$

3. Frequency Shifting :

$$W_N^{k_0 n} x[n] \leftrightarrow X[k - k_0]_{\text{mod } N}$$

... (3.26.3)

4. Conjugation :

$$x^*[n] \leftrightarrow X^*[-k]_{\text{mod } N}$$

... (3.26.4)

where * denotes the complex conjugate.

5. Time Reversal :

$$x[-n]_{\text{mod } N} \leftrightarrow X^*[-k]_{\text{mod } N}$$

... (3.26.5)

6. Duality : If $x[n] \leftrightarrow X[k]$

$$\text{then } X[n] \leftrightarrow Nx[-k]_{\text{mod } N}$$

... (3.26.6)

7. Circular convolution :

$$x_1[n] \otimes x_2[n] \leftrightarrow X_1[k] X_2[k]$$

... (3.26.7)

$$\text{where } x_1[n] \otimes x_2[n] = \sum_{i=0}^{N-1} x_1[i] x_2[n-i]_{\text{mod } N}$$

... (3.26.8)

The convolution sum in eq. (3.26.8) is known as the circular convolution of $x_1[n]$ and $x_2[n]$.**8. Multiplication :**

$$x_1[n] x_2[n] \leftrightarrow \frac{1}{N} X_1[k] \otimes X_2[k]$$

... (3.26.9)

$$\text{where } X_1[k] \otimes X_2[k] = \sum_{l=0}^{N-1} X_1[l] X_2[k-l]_{\text{mod } N}$$

... (3.26.10)

9. Additional Properties : When $x[n]$ is real, let

$$x[n] = x_e[n] + x_o[n]$$

where $x_e[n]$ and $x_o[n]$ are the even and odd components of $x[n]$, respectively. Let

$$x[n] \leftrightarrow X[k] = A[k] + jB[k] = |X[k]| e^{j\theta[k]}$$

Then $|X[-k]|_{\text{mod } N} = X^*[k]$

$$x_e[n] \leftrightarrow \text{Re } |X[k]| = A[k]$$

$$x_o[n] \leftrightarrow j \text{Im } |X[k]| = jB[k]$$

$$A[-k]_{\text{mod } N} = A[k], \quad B[-k]_{\text{mod } N} = -B[k]$$

$$|X[-k]|_{\text{mod } N} = |X[k]|, \quad \theta[-k]_{\text{mod } N} = -\theta[k]$$

10. Parseval's relation :

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2$$

... (3.26.11)

Eq. (3.26.11) is known as Parseval's identity (or Parseval's theorem) for the DFT.

Que 3.27.

State and prove Parseval's theorem and Parseval's identity.

OR

Explain and proof Parseval's Theorem.

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Answer**i. Parseval's theorem :**

Parseval's theorem states that if

$$x_1(t) \xrightarrow{FT} X_1(\omega) \text{ and } x_2(t) \xrightarrow{FT} X_2(\omega)$$

$$\text{then } \int_{-\infty}^{\infty} x_1(t) x_2^*(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\omega) X_2^*(\omega) d\omega$$

ii. Parseval's identity :If $x_1(t) = x_2(t) = x(t)$, then Parseval's identity states that the energy content of $x(t)$ is

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

Proof :

$$1. \int_{-\infty}^{\infty} x_1(t) x_2^*(t) dt = \int_{-\infty}^{\infty} \left\{ \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\omega) e^{j\omega t} d\omega \right\} x_2^*(t) dt$$

Interchanging the order of integration, we have

$$\begin{aligned} \int_{-\infty}^{\infty} x_1(t) x_2^*(t) dt &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\omega) \left\{ \int_{-\infty}^{\infty} x_2^*(t) e^{j\omega t} dt \right\} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\omega) \left\{ \int_{-\infty}^{\infty} x_2(t) e^{-j\omega t} dt \right\} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\omega) [X_2(\omega)]^* d\omega \end{aligned}$$

$$\text{So, } \int_{-\infty}^{\infty} x_1(t) x_2^*(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\omega) X_2^*(\omega) d\omega$$

2. If $x_1(t) = x_2(t) = x(t)$, then the energy of the signal is given by

$$E = \int_{-\infty}^{\infty} x(t) x^*(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) X^*(\omega) d\omega$$

Since, $|x(t)|^2 = x(t)x^*(t)$ and $|X(\omega)|^2 = X(\omega)X^*(\omega)$, we get

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

Que 3.28.

Discuss signal space and orthogonal bases.

Answer**Signal space :**

1. The basic idea is to view waveforms, i.e., finite-energy functions, as vectors in a certain vector space that we call signal-space.
2. The set of waveforms in this space is the set of finite-energy complex functions $u(t)$ mapping each t on the real time axis into a complex value.
3. The energy in $u(t)$ is defined as

$$\|u\|^2 = \int_{-\infty}^{\infty} |u(t)|^2 dt$$

4. The set of functions that we are considering is the set for which this integral exists and is finite.
5. This set of waveforms is called the space of finite-energy complex functions, or, more briefly, the space of complex waveforms.

Orthogonal bases :

1. In an inner product space, a set of vectors $\phi_1, \phi_2, \dots$ is orthonormal if

$$(\phi_j, \phi_k) = \begin{cases} 0 & \text{for } j \neq k \\ 1 & \text{for } j = k \end{cases}$$

2. An orthonormal set is a set of orthogonal vectors where each vector is normalized in the sense of having unit length.
3. If a set of vectors $v_1, v_2, \dots$ is orthogonal, then the set is,

$$\phi_j = \frac{1}{\|v_j\|} v_j$$

4. If two vectors are orthogonal, then any scaling (including normalization) of each vector maintains the orthogonality.
5. If we project a vector u onto a normalized vector ϕ , then, the projection has the simplified form

$$u = (u, \phi)\phi$$

PART-5

The Laplace Transform, Notion of Eigen Functions of LSI Systems, a Basis of Eigen Functions, Region of Convergence, Poles and Zeros of System.

Que 3.29.

What are the different properties of unilateral Laplace transform ?

OR

Show that if $x_3(t) = ax_1(t) + bx_2(t)$ then $X_3(s) = aX_1(s) + bX_2(s)$.

Answer

There are following properties of unilateral Laplace transform :

1. **Linearity :** Considering two signals $x_1(t)$ and $x_2(t)$ with Laplace transforms $X_1(s)$ and $X_2(s)$ respectively,

$$\text{If } x_1(t) \xrightarrow{LT} X_1(s)$$

$$\text{and } x_2(t) \xrightarrow{LT} X_2(s)$$

$$\text{then, } L[a_1x_1(t) + a_2x_2(t)] = a_1X_1(s) + a_2X_2(s)$$

where, a_1 and a_2 are constants.

2. **Time shifting :**

If the $x(t)$ signal is delayed in time by t_0 unit then its Laplace transform is multiplied by e^{-st_0} factor.

$$\text{if } x(t) \xrightarrow{LT} X(s)$$

$$\text{then, } x(t-t_0) \xrightarrow{LT} e^{-st_0}X(s)$$

where, t_0 is a constant.

3. **Frequency shifting :** If there is any shift in $X(s)$ by a unit, it will cause a multiplication factor e^{at} in signal $x(t)$.

$$\text{if } x(t) \xrightarrow{LT} X(s)$$

$$\text{then, } e^{at}x(t) \xrightarrow{LT} X(s-a)$$

4. **Differentiation theorem :**

$$\text{If } x(t) \xrightarrow{LT} X(s)$$

then differentiation theorem gives,

$$L\left[\frac{d}{dt}x(t)\right] \xrightarrow{LT} sX(s) - x(0^-)$$

where, $x(0^-) \rightarrow$ value of $x(t)$ at $t = 0^-$

$$(t = 0^-) \rightarrow \text{indicates the time first before } t = 0$$

5. **Differentiation by s :** The differentiation in complex frequency-domain corresponds to multiplication by t in the time-domain.

$$\text{If } x(t) \xrightarrow{LT} X(s)$$

$$\text{then, } tx(t) \xrightarrow{LT} -\frac{d}{ds}X(s)$$

$$\text{then } t^n x(t) \xrightarrow{LT} (-1)^n \frac{d^n}{ds^n} X(s)$$

5. **Integration theorem :** This theorem states that,

$$\text{If } x(t) \xrightarrow{LT} X(s)$$

$$\text{then, } \int_0^t x(t)dt \xrightarrow{LT} \frac{X(s)}{s}$$

6. **Scaling property :** This theorem states that,

$$\text{If } x(t) \xrightarrow{LT} X(s)$$

$$\text{then, } x(at) \xrightarrow{\text{LT}} \frac{1}{|a|} X\left(\frac{s}{a}\right)$$

7. **Convolution theorem** : The convolution theorem can be given in two forms.

i. **Convolution in time-domain** : According to this theorem the Laplace transform of convolution of two functions in time-domain is equivalent to the multiplication of their Laplace transforms.

$$\text{If } x_1(t) \xrightarrow{\text{LT}} X_1(s) \text{ and } x_2(t) \xrightarrow{\text{LT}} X_2(s)$$

$$\text{then, } x_1(t) \otimes x_2(t) \xrightarrow{\text{LT}} X_1(s) \cdot X_2(s)$$

ii. **Convolution in frequency-domain** : According to this theorem, the inverse Laplace transform of convolution of two functions in frequency-domain is equivalent to the linear multiplication of these functions in the time-domain.

$$L^{-1}[X_1(s) \otimes X_2(s)] = x_1(t) \cdot x_2(t)$$

$$\text{or, } x_1(t) \cdot x_2(t) \xrightarrow{\text{LT}} X_1(s) \otimes X_2(s)$$

8. **Initial value theorem** : Using initial value theorem, we can directly find out the initial value of $x(t)$, i.e., $x(0)$ from $X(s)$, without finding the inverse Laplace transform of $X(s)$.

$$\text{If } x(t) \xrightarrow{\text{LT}} X(s) \text{ then the initial value of } x(t) \text{ can be calculated as,}$$

$$x(0^+) = \lim_{t \rightarrow 0^+} x(t) = \lim_{s \rightarrow \infty} [sX(s)]$$

9. **Final value theorem** : Using this theorem the final value of $x(t)$ (i.e., $x(\infty)$) can be calculated directly from $X(s)$ without finding the inverse Laplace transform of $X(s)$.

$$\text{If } x(t) \xrightarrow{\text{LT}} X(s) \text{ then the final value theorem gives,}$$

$$x(\infty) = \lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} [sX(s)]$$

Que 3.30. State and prove initial and final value theorem of Laplace transform.

Answer

AKTU 2018-19, Marks 07

Initial value theorem : Refer Q. 3.29, Page 3-26J, Unit-3.
Proof : Using differentiation in time property for unilateral Laplace transform,

$$L\left[\frac{dx(t)}{dt}\right] = \int_0^{\infty} \frac{dx(t)}{dt} e^{-st} dt = sX(s) - x(0^-)$$

$$sX(s) - x(0^-) = \int_0^{\infty} \frac{dx(t)}{dt} e^{-st} dt + \int_0^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

$$= [x(t)]_0^{\infty} + \int_0^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

$$= x(0^+) - x(0^-) + \int_0^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

$$\text{so, } sX(s) = x(0^+) + \int_0^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

$$\text{and } \lim_{s \rightarrow \infty} [sX(s)] = x(0^+) + \lim_{s \rightarrow \infty} \int_0^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

$$= x(0^+) + \int_0^{\infty} \frac{dx(t)}{dt} \left(\lim_{s \rightarrow \infty} e^{-st} \right) dt = x(0^+)$$

$$\text{hence, } x(0^+) = \lim_{s \rightarrow \infty} [sX(s)]$$

Final value theorem : Refer Q. 3.29, Page 3-26J, Unit-3.

Proof : Using differentiation in time property, for unilateral Laplace transform,

$$L\left[\frac{dx(t)}{dt}\right] = \int_0^{\infty} \frac{dx(t)}{dt} e^{-st} dt = sX(s) - x(0^-)$$

$$\lim_{s \rightarrow 0} [sX(s) - x(0^-)] = \int_0^{\infty} \frac{dx(t)}{dt} (\lim_{s \rightarrow 0} e^{-st}) dt$$

$$= \int_0^{\infty} \frac{dx(t)}{dt} dt = [x(t)]_0^{\infty} = x(\infty) - x(0^-)$$

hence,

$$x(\infty) = \lim_{s \rightarrow 0} [sX(s)]$$

Que 3.31. Explain basis of eigen function.

Answer

1. Eigen functions are special functions associated with a given type of system in which the output of the system has same functional form as the input but with a possible scale factor.
The scale factor is called eigen value.
For example, the derivative operator has exponential function for their eigen functions because
- 2.

$$\frac{d}{dx} e^{-ax} = -a e^{-ax}$$

- with the decay constant α as the eigen value.
3. Digitized sinusoidal functions are eigen functions of digital LSI systems.

Que 3.32. Describe the region of convergence of Laplace transform.

Answer

Region of convergence (ROC):

- The ROC is defined as the region where the value of $X(s)$ is finite.
- For example :

$$X(s) = L[e^{at}] = \frac{1}{(s-a)}$$

Here, the Laplace transform is defined only for $\text{Re}[X(s)] > a$. The region $\text{Re}[X(s)] > a$, i.e., $\sigma > a$ is called the region of convergence (ROC) as shown in Fig. 3.32.1.

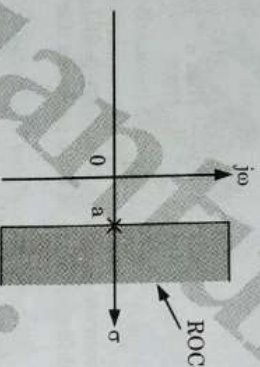


Fig. 3.32.1. ROC of $e^{at} u(t)$.

- ROC is required to compute the Laplace transform and inverse Laplace transform. If the ROC is not specified, the inverse Laplace transform is not unique.

Que 3.33. Explain poles and zeroes of the system.

Answer

- The most commonly encountered form of the Laplace transform in engineering is a ratio of two polynomials in s ; that is

$$X(s) = \frac{b_M s^M + b_{M-1} s^{M-1} + \dots + b_0}{s^N + a_{N-1} s^{N-1} + \dots + a_1 s + a_0}$$

- Factor $X(s)$ as a product of terms involving the roots of the denominator and numerator polynomials :

$$X(s) = \frac{b_M \prod_{k=1}^M (s - c_k)}{\prod_{k=1}^N (s - d_k)}$$

- The c_k are the roots of the numerator polynomial and are termed the zeros of $X(s)$.
- The d_k are the roots of the denominator polynomial and are termed the poles of $X(s)$.

- We denote the locations of zeros in the s -plane with the "O" symbol and the locations of poles with the "X" symbol, as illustrated in Fig. 3.42.1.
- The locations of poles and zeros in the s -plane uniquely specify $X(s)$, except for the constant gain factor b_M .

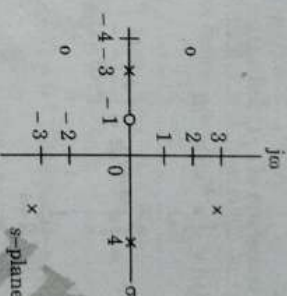


Fig. 3.33.1.

Que 3.34. If Laplace transform of $x(t)$ is $\frac{(s+2)}{(s^2+4s+5)}$. Determine

Laplace transform of $y(t) = x(2t-1)u(2t-1)$.

AKTU 2017-18, Marks 3.5

Answer

- The Laplace transform of $x(t) u(t)$ is $\frac{(s+2)}{(s^2+4s+5)}$
- Applying time shifting property.
 $x(t-1)u(t-1) \xrightarrow{\text{LT}} e^{-s} \frac{(s+2)}{(s^2+4s+5)}$
- Now, applying time scaling property.

$$x(2t-1)u(2t-1) \xrightarrow{\text{LT}} \frac{1}{2} e^{-\frac{s}{2}} \left[\left(\frac{s}{2} \right)^2 + 4 \left(\frac{s}{2} \right) + 5 \right]$$

$$= e^{-\frac{s}{2}} \frac{(s+4)}{(s^2+8s+20)}$$

- Hence, the Laplace transform of $y(t) = x(2t-1)u(2t-1)$ is,

$$Y(s) = \frac{e^{-\frac{s}{2}} (s+4)}{(s^2+8s+20)}$$

Que 3.35. Use the convolution theorem to find the Laplace transform of $y(t) = x_1(t) \otimes x_2(t)$, if $x_1(t) = e^{-3t} u(t)$ and $x_2(t) = u(t-2)$.

Answer

- From the convolution property, the convolution of two signals in time-domain results in the multiplication of those signals in frequency-domain.
- So taking Laplace transform of $x_1(t)$ and $x_2(t)$.

$$L\{x_1(t)\} = \int_{-\infty}^{\infty} e^{-st} u(t) e^{-st} dt$$

$$= \int_0^{\infty} e^{-(3+s)t} dt = \left[\frac{e^{-(3+s)t}}{-(3+s)} \right]_0^{\infty}$$

$$X_1(s) = \frac{1}{3+s}$$

3.

$$L\{x_2(t)\} = \int_{-\infty}^{\infty} u(t-2)e^{-st} dt$$

$$= \int_2^{\infty} e^{-st} dt = \left[\frac{e^{-st}}{-s} \right]_2^{\infty}$$

$$X_2(s) = \left[0 - \frac{e^{-2s}}{-s} \right] = \frac{e^{-2s}}{s}$$

4. So,

$$Y(s) = X_1(s) \cdot X_2(s) = \frac{e^{-2s}}{s(s+3)}$$

Que 3.36.

If $X(s) = \frac{5s-7}{(s-1)(s+2)}$ with $-2 < R\{s\} < -1$. Find $x(t)$.

AKTU 2017-18, Marks 07**Answer**

$$1. \text{ Given, } X(s) = \frac{5s-7}{(s-1)(s+2)}, -2 < R\{s\} < -1$$

$$\text{Now, } \frac{5s-7}{(s-1)(s+2)} = \frac{k_1}{(s-1)} + \frac{k_2}{(s+2)}$$

$$k_1 = \frac{5s-7}{s+2} \Big|_{s=-2} = \frac{-2}{3}$$

$$k_2 = \frac{5s-7}{s-1} \Big|_{s=-2} = \frac{-17}{-3} = \frac{17}{3}$$

$$2. \text{ So, } X(s) = \frac{-2}{3(s-1)} + \frac{17}{3(s+2)} = X_1(s) + X_2(s)$$

The given ROC is: $-2 < R\{s\} < -1$

Since, the pole $s = -2$ is to the left of the ROC, so,

$$X_2(s) = \frac{17}{3(s+2)}$$

then,

$$x_2(t) = \frac{17}{3} e^{-2t} u(t)$$

- And since, the pole $s = 1$ lies right to the ROC and so giving non-causal term, hence

$$X_1(-s) = \frac{-2}{3(-s-1)} = \frac{2}{3(s+1)}$$

then,

$$x_1(t) = \frac{2}{3} e^{-t} u(t)$$

So,

$$x(t) = L^{-1}\{X(s)\} = x_1(t) + x_2(t)$$

$$x(t) = \frac{2}{3} e^{-t} u(t) + \frac{17}{3} e^{-2t} u(t)$$

Que 3.37.

For a LTI system with unit impulse response $h(t) = e^{-2t} u(t)$. Determine the output to the input $x(t) = e^{-t} u(t)$.

AKTU 2018-19, Marks 07**Answer**

- For an LTI system,

$$y(t) = h(t) * x(t)$$

$$Y(s) = H(s) X(s)$$

2.

$$h(t) = e^{-2t} u(t)$$

$$L\{h(t)\} = H(s) = \frac{1}{s+2}$$

3.

$$x(t) = e^{-t} u(t)$$

$$L\{x(t)\} = X(s) = \frac{1}{s+1}$$

4.

$$y(t) = L^{-1} \left[\frac{1}{s+1} \times \frac{1}{s+2} \right]$$

$$y(t) = L^{-1} \left[\frac{1}{s+1} - \frac{1}{s+2} \right]$$

$$y(t) = L^{-1} \left(\frac{1}{s+1} \right) - L^{-1} \left(\frac{1}{s+2} \right)$$

$$y(t) = (e^{-t} - e^{-2t}) u(t)$$

Que 3.38.

Find impulse response of system described by the equation $2y'(t) + 3y(t) = x(t)$.

AKTU 2018-19, Marks 07

3-34 J (EC-Sem-4)

Answer

1. We have, $\frac{2}{s} \frac{dy(t)}{dt} + 3y(t) = x(t)$
2. Taking Laplace on both sides by considering zero initial conditions
 $2sY(s) + 3Y(s) = X(s)$
 $H(s) = \frac{Y(s)}{X(s)} = \frac{1}{2s+3} = \frac{1/2}{s+3/2}$
3. Taking Inverse Laplace transform
 $h(t) = \frac{1}{2} e^{-\frac{3}{2}t} u(t)$

Que 3.39. Find Laplace transform of following signal and draw

ROC $x(t) = \cos(3t + \pi/4) u(t)$.

AKTU 2018-19, Marks 07

Answer

1. $x(t) = \cos(3t + \pi/4) u(t)$
 $x(t) = (\cos 3t \cos \pi/4 - \sin 3t \sin \pi/4) u(t)$
 $= \left(\frac{1}{\sqrt{2}} \cos 3t - \frac{1}{\sqrt{2}} \sin 3t \right) u(t)$
 $x(t) = \frac{1}{\sqrt{2}} (\cos 3t - \sin 3t) u(t)$
2. Taking Laplace transform on both sides
 $L[x(t)] = \frac{1}{\sqrt{2}} \left[\frac{s}{s^2 + 3^2} - \frac{3}{s^2 + 3^2} \right]$
 $X(s) = \frac{1}{\sqrt{2}} \left[\frac{s-3}{s^2 + 3^2} \right]$
3. The location of the poles are $(3j)$ and $(-3j)$.
 The ROC and location of poles of $X(s)$ is shown in Fig. 3.39.1.

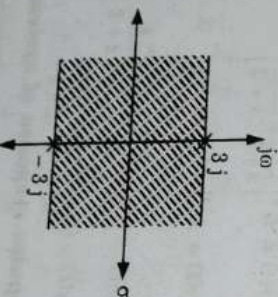


Fig. 3.39.1.

Signal System

3-35 J (EC-Sem-4)

Que 3.40. Find the inverse Laplace of the following :

- $X(S) = \frac{2}{(s+4)(s-1)}$ if the region of convergence is $-4 < \text{Re}(s) < 1$
- $\text{Re}(s) > 1$

AKTU 2021-22, Marks 10

Answer

1. Given, $X(s) = \frac{1}{(s+4)(s-1)}$

$$X(s) = \frac{A}{(s+4)} + \frac{B}{(s-1)}$$

On solving partial fraction then we get

$$A = -\frac{1}{5}, B = \frac{1}{5}$$

$$\text{So } X(s) = -\frac{1}{5} \frac{1}{(s+4)} + \frac{1}{5} \frac{1}{(s-1)}$$

- The $X(s)$ has pole at $s = -4$ and $s = 1$. The ROC for $-4 < \text{Re}(s) < 1$ is shown in Fig. 3.40.1(a). The pole at $s = -4$ which is at the left of the strip of ROC corresponds to the causal signal and the pole at $s = 1$ which is at the right of the strip of ROC corresponds to the anticausal signal.

$$\text{Therefore, } x(t) = -\frac{1}{5} e^{-4t} u(-t) - \frac{1}{5} e^t u(t)$$

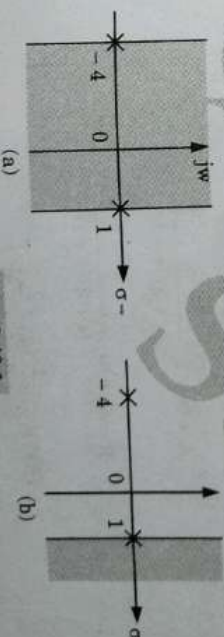


Fig. 3.40.1.

- The ROC is $\text{Re}(s) > 1$ as shown in Fig. 3.40.1(b). Both the poles lie to the left of ROC, so both the poles correspond to causal signal.
 Therefore, $x(t) = -\frac{1}{5} e^{-4t} u(t) + \frac{1}{5} e^t u(t)$

PART-6

Laplace Domain Analysis, Solution to Differential Equations and System Behaviour.

Que 3.41. LTI System, which is initially at rest is described by

$$\text{differential equation, } \frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt}.$$

Calculate system transfer function and impulse response.

Answer

1. Given, $\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt}$
2. Taking Laplace transform on both sides,
 $s^2 Y(s) + 3sY(s) + 2Y(s) = sX(s)$
 $[s^2 + 3s + 2] Y(s) = sX(s)$

$$\text{3. Transfer function, } H(s) = \frac{Y(s)}{X(s)} = \frac{s}{s^2 + 3s + 2} \quad \dots(3.41.1)$$

4. Now impulse response can be calculated for $x(t) = \delta(t) \Rightarrow X(s) = 1$

$$\text{From eq. (3.41.1), } H(s) = \frac{s}{(s+2)(s+1)}$$

5. Using partial fraction expansion,

$$H(s) = \frac{A}{(s+1)} + \frac{B}{(s+2)}$$

$$A = (s+1) \left. \frac{s}{(s+2)(s+1)} \right|_{s=-1} = -1$$

$$B = (s+2) \left. \frac{s}{(s+2)(s+1)} \right|_{s=-2} = 2$$

$$6. H(s) = \frac{-1}{(s+1)} + \frac{2}{(s+2)}$$

Taking inverse Laplace transform, we get impulse response,

$$h(t) = [-e^{-t} + 2e^{-2t}] u(t)$$

Que 3.42.

Find the energy spectral density of $f(t) = e^{-st} u(t)$.

AKTU 2018-19, Marks 07

Answer

Taking Fourier transform of

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-st} u(t) e^{-j\omega t} dt \\ &= \int_0^{\infty} e^{-(s+j\omega)t} dt = \frac{-1}{s+j\omega} \left[e^{-(s+j\omega)t} \right]_0^{\infty} \end{aligned}$$

$$X(\omega) = \frac{1}{s+j\omega}$$

$$EDS = |X(\omega)|_{s=j\omega}^2$$

$$EDS = \left| \frac{1}{2j\omega} \right|^2 = \frac{1}{4\omega^2}$$

Que 3.43. Using Laplace transform obtain the impulse response of the given second-order system

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = x(t)$$

AKTU 2021-22, Marks 10

Answer

1. Given, $\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = x(t)$

2. Taking Laplace transform

$$s^2 Y(s) - sy(0) - y'(0) + 3[sY(s) - y'(0)] + 2Y(s) = X(s)$$

$$Y(s)[s^2 + 3s + 2] - s.3 - 4 - 3 \times 3 = X(s)$$

$$Y(s)[s^2 + 3s + 2] - 3s - 13 = X(s) = \frac{4}{s+2}$$

$$3. Y(s) = \frac{\frac{4}{s+2} + 3s + 13}{s^2 + 3s + 2}$$

$$Y(s) = \frac{4 + (3s + 13)(s + 2)}{(s + 2)(s^2 + 3s + 2)} = \frac{4 + 3s^2 + 19s + 26}{(s + 2)(s^2 + 3s + 2)}$$

$$4. Y(s) = \frac{3s^2 + 19s + 30}{(s + 2)^2 (s + 1)}$$

$$Y(s) = \frac{3s^2 + 19s + 30}{(s + 2)^2 (s + 1)} = \frac{A}{s + 2} + \frac{B}{(s + 2)^2} + \frac{C}{s + 1}$$

6. On solving partial fraction, we get

$$A = -11, B = -4, C = 14$$

$$7. Y(s) = \frac{-11}{s + 2} + \frac{(-4)}{(s + 2)^2} + \frac{14}{s + 1}$$

$$y(t) = -11e^{-2t}u(t) - 4te^{-2t}u(t) + 14e^{-t}u(t)$$

$$y(t) = -e^{-2t}(11 + 4t)u(t) + 14e^{-t}u(t).$$

[Note : Assume : $y(0) = 3$, $y'(0) = 4$, $x(t) = 4e^{-2t}$ at $t \geq 0$]

Que 3.44. If $X(s) = \frac{2s+3}{(s+1)(s+2)}$, Find $x(t)$ for

- System is stable.
- System is causal.
- System is non-causal.

Answer

$$\text{Given, } X(s) = \frac{2s+3}{(s+1)(s+2)}$$

Using partial fraction expansion,

$$X(s) = \frac{A}{(s+1)} + \frac{B}{(s+2)}$$

on solving partial fraction, $A = 1$, $B = 1$

so,

$$X(s) = \frac{1}{(s+1)} + \frac{1}{(s+2)} \quad \dots (3.44.1)$$

- If system is stable, its ROC should include $j\omega$ axis, thus ROC is $\text{Re}(s) > -1$.

Now taking inverse Laplace transform of eq. (3.44.1)

$$x(t) = e^{-t}u(t) + e^{-2t}u(t)$$

- If system is causal then ROC is $\text{Re}(s) > -1$, hence taking inverse Laplace transform of eq. (3.44.1),

$$x(t) = e^{-t}u(t) + e^{-2t}u(t)$$

- If system is non-causal, then ROC is $\text{Re}(s) < -2$. Hence taking inverse Laplace transform of eq. (3.44.1)

$$x(t) = -e^{-t}u(-t) - e^{-2t}u(-t)$$

Que 3.45. $\frac{5}{5} \frac{d^2 y(t)}{dt^2} + 8 \frac{dy(t)}{dt} + 4y(t) = 3x(t)$ for the system described

by the above differential equation, determine whether the system is underdamped, overdamped or critically damped. Also find the impulse response of the system.

Answer

- Given, $5 \frac{d^2 y(t)}{dt^2} + 8 \frac{dy(t)}{dt} + 4y(t) = 3x(t)$...(3.45.1)

- Taking Laplace transform of both sides of eq. (3.45.1)

$$5s^2 Y(s) + 8s Y(s) + 4Y(s) = 3X(s)$$

$$(5s^2 + 8s + 4) Y(s) = 3X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{3}{5s^2 + 8s + 4}$$

$$\text{Comparing with } (s^2 + 2\zeta\omega_n s + \omega_n^2) \quad s^2 + \frac{8}{5}s + \frac{4}{5}$$

thus,

$$\omega_n^2 = 4/5 = 0.8, \quad 2\zeta\omega_n = \frac{8}{5}$$

$$\xi = 0.894$$

As, $\xi < 1$, thus system is underdamped.

- We can also write $H(s)$ as

$$H(s) = \frac{3/5}{\left(s + \frac{4}{5} - \frac{2}{5}i\right)\left(s + \frac{4}{5} + \frac{2}{5}i\right)}$$

Doing partial fraction expansion,

$$\begin{aligned} H(s) &= \frac{A}{\left(s + \frac{4}{5} - \frac{2}{5}i\right)} + \frac{B}{\left(s + \frac{4}{5} + \frac{2}{5}i\right)} \\ &= \frac{3/4i}{\left(s + \frac{4}{5} - \frac{2}{5}i\right)} - \frac{3/4i}{\left(s + \frac{4}{5} + \frac{2}{5}i\right)} \\ &= \frac{3}{4i} \left[\frac{1}{\left(s + \frac{4}{5} - \frac{2}{5}i\right)} - \frac{1}{\left(s + \frac{4}{5} + \frac{2}{5}i\right)} \right] \end{aligned}$$

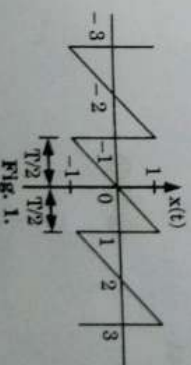
Taking inverse Laplace transform, the impulse response

$$\begin{aligned} h(t) &= \frac{3}{4i} \left[e^{-\left(\frac{4}{5} - \frac{2}{5}i\right)t} - e^{-\left(\frac{4}{5} + \frac{2}{5}i\right)t} \right] u(t) \\ &= \frac{3}{4i} \left[e^{-\frac{4}{5}t} \left(e^{\frac{2}{5}it} - e^{-\frac{2}{5}it} \right) \right] u(t) \\ &= \frac{-3i}{4} e^{-\frac{4}{5}t} \left(2i \sin\left(\frac{2}{5}t\right) \right) u(t) = \frac{3}{2} e^{-\frac{4}{5}t} \sin\left(\frac{2}{5}t\right) u(t) \end{aligned}$$

VERY IMPORTANT QUESTIONS

Following questions are very important. These questions may be asked in your SESSIONALS as well as UNIVERSITY EXAMINATION.

- Determine the exponential fourier series for the waveform shown



Ans. Refer Q. 3.4.

Q. 2. Explain Fourier transform of single sided exponential pulse.

Ans. Refer Q. 3.10.

Q. 3. Find the fourier transform of the signal $x(t) = e^{-a|t|} \text{sgn}(t)$.

Ans. Refer Q. 3.12.

Q. 4. Using Fourier transform, find the convolution of :

$$x_1(t) = e^{-2t} u(t) \\ x_2(t) = e^{-3t} u(t)$$

Ans. Refer Q. 3.13.

Q. 5. Find the fourier transform of a signal $x(t) = \frac{1}{a^2 + t^2}$.

Ans. Refer Q. 3.16.

Q. 6. State and prove frequency shifting theorem of DTFT.

Ans. Refer Q. 3.22.

Q. 7. Analyze the Discrete Time Fourier Transform of the following

$$x(n) = 0.5^n u(n) + 2^{-n} u(-n-1)$$

Ans. Refer Q. 3.23.

Q. 8. State and prove initial and final value theorem of Laplace transform.

Ans. Refer Q. 3.30.

Q. 9. For a LTI system with unit impulse response $h(t) = e^{-2t} u(t)$. Determine the output to the input $x(t) = e^{-t} u(t)$.

Ans. Refer Q. 3.37.

Q. 10. Using Laplace transform obtain the impulse response of the given second-order system

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = x(t)$$

Ans. Refer Q. 3.43.



4

UNIT

Z-Transform

CONTENTS

Part-1 : The z-Transform for Discrete Time Signals and Systems-Eigen Functions	4-2J to 4-7J
Part-2 : Region of Convergence, z-Domain Analysis.	4-7J to 4-20J

PART-1

The z-Transform for Discrete Time
Signals and Systems-Eigen Functions.

Que 4.1. Explain unilateral and bilateral z-transform.

Answer

1. The unilateral or one sided z-transform is evaluated for the non-negative values of time index ($n \geq 0$). This form of the z-transform is appropriate for problems involving causal and LTI systems.
2. The unilateral z-transform of a signal $x[n]$ is defined as,

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

which depends only on $x[n]$ for $n \geq 0$.

3. For a discrete-time signal $x[n]$, the bilateral z-transform is defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

4. The bilateral and unilateral z-transform are equivalent if $x[n] = 0$ for $n < 0$.

Que 4.2. Determine the z-transform of

- i. $x[n] = \sin(\omega n) u[n]$
- ii. $x[n] = \cos(\omega n) u[n]$

OR

Determine Z-transform of $x(n) = \sin \omega_0 n u(n)$.

AKTU 2018-19, Marks 07

Answer

- i. Given, $x(n) = \sin(\omega n) u(n) = \begin{cases} \sin(\omega n) & ; \text{ for } n \geq 0 \\ 0 & ; \text{ for } n < 0 \end{cases}$

$$\therefore X(z) = Z[\sin(\omega n) u(n)] = \sum_{n=0}^{\infty} \sin(\omega n) z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{e^{j\omega n} - e^{-j\omega n}}{2j} \right) z^{-n} = \frac{1}{2j} \sum_{n=0}^{\infty} (e^{j\omega n} z^{-n} - e^{-j\omega n} z^{-n})$$

$$= \sum_{n=0}^{\infty} \frac{(z^{-1} e^{j\omega})^n - (z^{-1} e^{-j\omega})^n}{2j} = \frac{1}{2j} \left[\sum_{n=0}^{\infty} (z^{-1} e^{j\omega})^n - \sum_{n=0}^{\infty} (z^{-1} e^{-j\omega})^n \right]$$

$$\begin{aligned} \therefore X(z) &= \frac{1}{2j} \left(\frac{1}{1 - z^{-1} e^{j\omega}} - \frac{1}{1 - z^{-1} e^{-j\omega}} \right) = \frac{1}{2j} \left(\frac{z}{z - e^{j\omega}} - \frac{z}{z - e^{-j\omega}} \right) \\ &= \frac{1}{2j} \left[\frac{z(z - e^{-j\omega}) - z(z - e^{j\omega})}{(z - e^{j\omega})(z - e^{-j\omega})} \right] = \frac{1}{2j} \left[\frac{z(e^{j\omega} - e^{-j\omega})}{z^2 - z(e^{j\omega} + e^{-j\omega}) + 1} \right] \\ &= \frac{z \sin \omega}{z^2 - 2z \cos \omega + 1} \end{aligned}$$

- ii. Given, $x(n) = \cos(\omega n) u(n) = \begin{cases} \cos(\omega n) & ; \text{ for } n \geq 0 \\ 0 & ; \text{ for } n < 0 \end{cases}$

$$\begin{aligned} Z[\cos(\omega n) u(n)] &= Z \left[\frac{e^{j\omega n} + e^{-j\omega n}}{2} u(n) \right] \\ &= \frac{1}{2} [Z[e^{j\omega n} u(n)] + Z[e^{-j\omega n} u(n)]] \\ &= \frac{1}{2} \left[\frac{z}{z - e^{j\omega}} + \frac{z}{z - e^{-j\omega}} \right] = \frac{1}{2} \left[\frac{z(z - e^{-j\omega}) + z(z - e^{j\omega})}{(z - e^{j\omega})(z - e^{-j\omega})} \right] \\ &= \frac{1}{2} \left[\frac{z[2z - (e^{j\omega} + e^{-j\omega})]}{z^2 - z(e^{j\omega} + e^{-j\omega}) + 1} \right] = \frac{z(2 - \cos \omega)}{z^2 - 2z \cos \omega + 1} \end{aligned}$$

Que 4.3. What are the different properties of z-transform?

Answer

1. **Linearity:** This property states that the z-transform of weighted sum of two signals is equal to the weighted sum of the individual z-transforms.

$$\text{If } x_1[n] \xrightarrow{ZT} X_1(z) \text{ and } x_2[n] \xrightarrow{ZT} X_2(z)$$

$$\text{then, } Z[a_1 x_1[n] + a_2 x_2[n]] \xrightarrow{ZT} a_1 X_1(z) + a_2 X_2(z)$$

where, a_1 and a_2 are constants.

2. **Time reversal:**

$$\text{If } x[n] \xrightarrow{ZT} X(z)$$

$$\text{ROC: } r_1 < |z| < r_2$$

$$\text{then, } x[-n] \xrightarrow{ZT} X(z^{-1})$$

$$\text{ROC: } \frac{1}{r_2} < |z| < \frac{1}{r_1}$$

3. **Time shifting:** According to this property a shift of ' k ' unit in time-domain signal causes a multiplication factor ' z^{-k} ' in z-transform.

$$\text{If } x[n] \xrightarrow{ZT} X(z)$$

$$\text{then, } x[n - k] \xrightarrow{ZT} z^{-k} X(z)$$

The ROC will remain same as that of $X(z)$ except for $n = 0$.

4. **Differentiation in z-domain of multiplication by n :**

$$\text{If } x[n] \xrightarrow{ZT} X(z)$$

then, $n x[n] \xrightarrow{ZT} -z \frac{dX(z)}{dz}$

5. **Accumulation property** : The accumulation property is given as,

If $x[n] \xrightarrow{ZT} X(z)$
then, $\sum_{k=-\infty}^n x[k] \xrightarrow{ZT} \frac{1}{1-z^{-1}} X(z)$

6. **Scaling property** :

If $x[n] \xrightarrow{ZT} X(z)$ ROC : $r_1 < |z| < r_2$
then, $a^n x[n] \xrightarrow{ZT} X(az^{-1})$ ROC : $|a| r_1 < |z| < |a| r_2$

Note : Constant 'a' may be real or complex quantity.

7. **Convolution** : This property gives that the convolution of two discrete-time signal can be expressed as simple multiplication of their z-transforms.

8. **Conjugate property** : $x_1[n] \otimes x_2[n] \xrightarrow{ZT} X_1(z) \cdot X_2^*(z)$

If $x[n] \xrightarrow{ZT} X(z)$ ROC = R
then, $x^*[n] \xrightarrow{ZT} X^*(z^*)$ ROC = R

It means that if z-transform $X(z)$ has pole or zero at $z = z_0$, it must also have pole or zero at $z = z_0^*$.

9. **Initial value theorem** : For a causal discrete-time signal, the initial value may be determined by,

$$x(0) = \lim_{n \rightarrow 0} x[n] = \lim_{z \rightarrow \infty} X(z)$$

10. **Final value theorem** : The final value of a discrete-time signal can be determined as,

$$x(\infty) = \lim_{n \rightarrow \infty} x[n] = \lim_{z \rightarrow 1} [(z-1) X(z)]$$

11. **Parseval's theorem** : Let $x_1[n]$ and $x_2[n]$ are two complex valued sequences then Parseval's theorem gives,

$$\sum_{n=-\infty}^{\infty} x_1[n] \cdot x_2^*[n] = \frac{1}{2\pi j} \oint X_1(z) \cdot X_2^*[(z^*)^{-1} z^{-1}] dz$$

Que 4.4. State and prove initial value theorem and final value theorem for z-transform.

Answer

Initial value theorem : For a causal signal $x[n]$,

If $x[n] \xrightarrow{ZT} X(z)$

then $\lim_{n \rightarrow 0} x[n] = x[0] = \lim_{z \rightarrow \infty} X(z)$

Proof : We know that for a causal signal $x[n]$

$$Z\{x[n]\} = X(z)$$

$$\begin{aligned} &= \sum_{n=0}^{\infty} x[n] z^{-n} = x[0] + x[1] z^{-1} + x[2] z^{-2} + \dots \\ &= x[0] + \frac{x[1]}{z} + \frac{x[2]}{z^2} + \dots \end{aligned}$$

Taking limit ($z \rightarrow \infty$) on both sides, we have

$$\begin{aligned} \lim_{z \rightarrow \infty} X(z) &= \lim_{z \rightarrow \infty} \left\{ x[0] + \frac{x[1]}{z} + \frac{x[2]}{z^2} + \dots \right\} \\ &= x[0] + 0 + 0 + \dots = x[0] \end{aligned}$$

Hence, $\lim_{n \rightarrow 0} x[n] = x[0] = \lim_{z \rightarrow \infty} X(z)$

Final value theorem : For a causal signal $x[n]$,

if $x[n] \xrightarrow{ZT} X(z)$
and if $X(z)$ has no poles outside the unit circle and it has no double or higher order poles on the unit circle centred at the origin of the z-plane, then

$$\lim_{n \rightarrow \infty} x[n] = x[\infty] = \lim_{z \rightarrow 1} (z-1) X(z)$$

Proof : We know, for a causal signal $x[n]$,

$$\begin{aligned} Z\{x[n]\} &= X(z) = \sum_{n=0}^{\infty} x[n] z^{-n} \\ Z\{x[n+1]\} &= zX(z) - zx[0] = \sum_{n=0}^{\infty} x[n+1] z^{-n} \\ &= zX(z) - zx[0] - X(z) \\ &= \sum_{n=0}^{\infty} x[n+1] z^{-n} - \sum_{n=0}^{\infty} x[n] z^{-n} \end{aligned}$$

$$\begin{aligned} \text{or } (z-1) X(z) - zx[0] &= \sum_{n=0}^{\infty} [x[n+1] - x[n]] z^{-n} \\ &= [x[1] - x[0]] + [x[2] - x[1]] z^{-1} + [x[3] - x[2]] z^{-2} + \dots \end{aligned}$$

Taking limit ($z \rightarrow 1$) on both sides, we have

$$\begin{aligned} \lim_{z \rightarrow 1} (z-1) X(z) - x[0] &= x[1] - x[0] + x[2] - x[1] + x[3] - x[2] + \dots + x[\infty] \\ \text{Hence, } \lim_{n \rightarrow \infty} x[n] &= x[\infty] = \lim_{z \rightarrow 1} (z-1) X(z) \end{aligned}$$

Que 4.5. Find the convolution of $x_1[n]$ and $x_2[n]$ using z-transform.

i. $x_1[n] = [1, 3, 4, 5]$

ii. $x_2[n] = [5, 1, 2, 6, 3, 4, 5]$

Answer

- Given,
 $x_1[n] = [1, 3, 4, 5]$... (4.5.1)
 $x_2[n] = [5, 1, 2, 6, 3, 4, 5]$... (4.5.2)

- Let convolution of $x_1[n]$ and $x_2[n]$ is $y[n]$

$$y[n] = x_1[n] \otimes x_2[n]$$

$$Y(z) = X_1(z) X_2(z) \quad \dots (4.5.3)$$

- Taking z-transform of eq. (4.5.1) and (4.5.2), we get

$$X_1(z) = 1 + 3z^{-1} + 4z^{-2} + 5z^{-3}$$

$$X_2(z) = 5 + z^{-1} + 2z^{-2} + 6z^{-3} + 3z^{-4} + 4z^{-5} + 5z^{-6}$$

Now substituting in eq. (4.5.3),

$$Y(z) = X_1(z) X_2(z) = (1 + 3z^{-1} + 4z^{-2} + 5z^{-3})$$

$$(5 + z^{-1} + 2z^{-2} + 6z^{-3} + 3z^{-4} + 4z^{-5} + 5z^{-6})$$

$$= 5 + (1 + 15)z^{-1} + (2 + 3 + 20)z^{-2} +$$

$$(6 + 6 + 4 + 25)z^{-3} + (3 + 18 + 8 + 5)z^{-4}$$

$$+ (4 + 9 + 24 + 10)z^{-5} + (5 + 12 + 12 + 30)z^{-6}$$

$$+ (15 + 16 + 15)z^{-7} + (20 + 20)z^{-8} + 25z^{-9}$$

$$Y(z) = 5 + 16z^{-1} + 25z^{-2} + 41z^{-3} + 34z^{-4} + 47z^{-5} + 59z^{-6}$$

$$+ 46z^{-7} + 40z^{-8} + 25z^{-9} \quad \dots (4.5.4)$$

- Thus, by taking inverse z-transform of eq. (4.5.4)

$$y[n] = [5, 16, 25, 41, 34, 47, 59, 46, 40, 25]$$

Que 4.6. Discuss about the eigen function and eigen value associated with the discrete-time LTI system.

Answer

- The complex exponential sequences are eigen functions of discrete-time LTI systems.
- Suppose that an LTI system with impulse response $h[n]$ has its input sequence as $x[n] = z^n$,
 $\dots (4.6.1)$
 where z is a complex number.
- Then the output of the system can be determined from the convolution sum as

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

$$= \sum_{k=-\infty}^{\infty} h[k]z^{n-k} = z^n \sum_{k=-\infty}^{\infty} h[k]z^{-k} \quad \dots (4.6.2)$$

- If the input $x[n]$ is the complex exponential given by eq. (4.6.1), then, assuming that the summation on the right-hand side of eq. (4.6.2) converges, the output is the same complex exponential multiplied by a constant that depends on the value of z . That is,

$$y[n] = H(z)z^n, \quad \dots (4.6.3)$$

where

$$H(z) = \sum_{k=-\infty}^{\infty} h[k]z^{-k} \quad \dots (4.6.4)$$

- Complex exponentials are eigen functions of discrete-time LTI systems. The constant $H(z)$ for a specified value of z is the eigen value associated with the eigen function z^n .

- If the input to a discrete-time LTI system is represented as a linear combination of complex exponentials, that is, if

$$x[n] = \sum_k a_k z_k^n, \quad \dots (4.6.5)$$

then the output will be

$$y[n] = \sum_k a_k H(z_k)$$

PART-2*Region of Convergence, z-Domain Analysis.*

Que 4.7. What is region of convergence? What are the important properties of ROC for z-transform?

Answer

ROC : ROC is defined as the set of values of z in the z -plane for which the magnitude of $X(z)$ is finite.

Characteristics of ROC :

- The condition for z-transform $X(z)$ to be finite is $|z| > 1$.
- ROC for $X(z)$ is the area outside the unit circle in the z -plane.
- The ROC of a rational z-transform is bounded by the location of its poles.

Properties of ROC for z-transform :

- The ROC is a ring in the z -plane centred at the origin.
- The ROC cannot contain any poles.
- The ROC must be connected region.

4-8 J (EC-Sem-4)

4. If $x(n)$ is a finite duration sequence, then the ROC is the entire z -plane except possibly $z = 0$ or $z = \infty$.
5. If $x(n)$ is right-sided sequence, then the ROC exist outward from the outermost finite pole to $z = \infty$.
6. If $x(n)$ is a left-sided sequence, then the ROC extends inward from the innermost pole to $z = 0$.
7. If two sided sequence, then the ROC will consist a ring in the z -plane, bounded on the interior and exterior by a pole and consistent with property (2).

Example : Let $X(z) = 1/(z-1)$

The $X(z)$ has a pole at $z = 1$, therefore the ROC will be $|z| > 1$ as shown in Fig. 4.7.1.

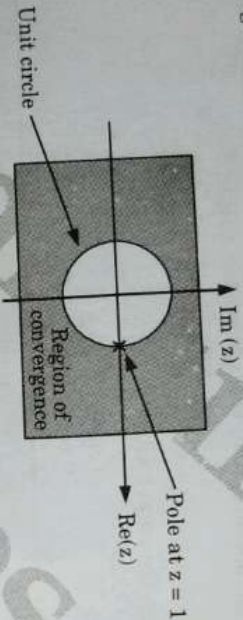


Fig. 4.7.1.

Que 4.8. Find the Z-transform and ROC of $X(z)$ for

$$x(n) = 3\left(\frac{5}{7}\right)^n u(n) + 2\left(-\frac{1}{3}\right)^n u(n)$$

AKTU 2022-23, Marks 10

Answer

$$\text{Given : } x(n) = 3\left(\frac{5}{7}\right)^n u(n) + 2\left(-\frac{1}{3}\right)^n u(n)$$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=-\infty}^{\infty} \left[3\left(\frac{5}{7}\right)^n u(n) + 2\left(-\frac{1}{3}\right)^n u(n) \right] z^{-n} \\ &= \sum_{n=0}^{\infty} \left[3\left(\frac{5}{7}\right)^n z^{-n} + 2\left(-\frac{1}{3}\right)^n z^{-n} \right] \\ &= 3 \sum_{n=0}^{\infty} \left(\frac{5}{7}\right)^n z^{-n} + 2 \sum_{n=0}^{\infty} \left(-\frac{1}{3}\right)^n z^{-n} \end{aligned}$$

$$= 3 \sum_{n=0}^{\infty} \left(\frac{5}{7} z^{-1}\right)^n + 2 \sum_{n=0}^{\infty} \left(-\frac{1}{3} z^{-1}\right)^n$$

The first summation converges for $\left|\frac{5}{7} z^{-1}\right| < 1$ or $|z| > \frac{5}{7}$ and second summation converges for $\left|-\frac{1}{3} z^{-1}\right| < 1$ or $|z| > \frac{1}{3}$

$\therefore X(z)$ converges for $|z| > \frac{5}{7}$

$$\begin{aligned} X(z) &= 3 \frac{1}{1 - \left(\frac{5}{7}\right) z^{-1}} + 2 \frac{1}{1 + \left(\frac{1}{3}\right) z^{-1}} \\ &= 3 \frac{z}{z - \left(\frac{5}{7}\right)} + 2 \frac{z}{z + \left(\frac{1}{3}\right)} = \frac{3z \left[z + \left(\frac{1}{3}\right) \right] + 2z \left[z - \left(\frac{5}{7}\right) \right]}{\left[z - \left(\frac{5}{7}\right) \right] \left[z + \left(\frac{1}{3}\right) \right]} \\ &= \frac{5z^2 - \left(\frac{3}{7}\right)z}{\left[z - \left(\frac{5}{7}\right) \right] \left[z + \left(\frac{1}{3}\right) \right]} = \frac{5z \left[z - \left(\frac{3}{35}\right) \right]}{\left[z - \left(\frac{5}{7}\right) \right] \left[z + \left(\frac{1}{3}\right) \right]} \end{aligned}$$

The poles of $X(z)$ are at $z = \frac{5}{7}$

and $z = -\left(\frac{1}{3}\right)$ so ROC is $|z| > \frac{5}{7}$

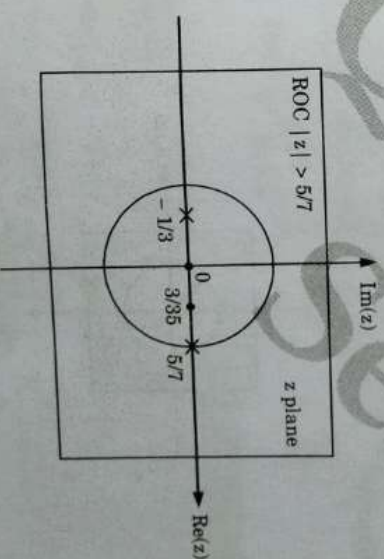


Fig. 4.8.1.

Que 4.9. Determine the z-transform of $x(n) = \cos(\omega_0 n) u(n)$ and sketch the ROC.

AKTU 2021-22, Marks 10

Answer

z-transform of $\cos \omega_0 n u[n]$:

$$x[n] = (\cos \omega_0 n) u[n]$$

$$= \frac{1}{2} e^{j\omega_0 n} u[n] + \frac{1}{2} e^{-j\omega_0 n} u[n]$$

$$X(z) = \frac{1}{2} Z \{ e^{j\omega_0 n} u[n] \} + \frac{1}{2} Z \{ e^{-j\omega_0 n} u[n] \}$$

1.

$$e^{j\omega_0 n} u[n] \xrightarrow{ZT} \frac{1}{1 - e^{j\omega_0} z^{-1}}$$

$$\text{ROC: } |z| > 1$$

and

$$e^{-j\omega_0 n} u[n] \xrightarrow{ZT} \frac{1}{1 - e^{-j\omega_0} z^{-1}}$$

$$\text{ROC: } |z| > 1$$

2.

$$X(z) = \frac{1}{2} \frac{1}{1 - e^{j\omega_0} z^{-1}} + \frac{1}{2} \frac{1}{1 - e^{-j\omega_0} z^{-1}} \quad \text{ROC: } |z| > 1$$

$$X(z) = \frac{1}{2} \left[\frac{z(2 - e^{-j\omega_0}) + z(2 - e^{j\omega_0})}{(z - e^{j\omega_0})(z - e^{-j\omega_0})} \right]$$

$$= \frac{1}{2} \left[\frac{z(2 - (e^{-j\omega_0} + e^{j\omega_0}))}{z^2 - z(e^{j\omega_0} + e^{-j\omega_0}) + 1} \right] = \frac{z(2 - \cos \omega_0)}{z^2 - 2z \cos \omega_0 + 1}$$

$$X(z) = \frac{1 - z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$$

$$\text{ROC: } |z| > 1$$

ROC:

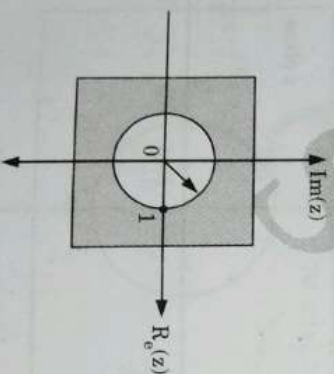


Fig. 4.9.1.

Que 4.10. Using properties of z-transform, find z-transform and ROC of signal.

$$x[n] = n 2^n \sin\left(\frac{n\pi}{2}\right) u[n]$$

Answer

1. Given,

$$x[n] = n 2^n \sin\left(\frac{n\pi}{2}\right) u[n]$$

2. We know that

$$Z \left[\sin\left(\frac{\pi}{2} n\right) u[n] \right] = \frac{z \sin(\pi/2)}{z^2 - 2z \cos(\pi/2) + 1} = \frac{z}{z^2 + 1}, |z| > 1$$

3. Using the multiplication by an exponential property, we have

$$\begin{aligned} Z \left[2^n \sin\left(\frac{\pi}{2} n\right) u[n] \right] &= Z \left[\sin\left(\frac{\pi}{2} n\right) u[n] \right]_{z \rightarrow z/2} \\ &= \frac{z}{z^2 + 1} \bigg|_{z \rightarrow z/2} = \frac{z/2}{(z/2)^2 + 1} = \frac{2z}{z^2 + 4}, |z| > 2 \end{aligned}$$

4. Using differentiation in z-domain property, we have

$$X(z) = Z \left[n 2^n \sin\left(\frac{\pi}{2} n\right) u[n] \right] = -z \frac{dZ}{dz} \left[2^n \sin\left(\frac{\pi}{2} n\right) u[n] \right]$$

$$X(z) = -z \frac{d}{dz} \left(\frac{2z}{z^2 + 4} \right) = -z \left[\frac{(z^2 + 4)(2) - 2z(2z)}{(z^2 + 4)^2} \right], |z| > 2$$

$$X(z) = -z \left[\frac{-2z^2 + 8}{(z^2 + 4)^2} \right] = \frac{2z(z^2 - 4)}{(z^2 + 4)^2} \text{ and ROC: } |z| > 2.$$

Que 4.11. Find inverse Z-Transform of

$$X(z) = \frac{2z - 7}{z^3 - 5z + 6}; \text{ROC: } |z| < 2$$

AKTU 2022-23, Marks 10

Answer

Given:

$$X(z) = \frac{2z - 7}{z^3 - 5z + 6}; \text{ROC: } |z| < 2$$

$$\frac{X(z)}{z} = \frac{2z - 7}{z(z^3 - 5z + 6)} = \frac{2z - 7}{z(z^3 - 5z + 6)(z - 2)(z - 3)}$$

$$= \frac{C_1}{z} + \frac{C_2}{z - 2} + \frac{C_3}{z - 3}$$

where C_1 , C_2 and C_3 are determined as follows

$$C_1 = \frac{zX(z)}{z} \Big|_{z=0} = \frac{2z-7}{(z-2)(z-3)} \Big|_{z=0} = -\frac{7}{6}$$

$$C_2 = (z-2) \frac{X(z)}{z} \Big|_{z=2} = \frac{2z-7}{z(z-3)} \Big|_{z=2} = \frac{3}{2}$$

$$C_3 = (z-3) \frac{X(z)}{z} \Big|_{z=3} = \frac{2z-7}{z(z-2)} \Big|_{z=3} = \frac{1}{3}$$

$$\frac{X(z)}{z} = -\frac{7}{6} \left(\frac{1}{z} \right) + \frac{3}{2} \left(\frac{1}{z-2} \right) - \frac{1}{3} \left(\frac{1}{z-3} \right)$$

$$X(z) = -\frac{7}{6} + \frac{3}{2} \frac{z}{z-2} - \frac{1}{3} \frac{z}{z-3}; \text{ ROC } |z| < 2$$

Que 4.12. Determine the inverse z-transform of the following function

$$H(z) = \frac{0.2z}{(z+0.4)(z-0.2)} \quad \text{ROC: } |z| > 0.4$$

AKTU 2021-22, Marks 10

Answer

1. Given,

$$H(z) = \frac{0.2z}{(z+0.4)(z-0.2)}$$

$$\frac{H(z)}{z} = \frac{0.2}{(z+0.4)(z-0.2)}$$

$$\frac{H(z)}{z} = \frac{A}{(z+0.4)} + \frac{B}{(z-0.2)}$$

$$A = \frac{H(z)}{z} \Big|_{z=-0.4}$$

$$= \frac{0.2z(z+0.4)}{(z+0.4)(z-0.2)} \Big|_{z=-0.4}$$

$$= \frac{0.2(-0.4)}{(-0.4-0.2)}$$

$$= \frac{-0.08}{-0.6} = \frac{2}{15}$$

$$B = \frac{H(z)}{z} \Big|_{z=0.2}$$

2. For $|z| > 0.4$

$$= \frac{0.2z(z-0.2)}{(z+0.4)(z-0.2)} \Big|_{z=0.2} = \frac{0.2 \times 0.2}{0.2+0.4} = \frac{0.4}{0.6} = \frac{4}{15}$$

$$H(z) = \frac{2}{15} \frac{1}{(z+0.4)} + \frac{1}{15} \frac{1}{(z-0.2)}$$

$$h(n) = \frac{2}{15} (-0.4)^n u(n) + \frac{1}{15} (0.2)^n u(n)$$

Que 4.13. Determine inverse z-transform of the following function.

$$H(z) = \frac{3 + 3.6z^{-1} + 0.6z^{-2}}{1 + 0.1z^{-1} - 0.2z^{-2} + z^{-3}}$$

Answer

1. Given,

$$H(z) = \frac{3 + 3.6z^{-1} + 0.6z^{-2}}{1 + 0.1z^{-1} - 0.2z^{-2} + z^{-3}} \quad \dots (4.13.1)$$

2. From eq. (4.13.1), $H(z) = \frac{3z^3 + 3.6z^2 + 0.6z}{z^3 + 0.1z^2 - 0.2z + 1}$

3. We can find inverse z-transform by long division method.

$$\begin{array}{r} z^3 + 0.1z^2 - 0.2z + 1 \overline{) 3z^3 + 3.6z^2 + 0.6z} \\ \underline{3z^3 + 0.3z^2 - 0.6z + 3} \\ 3.3z^2 + 1.2z - 3 \\ \underline{3.3z^2 + 0.33z - 0.66 + 3.3z^{-1}} \\ 0.87z - 2.34 - 3.3z^{-1} \\ \underline{0.87z - 0.087 - 0.174z^{-1} + 0.87z^{-2}} \\ -2.253 - 3.126z^{-1} - 0.87z^{-2} \\ \underline{-2.253 - 0.2253z^{-1} + 0.4506z^{-2} - 2.253z^{-3}} \\ -3.3513z^{-1} - 0.87z^{-2} - 2.253z^{-3} \dots \end{array}$$

so,

$$H(z) = 3 + 3.3z^{-1} + 0.87z^{-2} - 2.253z^{-3} - 3.3513z^{-4} \dots (4.13.2)$$

4. Taking inverse z-transform of eq. (4.13.2)

$$h[n] = [3, 3.3, 0.87, -2.253, -3.3513, \dots]$$

Que 4.14. Determine inverse z-transform of the following signal

$$X(z) = \frac{z^3 - z^2 + z}{(z - 0.5)(z - 2)(z - 1)}; \text{ROC: } 1 < |z| < 2$$

Answer

$$1. \text{ Given, } X(z) = \frac{z^3 - z^2 + z}{(z - 0.5)(z - 2)(z - 1)}; \text{ROC: } 1 < |z| < 2 \quad \dots(4.14.1)$$

2. Now, eq. (4.14.1) can be written as

$$\frac{X(z)}{z} = \frac{z^2 - z + 1}{(z - 0.5)(z - 2)(z - 1)} = X_1(z)$$

3. Doing partial fraction expansion,

$$\frac{X(z)}{z} = \frac{A}{(z - 0.5)} + \frac{B}{(z - 2)} + \frac{C}{(z - 1)}$$

$$A = 1, B = 2, C = -2$$

$$\text{So, } \frac{X(z)}{z} = \frac{1}{(z - 0.5)} + \frac{2}{(z - 2)} - \frac{2}{(z - 1)}$$

4. Thus the terms with pole at $z = 0.5$ and 1 must be causal and term with pole $z = 2$ must be non-causal, therefore

$$X(z) = \frac{z}{(z - 0.5)} + \frac{2z}{(z - 2)} - \frac{2z}{(z - 1)} \quad \dots(4.14.2)$$

5. Taking inverse z -transform, of eq. (4.14.2)

$$x[n] = (0.5)^n u[n] - 2(2)^n u[-n-1] - 2 u[n]$$

Que 4.15. Determine the inverse z -transform using partial fraction method for

$$X(z) = \frac{\left(\frac{1}{4}\right)z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1}\right)}$$

i. $|z| > \frac{1}{2}$

ii. $|z| < \frac{1}{4}$

iii. $\frac{1}{4} < |z| < \frac{1}{2}$

Answer

1. Given,

$$X(z) = \frac{\left(\frac{1}{4}\right)z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1}\right)}$$

$\dots(4.15.1)$

Signal System

4-15 J (EC-Sem-4)

2. Multiplying the numerator and denominator with z^2 in eq. (4.15.1), we obtain

$$X(z) = \frac{(1/4)z}{[z - (1/2)][z - (1/4)]}$$

3. Now, $X(z)$ has two poles, one at $z = (1/2)$ and the other at $z = (1/4)$ as shown in Fig. 4.15.1.

or

$$X(z) = \frac{z}{z - (1/2)} - \frac{z}{z - (1/4)}$$

i. **For ROC: $|z| > \frac{1}{2}$**

Here both the poles, i.e., $z = (1/2)$ and $z = (1/4)$ correspond to causal terms.

$$\text{So, } x[n] = \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{4}\right)^n u[n]$$

ii. **For ROC: $|z| < \frac{1}{4}$**

Here both the poles must correspond to anticausal terms.

$$\text{So, } x[n] = -\left(\frac{1}{2}\right)^n u[-n-1] + \left(\frac{1}{4}\right)^n u[-n-1]$$

iii. **For ROC: $\frac{1}{4} < |z| < \frac{1}{2}$**

Here the pole at $z = (1/4)$ must correspond to causal term and the pole at $z = (1/2)$ must correspond to non-causal term.

$$x[n] = -\left(\frac{1}{2}\right)^n u[-n-1] - \left(\frac{1}{4}\right)^n u[n]$$

4. The ROCs are shown in Fig. 4.15.1.

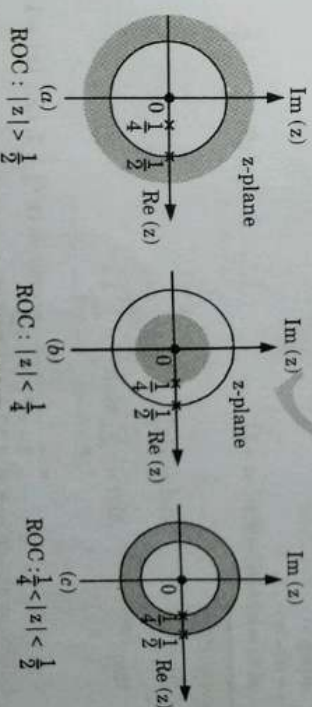


Fig. 4.15.1.

Que 4.16. Using properties of z-transform, find z-transform and ROC of signal.

$$x[n] = n2^n \sin\left(\frac{n\pi}{2}\right) u[n]$$

Answer

1. Given,

$$x[n] = n2^n \sin\left(\frac{n\pi}{2}\right) u[n]$$

2. We know that

$$\left[\sin\left(\frac{\pi}{2}n\right) u[n] \right] = \frac{z \sin(\pi/2)}{z^2 - 2z \cos(\pi/2) + 1} = \frac{z}{z^2 + 1}, \quad |z| > 1$$

3. Using the multiplication by an exponential property, we have

$$\begin{aligned} Z\left[2^n \sin\left(\frac{\pi}{2}n\right) u[n]\right] &= Z\left[\sin\left(\frac{\pi}{2}n\right) u[n]\right]_{z \rightarrow z/2} \\ &= \frac{z}{z^2 + 1} \bigg|_{z \rightarrow z/2} = \frac{z/2}{(z/2)^2 + 1} = \frac{2z}{z^2 + 4}, \quad |z| > 2 \end{aligned}$$

4. Using differentiation in z-domain property, we have

$$X(z) = Z\left[n2^n \sin\left(\frac{\pi}{2}n\right) u[n]\right] = -z \frac{dZ}{dz} \left[\frac{2^n \sin\left(\frac{\pi}{2}n\right) u[n]}{z^2 + 4} \right]$$

$$X(z) = -z \frac{d}{dz} \left(\frac{2z}{z^2 + 4} \right) = -z \left[\frac{(z^2 + 4)(2) - 2z(2z)}{(z^2 + 4)^2} \right], \quad |z| > 2$$

$$X(z) = -z \left[\frac{-2z^2 + 8}{(z^2 + 4)^2} \right] = \frac{2z(z^2 - 4)}{(z^2 + 4)^2} \text{ and ROC: } |z| > 2.$$

Que 4.17. Determine the z-transform of following sequences with

ROC:

i. $u[n]$

ii. $-u[-n-1]$

iii. $x[n] = a^n u[n] - b^n u[-n-1]$

Answer

i. Given,

$$x[n] = u[n]$$

then,

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n]z^{-n} = \sum_{n=0}^{+\infty} 1 \cdot z^{-n}$$

$$= \frac{1}{1 - z^{-1}} = \frac{z}{z - 1} \text{ provided } |z^{-1}| < 1, \text{ i.e.; } |z| > 1$$

hence, as shown in Fig. 4.17.1, the ROC of $X(z)$ is the exterior of the unit circle.

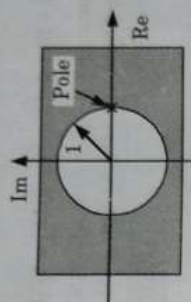


Fig. 4.17.1. ROC of $X(z)$ when $x[n] = u[n]$.

ii. Given, $x[n] = -u[-n-1]$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = -\sum_{n=-\infty}^{-1} z^{-n}$$

put

$$k = -n$$

$$X(z) = -\sum_{k=1}^{\infty} z^k = 1 - \sum_{k=0}^{\infty} z^k = 1 - \frac{1}{1-z} = \frac{z}{z-1}; \text{ provided } |z| < 1$$

so, $X(z) = \frac{z}{z-1}; |z| < 1$

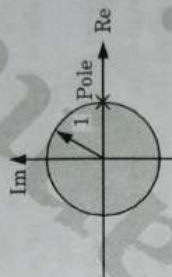


Fig. 4.17.2. ROC and pole location of $X(z)$ when $x[n] = -u[-n-1]$.

iii. Given, $x[n] = a^n u[n] - b^n u[-n-1]$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} a^n u[n] \cdot z^{-n} - \sum_{n=-\infty}^{\infty} b^n u[-n-1] \cdot z^{-n} \\ &= \sum_{n=0}^{\infty} (az^{-1})^n - \sum_{n=-\infty}^{-1} (bz^{-1})^n \\ &= \frac{1}{1 - az^{-1}} + \frac{1}{1 - bz^{-1}} \end{aligned}$$

$$; \text{ ROC: } |a| < |z| < |b|$$

$$; \text{ ROC: } |a| < |z| < |b|$$

$$X(z) = \frac{z}{z-a} + \frac{z}{z-b}$$

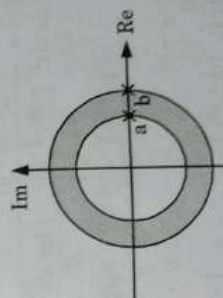


Fig. 4.17.3. ROC and pole location of $X(z)$ when $x[n] = a^n u[n] - b^n u[-n-1]$.

Que 4.18. Solve $y[k+2] + 5y[k+1] + 6y[k] = 3f[k+1] + 5f[k]$

If the initial conditions are $y[-1] = 11/6$, $y[-2] = 37/36$ and the input $f[k] = 2^{-k} V[k]$ and indicate zero input and zero state component in the output $y[k]$.

AKTU 2022-23, Marks 10

Answer

$$y[k+2] + 5y[k+1] + 6y[k] = 3f[k+1] + 5f[k]$$

Given initial conditions

$$y[-1] = 11/6, y[-2] = 37/36 \text{ and the input } f[k] = 2^{-k} V[k]$$

Using z-transform we get

$$Y[z] = \frac{z(3z^2 - 9.5z + 10.5)}{(z - 0.5)(z^2 - 5z + 6)}$$

$$Y[z] = \frac{26}{15} \left(\frac{z}{z - 0.5} \right) - \frac{7}{3} \left(\frac{z}{z - 2} \right) + \frac{18}{5} \left(\frac{z}{z - 3} \right)$$

$$y[n] = \left[\frac{26}{15} (0.5)^n - \frac{7}{3} (2)^n + \frac{18}{5} (3)^n \right] u(n)$$

Que 4.19. A causal LTI system is described by difference equation.

$$y'(n) = y'(n-1) + y'(n-2) + x'(n-1)$$

Find the system function $H(z)$ for this system. Plot the poles zeros of $H(z)$ and indicate the region of convergence.

AKTU 2017-18, Marks 07

Answer

- Given, $y'(n) = y'(n-1) + y'(n-2) + x'(n-1)$
- Taking z-transform

$$Y(z) = z^{-1}Y(z) + z^{-2}Y(z) + z^{-1}X(z)$$

$$Y(z)(1 - z^{-1} - z^{-2}) = z^{-1}X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z^{-1}}{1 - z^{-1} - z^{-2}}$$

$$\text{or } H(z) = \frac{z}{z^2 - z - 1} = \frac{z}{z - \left(\frac{1+\sqrt{5}}{2}\right)} \left(z - \left(\frac{1-\sqrt{5}}{2}\right) \right)$$

$$\text{Poles: } z = \left(\frac{1+\sqrt{5}}{2}\right) \text{ and } z = \left(\frac{1-\sqrt{5}}{2}\right)$$

Zeros:

$$z = 0$$

ROC:

$$|z| < \left(\frac{1+\sqrt{5}}{2}\right) \text{ and } |z| < \left(\frac{-1+\sqrt{5}}{2}\right)$$

Since, ROC should not contain any pole, hence, ROC will be

$$|z| < \left(\frac{-1+\sqrt{5}}{2}\right)$$

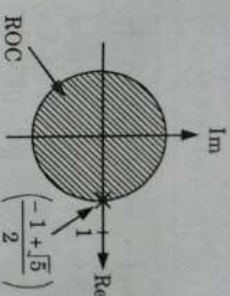


Fig. 4.19.1.

Que 4.20. Determine the total response of the differential

equation

$$y''[n] + 4y'[n-1] + 4y[n-2] = (-2)^n u[n]$$

$$\text{where } y[-1] = 0 \text{ and } y[-2] = 1$$

Answer

- Given, $y''[n] + 4y'[n-1] + 4y[n-2] = (-2)^n u[n]$
 $y[-1] = 0, y[-2] = 1$

2. Taking Z-transform, we get

$$Y(z) + 4[z^{-1}Y(z) + y(-1)] + 4[z^{-2}Y(z) + z^{-1}y(-1) + y(-2)] = \frac{z}{z - (-2)}$$

$$Y(z) [1 + 4z^{-1} + 4z^{-2}] + 4 \times 0 + 4 \times z^{-1} \times 0 + 4 \times 1 = \frac{z}{z + 2}$$

$$3. Y(z) [1 + 4z^{-1} + 4z^{-2}] + 4 = \frac{z}{z + 2}$$

$$Y(z) = \frac{\frac{z}{z+2} - 4}{1 + 4z^{-1} + 4z^{-2}} = \frac{\frac{z-4}{z+2}}{(z+2)(z^2+4z+4)}$$

$$Y(z) = \frac{-z^2(3z+8)}{(z+2)(z^2+4z+4)}$$

$$\frac{Y(z)}{z} = \frac{-2(3z+8)}{(z+2)(z^2+4z+4)}$$

$$\frac{Y(z)}{z} = \frac{-3z^2-8z}{(z+2)^3} = \frac{A}{z+2} + \frac{B}{(z+2)^2} + \frac{C}{(z+2)^3}$$

- On solving partial fraction, we get
 $A = -3, B = 4, C = 4$

$$5. \frac{Y(z)}{z} = \frac{-3}{(z+2)} + \frac{4}{(z+2)^2} + \frac{4}{(z+2)^3}$$

6. Taking inverse Z-transform
 $y[n] = -3(-2)^n u[n] + n(-2)^{n+1} u[n] + (-1)^n 2^{n-1} (n-1)u[n]$

VERY IMPORTANT QUESTIONS

Following questions are very important. These questions may be asked in your SESSIONALS as well as UNIVERSITY EXAMINATION.

- Q. 1. Find the Z-transform and ROC of $X(z)$ for

$$x(n) = 3\left(\frac{5}{7}\right)^n u(n) + 2\left(-\frac{1}{3}\right)^n u(n)$$

Ans. Refer Q. 4.8.

- Q. 2. Determine the z-transform of $x(n) = \cos(\omega_0 n) u(n)$ and sketch the ROC.

Ans. Refer Q. 4.9.

- Q. 3. Find inverse Z-Transform of

$$X(z) = \frac{2z-7}{z^2-5z+6}; \text{ROC}; |z| < 2$$

Ans. Refer Q. 4.11.

- Q. 4. Determine the inverse z-transform of the following function

$$H(z) = \frac{0.2z}{(z+0.4)(z-0.2)} \quad \text{ROC: } |z| > 0.4$$

Ans. Refer Q. 4.12.

- Q. 5. Solve $y[k+2] + 5y[k+1] + 6y[k] = 3f[k+1] + 5f[k]$

If the initial conditions are $y[-1] = 11/6$, $y[-2] = 37/36$ and the input $f[k] = 2^k V[k]$ and indicate zero input and zero state component in the output $y[k]$.

Ans. Refer Q. 4.18.



5 UNIT

Sampling of Time Signals

CONTENTS

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PART-1
The Sampling Theorem and its Implications-Spectra of Sampled Signals.

Que 5.1. What do you mean by sampling of a signal? Also explain Nyquist rate of sampling.

Answer

Sampling :

Sampling is the process of converting a continuous-time signal $x(t)$ into a discrete-time signal $x[n]$ by measuring the amplitudes of the continuous-time signal $x(t)$ at integer multiples of a sampling interval T as shown in Fig. 5.1.1.

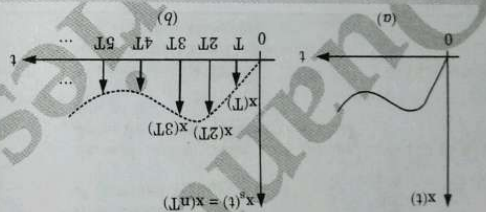


Fig. 5.1.1. (a) Continuous-time signal and (b) Its sampled version.

1. Nyquist rate of sampling is the theoretical minimum sampling rate at which a signal can be sampled and still be reconstructed from its samples without any distortion.
2. It is theoretically by minimum, because when the Nyquist rate of sampling is used, only an ideal LPF can be used to extract $X(\omega)$ from $X_s(\omega)$, i.e., to recover $x(t)$ from $x_s(t)$.
3. It is always equal to $2f_m$ where, f_m is the maximum frequency component present in the signal.
4. The minimum sampling rate, or minimum sampling frequency $\omega_s = 2\omega_m$ (or $f_s = 2f_m$), is referred as the Nyquist rate; its reciprocal $T_s = 2\pi/\omega_s = 1/f_s \leq 1/2f_m$ (measured in seconds) is called the Nyquist interval.
5. Sampling a signal at a rate less than the Nyquist rate is referred as undersampling and greater than the Nyquist rate is referred as oversampling.

1. Nyquist interval is the time interval between any two adjacent samples when sampling rate is Nyquist rate.
2. Nyquist rate, $f_s = 2f_m$ Hz
3. Nyquist interval, $T_s = \frac{1}{2f_m}$ sec

Que 5.2. State and prove sampling theorem. Also give some applications of sampling theorem.

OR

State and prove sampling theorem.

AKTU 2022-23, Marks 10

AKTU 2018-19, Marks 07

Answer

Statement :

Sampling theorem states that a band-limited signal of finite energy, which has no frequency components higher than ω_m rad/s (or f_m Hz), may be completely recovered from a knowledge of its samples if the sampling frequency $\omega_s \geq 2\omega_m$ samples/rads (or $f_s \geq 2f_m$ samples/s). Thus, the sampling frequency $\omega_s \geq 2\omega_m$ or $f_s \geq 2f_m$.

Proof :

1. Let $x(t)$ is a continuous-time band limited signal and it has $X(\omega) = 0$ for $\omega > \omega_m$.
2. $\delta_T(t)$ is an impulse train which samples at a rate of f_s Hz and $x_s(t)$ is the sampled signal

$$x_s(t) = x(t) \cdot \delta_T(t)$$

$$\text{where, } \delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

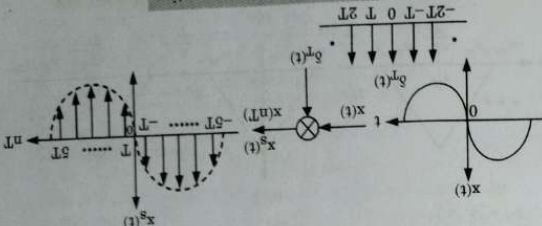


Fig. 5.2.1. Sampling operation.

3. Since, the exponential form of Fourier series of $\delta_T(t)$ is

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) = \frac{1}{T} \sum_{n=-\infty}^{\infty} e^{jn\omega_s t} \quad \dots(5.2.1)$$

$$x_s(t) = \frac{1}{T} \sum_{n=-\infty}^{\infty} x(t) e^{jn\omega_s t} \quad \dots(5.2.1)$$

4. Taking Fourier transform on both sides of eq. (5.2.1), we have

$$X_s(\omega) = \frac{1}{T} \sum_{n=-\infty}^{\infty} X(\omega - n\omega_s)$$

$$= \frac{1}{T} \sum_{n=-\infty}^{\infty} X\left(\omega - \frac{T}{2\pi} n\right)$$

$$X_s(f) = f_s \sum_{n=-\infty}^{\infty} X(f - nf_s)$$

5. Thus, the spectrum of the sampled signal $X_s(\omega)$ is the sum of shifted replicas $\frac{1}{T} X(\omega)$ of centering at $\left(\frac{T}{2\pi}\right) n$, where $n = 0, \pm 1, \pm 2, \dots$

6. For signal recovery, $\omega_s - \omega_m > \omega_m$

or $\omega_s > 2\omega_m$

or $f_s > 2f_m$

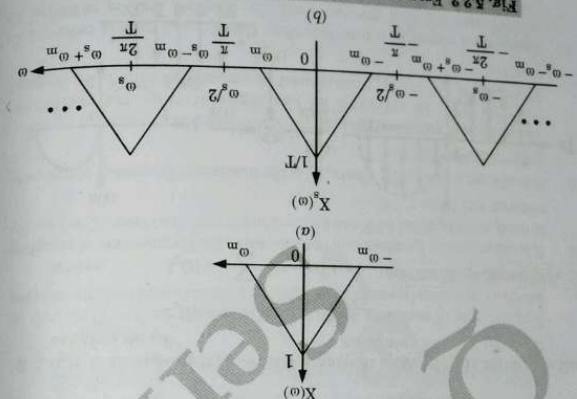


Fig. 5.2.2. Frequency-domain representation of sampling.

Applications of sampling theorem :

1. The sampling theorem is very important in signal analysis, processing and transmission because it allows us to replace a continuous-time signal by a discrete sequence of numbers. This leads us directly into the area of digital filtering.
2. This opens doors to many new techniques of communicating continuous time signals by pulse train. In this we may vary the amplitude, width or positions. We can have pulse amplitude modulation (PAM), pulse width modulation (PWM) and pulse position modulation (PPM).

Que 5.3. State and prove the sampling theorem and discuss the effect of under sampling.

Answer

A. Sampling theorem : Refer Q. 5.2, Page 5-3J, Unit-5.

B. Effect of under sampling :

1. Modulation Transfer Function (MTF) and Noise Power Spectra (NPS) do not behave as transfer amplitude and variance, respectively of a single sinusoid.
2. The response of a digital system to a delta function is not spatially invariant and therefore does not fulfill certain technical requirements of classical analysis.
3. Noise Equivalent Quantita (NEQ) loses its common meaning as maximum available signal-to-noise ratio at a particular frequency.

Que 5.4. Determine the Nyquist sampling rate and Nyquist sampling interval for

$$x(t) = \sin(100\pi t) + 3\sin^2(60\pi t)$$

Answer

$$\text{Given } x(t) = \sin(100\pi t) + 3\sin^2(60\pi t)$$

$$= x_1(t) + x_2(t)$$

$$x_1(t) = \sin(100\pi t) = \frac{\sin(100\pi t)}{100\pi t} = \frac{\omega_m t}{100\pi t}$$

$$\omega_{m1} = 100\pi \text{ rad/sec}$$

$$x_2(t) = 3\sin^2(60\pi t)$$

$$= 3\sin^2(60\pi t)$$

$$= 3\left[\frac{\sin(60\pi t)}{60\pi t}\right]^2 = 3\left[\frac{\sin(60\pi t)}{60\pi t}\right]^2$$

$$\omega_{m2} = 60\pi + 60\pi = 120\pi \text{ rad/sec}$$

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$$\omega_m = \omega_{m2} = 120 \pi$$

$$f_m = \frac{\omega_m}{2\pi} = \frac{120 \pi}{2\pi} = 60 \text{ Hz}$$

$$\text{Nyquist sampling rate } f_N = 2f_m = 2 \times 60 = 120 \text{ Hz}$$

$$\text{Nyquist interval} = \frac{1}{f_N} = \frac{1}{120} = \frac{2}{f_m} = \frac{1}{60} \text{ sec}$$

Que 5.5. Determine the Nyquist rate corresponding to each of the following signals :

- $x(t) = \sin(200 \pi t)$
- $x(t) = \sin^2(200 \pi t)$
- $x(t) = 1 + \cos(200 \pi t) + \sin(400 \pi t)$

Answer

- Given, $x(t) = \sin(200 \pi t)$

Comparing with

$$x(t) = \sin(\omega_1 t)$$

we have $\omega_1 = 200 \pi$ or $f_1 = 100$

Therefore, the Nyquist rate is given by

$$\omega_s = 2\omega_1 = 400 \pi \text{ or } f_s = 2f_1 = 200 \text{ Hz}$$

- Given,

$$x(t) = \sin^2(200 \pi t) = \frac{1}{2} - \frac{1}{2} \cos(400 \pi t)$$

Comparing with

$$x(t) = A_0 - A_1 \cos(\omega_1 t)$$

we have $\omega_1 = 400 \pi$ or $f_1 = 200$

Therefore, the Nyquist rate is given by

$$\omega_s = 2\omega_1 = 800 \pi \text{ or } f_s = 2f_1 = 400 \text{ Hz}$$

- Given,

$$x(t) = 1 + \cos(200 \pi t) + \sin(400 \pi t)$$

Comparing with

$$x(t) = A_0 + A_1 \cos(\omega_1 t) + A_2 \sin(\omega_2 t)$$

we have

$$\omega_1 = 200 \pi \text{ and } \omega_2 = 400 \pi$$

Maximum frequency present in the signal $x(t)$ is

$$\omega_{\max} = \omega_2 = 400 \pi \text{ or } f_{\max} = 200$$

therefore, the Nyquist rate is given by

$$\omega_s = 2\omega_{\max} = 800 \pi \text{ or } f_s = 2f_{\max} = 400 \text{ Hz}$$

Que 5.6. What are the different techniques of sampling? Discuss any one of them in detail.

OR

Explain the impulse train sampling of discrete time signals.

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Answer

Basically there are three types of sampling techniques :

- Instantaneous sampling or impulse sampling
- Natural sampling
- Flat-top sampling.

Out of these three methods, instantaneous or impulse sampling is also called ideal sampling, whereas the natural sampling and flat-top sampling are called practical sampling methods.

Ideal or impulse sampling :

1. Ideally, sampling should be done instantaneously so that the k^{th} element of the sequence obtained by sampling represents the value of $x(t)$ at $t = kT$. The operation is shown in Fig. 5.6.1(a).

2. Assume that the fictitious sampler closes almost for zero time once in every T sec. It is equivalent to transmitting the input signal to the output for a very very short time (almost zero time) once in every

T sec.

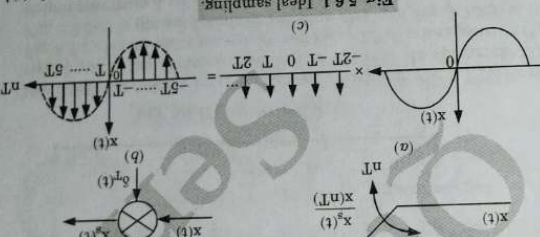


Fig. 5.6.1. Ideal sampling.

3. Now, the operation is equivalent to multiplying the input signal $x(t)$ by an impulse train $\delta_T(t)$ as shown in Fig. 5.6.1(b). So the output of the sampler is a train of impulses of height equal to the instantaneous value of the input signal at the sampling instant.

4. The impulse train, also called the sampling function is represented as :

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

5. The sampled signal is given by

$$x_s(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT)$$

$$\therefore X_s(\omega) = \frac{1}{T} \sum_{n=-\infty}^{\infty} X(\omega - n\omega_s)$$

$$\text{or } X_s(f) = f_s \sum_{n=-\infty}^{\infty} X(f - nf_s) \quad \dots(5.6.1)$$

6. This eq. (5.6.1) gives the spectrum of ideally sampled signal. It shows that the spectrum $X_s(\omega)$ is an infinite sum of shifted replicas of $X(\omega)$ spaced ω_s apart, where $n = \pm 1, \pm 2$, etc., and scaled by a factor $1/T$.

Que 5.7. Explain natural sampling in detail.

Answer

1. Natural sampling, is achieved by multiplying the signal $x(t)$ with a

2. Each pulse of $p_T(t)$ is of short duration τ and occurs at a sampling period of T sec. The output of the sampler is same as the input during that

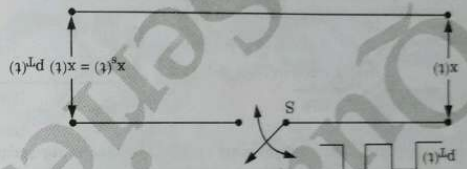


Fig. 5.7.1. Natural sampler.

- Fig. 5.7.2 explains the process of natural sampling. Fig. 5.7.2(a) is the signal $x(t)$ to be sampled, and Fig. 5.7.2(b) is its spectrum $X(f)$. Fig. 5.7.2(c) is the pulse train $p_T(t)$, and Fig. 5.7.2(d) is its spectrum $P(f)$. From Fig. 5.7.2(e), it is clear that $X(f)$ can be recovered from $X_s(f)$, i.e., $x(t)$ can be recovered from $x_s(t)$, if $f_s > 2f_m$ by using an LPF whose gain is constant at least upto $f = f_m$ and whose cut-off frequency B is such that $f_m < B < f_s - f_m$.
- The output of the sampler is $x_s(t) = x(t) p_T(t)$

where

$$p_T(t) = \sum_{n=-\infty}^{\infty} p(t - nT)$$

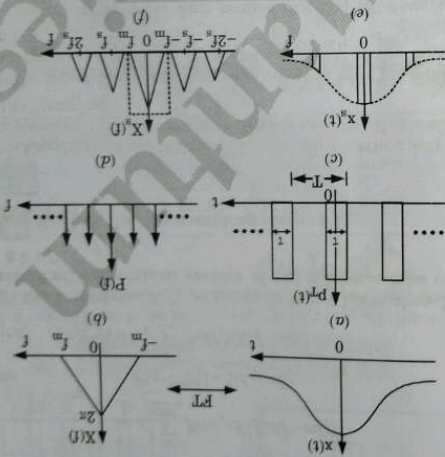


Fig. 5.7.2. Natural sampling.

6. As $p_T(t)$ is a periodic pulse train, its Fourier series expansion will be

$$p_T(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_s t} = \sum_{n=-\infty}^{\infty} C_n e^{jn2\pi f_s t}$$

where,

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} p_T(t) e^{-jn\omega_s t} dt$$

7. Since τ the width of $p(t)$, single pulse in $p_T(t)$ is very much less than T and $p(t) = 0$ for $|t| \geq T/2$, we may write

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} p_T(t) e^{-jn\omega_s t} dt = \frac{1}{T} \int_{-T/2}^{T/2} p(t) e^{-jn\omega_s t} dt$$

$$C_n = f_s P(nf_s)$$

$$\text{where, } P(nf_s) = F\{p(t)\}|_{f=nf_s}$$

$$p_T(t) = f_s \sum_{n=-\infty}^{\infty} P(nf_s) e^{jn2\pi f_s t}$$

and

$$X_g(f) = F[x_g(t)] = F\left[\sum_{n=-\infty}^{\infty} P(nf_s)x(t)e^{j2\pi n f_s t}\right]$$

$$= f_s \sum_{n=-\infty}^{\infty} P(nf_s)X(f)\delta(f - nf_s)$$

$$\text{Since } F[e^{j2\pi n f_s t}] = \delta(f - nf_s)$$

$$\text{Hence, } X_g(f) = f_s \sum_{n=-\infty}^{\infty} P(nf_s)X(f - nf_s)$$

8. If $x(t)$ has a spectrum $X(f)$, as shown in Fig. 5.7.2(b), then $X_g(f)$, the spectrum of the sampled version of $x(t)$ will appear as shown in Fig. 5.7.2(f).

Que 5.8. Explain flat-top sampling method.

Answer

1. This is the simplest and most popular sampling method that uses the sample and hold (S/H) circuit with flat-top samples. This is also called practical sampling.
2. Here the top of the samples remain constant which is equal to the instantaneous value of the baseband signal $x(t)$ at the beginning of sampling.
3. The duration or width of each sample is τ and the sampling rate, $f_s = 1/T$.
4. The schematic of a sample and hold (S/H) circuit is shown in Fig. 5.8.1(a), and a typical output waveform from an S/H circuit is shown in Fig. 5.8.1(b).
5. The S/H circuit essentially consists of two switches S_1 and S_2 and a capacitor C connected as shown in Fig. 5.8.1(a).
6. With S_2 open, S_1 is closed for a very brief period at each sampling instant. The capacitor C then gets charged to a voltage equal to the value of the input signal $x(t)$ at the sampling instant and holds it for a period τ at the end of which S_2 is closed to allow the capacitor to discharge.
7. This sequence of operations is repeated at the next and all subsequent sampling instant.
8. The switches S_1 and S_2 are generally FET switches and are operated by giving appropriate pulses to their gates. The voltage across C appears as $x_s(t)$ and is sketched in Fig. 5.8.1(b).
9. From the Fig. 5.8.1, it is the obvious that the sampled version, $x_s(t)$ consists of a sequence of rectangular pulses, the leading edge of the k th pulse being at $t = kT$ and the amplitude of the pulse being the value of $x(t)$ at $t = kT$, i.e., $x(kT)$.

PART-2

Reconstruction : Ideal Interpolator, Zero-Order Hold, First-Order Hold, and so on.

Que 5.9.

Explain reconstruction of signal from its samples using interpolation.

Answer

1. Interpolation is a method for estimating the value of a function at points where it has not been explicitly measured.
 2. In signal processing interpolation is often used to reconstruct a continuous signal from a set of discrete samples.
 3. The most common methods of interpolation for signal processing are zero-order hold, first-order hold, sine interpolation, spline interpolation etc.
- Reconstruction of signal from its samples using zero-order hold interpolation :**
1. Zero-order hold interpolation is a method for reconstructing a continuous-time signal from its discrete-time samples.
 2. The basic idea is to assume that the continuous-time signal is constant between each pair of consecutive samples.
 3. Then, to reconstruct the continuous-time signal, we simply connect the dots between the samples using straight lines.
 4. The resulting signal is a piecewise constant approximation of the original continuous-time signal.

- Zero-order hold interpolation is also known as "piecewise constant interpolation" or "boxcar interpolation".
- However in practice, reconstruction of a signal from its samples using interpolation may never be perfect, because the original signal is being approximated and the reconstruction error always exists.

Que 5.10. How a band limited signal can be reconstructed from its sampled signal? Explain in detail.

Answer

- The process of obtaining the analog signal $x(t)$ from the sampled signal $x_s(t)$ is called data reconstruction or interpolation. We know that

$$x_s(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$\text{or } x_s(t) = \sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT)$$

- Since $\delta(t - nT)$ is zero except at the sampling instants $t = nT$. The reconstruction filter, which is assumed to be linear and time-invariant, has unit impulse response $h(t)$.
- The reconstruction filter output, $y(t)$ is given by the convolution,

$$y(t) = \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x(nT) \delta(\lambda - nT) h(t - \lambda) d\lambda$$

or, upon changing the order of summation and integration,

$$y(t) = \sum_{n=-\infty}^{\infty} x(nT) \int_{-\infty}^{\infty} \delta(\lambda - nT) h(t - \lambda) d\lambda$$

$$\text{i.e., } y(t) = \sum_{n=-\infty}^{\infty} x(nT) h(t - nT) \quad \dots (5.10.1)$$

Ideal reconstruction filter :

- If $x(t)$ is sampled at a frequency exceeding the Nyquist rate and if the sampled signal $x_s(t)$ is passed through an ideal LPF, with bandwidth greater than f_m but less than $f_s - f_m$ and a pass band amplitude response of T , the filter output is $x(t)$.
- We choose the bandwidth of the ideal reconstruction filter to be $0.5f_s$. The transfer function of this ideal reconstruction filter is, therefore,

$$H(f) = \begin{cases} T & ; |f| < 0.5f_s \\ 0 & ; \text{otherwise} \end{cases}$$

as shown in Fig. 5.10.1.

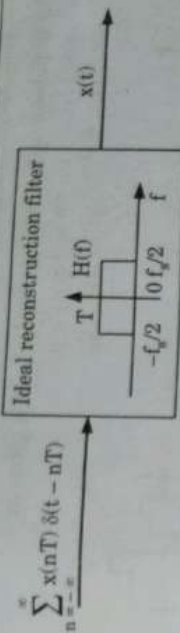


Fig. 5.10.1. Reconstruction filtering.

- The impulse response of the ideal reconstruction filter is given by

$$h(t) = \int_{-f_s/2}^{f_s/2} T e^{j2\pi ft} df$$

$$h(t) = T \left[\frac{e^{j2\pi ft}}{j2\pi t} \right]_{-f_s/2}^{f_s/2} = \frac{T}{j2\pi t} \left(e^{j\pi f_s t} - e^{-j\pi f_s t} \right)$$

$$= \frac{1}{\pi f_s t} \left(\frac{e^{j\pi f_s t} - e^{-j\pi f_s t}}{2j} \right) = \frac{\sin \pi f_s t}{\pi f_s t}$$

$$\text{or } h(t) = \text{sinc } f_s t$$

- Substituting this value of $h(t)$ in the eq. (5.10.1), we get

$$y(t) = x(t) = \sum_{n=-\infty}^{\infty} x(nT) \text{sinc } f_s(t - nT)$$

- A more convenient form for this expression, which is often referred to as an interpolation formula is

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT) \text{sinc} \left(\frac{t}{T} - n \right)$$

This shows that the original data signal can be reconstructed by weighing each sample by a sinc function, centred at the sample time and summing.

Que 5.11. What is zero order hold? Obtain the transfer function of zero order hold.

Answer

- One of the most widely used interpolator is the zero order hold (ZOH). The ZOH reconstructs the continuous-time signal from its samples by holding the given sample for an interval until the next sample is received as shown in Fig. 5.11.1. So the ZOH generates step approximations.
- Mathematically, $\hat{x}_a(t) = x(n)$ for $nT \leq t \leq (n+1)T$. In particular, $\hat{x}_a(t) = x(0)$ for $0 \leq t \leq T$
 $\hat{x}_a(t) = x(T)$ for $T \leq t \leq 2T$
 $\hat{x}_a(t) = x(2T)$ for $2T \leq t \leq 3T$
 $\vdots$

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3. The impulse response of a zero order hold is given by

$$h(t) = \begin{cases} 1 & ; \quad 0 \leq t \leq T \\ 0 & ; \quad \text{otherwise} \end{cases}$$

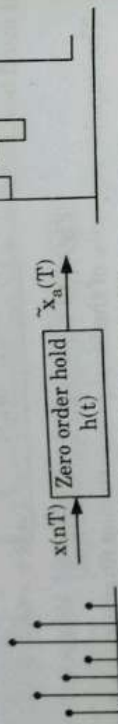


Fig. 5.11.1. Zero order hold.

Transfer function of a zero order hold :

1. The output $\tilde{x}_a(t)$ of a zero order hold is the convolution of its input $x(nT)$ and its impulse response $h(t)$, i.e.,

$$\tilde{x}_a(t) = x(nT) \otimes h(t) = \sum_{n=-\infty}^{\infty} x(nT) h(t - nT)$$

2. For zero order hold,

$$h(t) = u(t) - u(t - T)$$

$$\therefore h(t - nT) = u(t - nT) - u[t - (n + 1)T]$$

$$\therefore \tilde{x}_a(t) = \sum_{n=-\infty}^{\infty} x(nT) [u(t - nT) - u[t - (n + 1)T]]$$

Taking Laplace transform on both sides, we have

$$\begin{aligned} L[\tilde{x}_a(t)] &= \tilde{X}_a(s) = L \left[\sum_{n=-\infty}^{\infty} x(nT) [u(t - nT) - u[t - (n + 1)T]] \right] \\ &= \sum_{n=-\infty}^{\infty} x(nT) \left(\frac{e^{-nTs}}{s} - \frac{e^{-(n+1)Ts}}{s} \right) \\ &= \frac{1 - e^{-Ts}}{s} \sum_{n=-\infty}^{\infty} x(nT) e^{-nTs} = \left(\frac{1 - e^{-Ts}}{s} \right) X^*(s) \end{aligned}$$

$$\therefore \text{Transfer function of zero order hold} = \frac{\tilde{X}_a(s)}{X^*(s)} = \frac{1 - e^{-Ts}}{s}$$

3. Since the output of the ZOH consists of steps, it consists of higher order harmonics. To remove these harmonics, the output of ZOH is applied to an LPF. This filter tends to smooth the corners on the step approximations generated by the ZOH. Hence this filter is often called a smoothing filter.

Que 5.12. Discuss the first order hold interpolation (or linear interpolation).

Answer

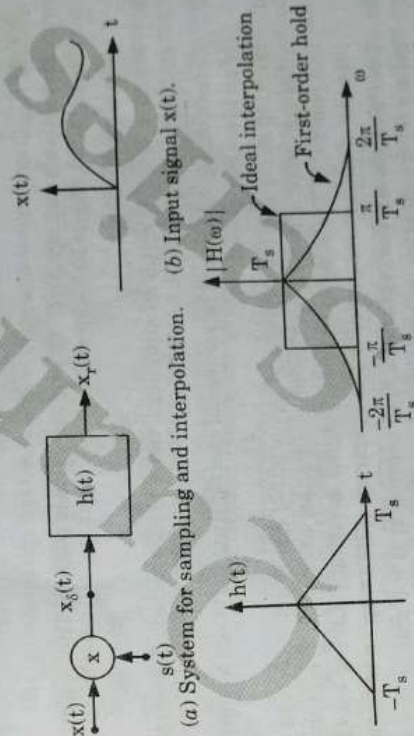
1. The first order hold can be viewed as a form of interpolation between sample values in which the impulse response $h(t)$ of the interpolating filter is a triangular pulse, as depicted in Fig. 5.12.1(c).

$$\begin{aligned} h(t) &= r(t + T_s) - 2r(t) + r(t - T_s) \\ &= \Delta\left(\frac{t}{T_s}\right) = \begin{cases} 1 + \frac{t}{T_s}, & -T_s \leq t \leq 0 \\ 1 - \frac{t}{T_s}, & 0 \leq t \leq T_s \\ 0, & \text{otherwise} \end{cases} \end{aligned} \quad \dots(5.12.1)$$

2. When the sampled signal $x_s(t)$ is applied to this filter, the output is $x_r(t)$.
 3. Each sample in $x_s(t)$, being an impulse, generates a triangular pulse of height equal to the strength of the sample.
 4. The output of the filter $x_r(t)$ is obtained by the convolution of the input $x_s(t)$ with the impulse response $h(t)$.

$$x_r(t) = x_s(t) \otimes h(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s) \otimes h(t)$$

$$x_r(t) = \sum_{n=-\infty}^{\infty} x(nT_s) h(t - nT_s) \quad \dots(5.12.2)$$



- (c) Impulse response of first order interpolating filter.
 (d) Frequency response for the first order hold and for the ideal interpolating filter.

Fig. 5.12.1.

5. Substitution of $h(t)$ from eq. (5.12.1) in eq. (5.12.2) yields

$$x_r(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \Delta\left(\frac{t - nT_s}{T_s}\right) \quad \dots(5.12.3)$$

6. The filter output is a linear approximation of $x(t)$. Fig. 5.12.1(d) shows the magnitude of the transfer function of the first order hold

interpolating filter, superimposed on the desired transfer function of the exact interpolating filter.

PART-3

Aliasing and its Effects, Relation between Continuous and Discrete Time Systems.

Que 5.13. Write a short note on aliasing effect.

OR

State and prove sampling theorem and discuss the effect of under sampling.

AKTU 2017-18, Marks 07

OR

What is Shannon's sampling theorem? Also discuss aliasing by taking an example.

AKTU 2018-19, Marks 07

Answer

Shannon's sampling theorem (Sampling theorem) :
Refer Q. 5.2, Page 5-3J, Unit-5.

Aliasing effect :

1. When $\omega_s < 2\omega_m$ (i.e., undersampling), there is an overlap between the shifted replicas of $X(\omega)$.
2. Consequently $x(t)$ cannot be recovered exactly from $x_s(t)$ even by means of an ideal low-pass filter.
3. Overlap in the shifted replicas of the original spectrum is termed as aliasing, which refers to the phenomenon of a high frequency component in the spectrum of the signal seemingly taking on the identity of a lower frequency in the spectrum of its sampled version.
4. Overlap between the replicas of $X(\omega)$ centred at $\omega = 0$ and at $\omega = \omega_s$ occurs for frequencies between $\omega_s - \omega_m$ and ω_m .
5. These replicas add, and thus the basic shape of the spectrum changes from portions of a triangle to a constant, as shown in Fig. 5.13.1.

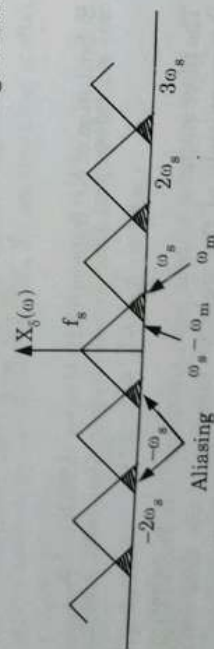


Fig. 5.13.1. Aliasing or spectrum folding.

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6. The spectra cross at frequency $\omega/2$. This frequency is called the folding frequency. The spectrum, therefore, folds onto itself at the folding frequency.
7. The components of frequencies above $\omega/2$ reappear as components of frequencies below $\omega/2$. This tail inversion is known as spectral folding or aliasing.
8. To combat the effect of aliasing we may use two corrective measures as described here :
 - i. Prior to sampling, a low-pass pre-alias filter is used to attenuate those high frequency components of the signal that are not essential to the information conveyed by the signal.
 - ii. The filtered signal is sampled at a rate slightly higher than the Nyquist rate.

Que 5.14. Discuss the relationship between the continuous and discrete time systems.

Answer

1. The relationship between the continuous-time signal $x_c(t)$ and its discrete-time representation $x_d[n]$ is helpful to represent C/D as a process of periodic sampling followed by a mapping of the impulse train to sequence.
2. These two steps are illustrated in Fig. 5.14.1. In the first step, representing the sampling process, the impulse train $x_p(t)$ corresponds to a sequence of impulses with amplitudes corresponding to the samples of $x_c(t)$ and with a time spacing equal to the sampling period T .
3. In the conversion from the impulse train to the discrete-time sequence, we obtain $x_d[n]$, corresponding to the same sequence of samples of $x_c(t)$, but with unity spacing in terms of the new independent variable n .
4. Thus, in effect, the conversion from the impulse train sequence of samples to the discrete-time sequence of samples can be thought of as normalization in time.
5. This normalization in converting $x_p(t)$ to $x_d[n]$ is evident in Fig. 5.14.1(b) and (c), in which $x_p(t)$ and $x_d[n]$ are respectively illustrated for sampling rates of $T = T_1$ and $T = 2T_1$.
6. We distinguish the continuous-time and discrete-time frequency variables by using ω in continuous time and Ω in discrete time.
7. The continuous-time Fourier transforms of $x_c(t)$ and $x_c(t)$ are $X_c(j\omega)$ and $Y_c(j\omega)$, respectively, while the discrete-time Fourier transforms of $x_d[n]$ and $y_d[n]$ are $X_d(e^{j\Omega})$ and $Y_d(e^{j\Omega})$, respectively.

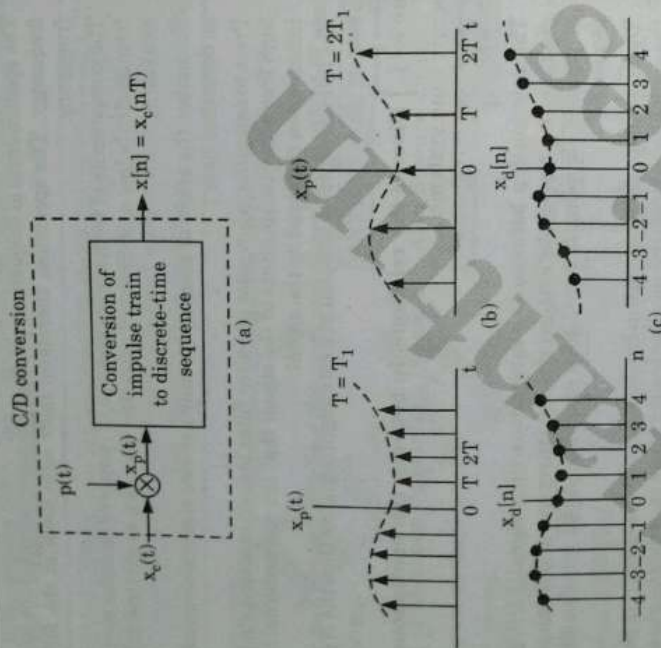


Fig. 5.14.1. Sampling with a periodic impulse train followed by conversion to a discrete-time sequence : (a) overall system; (b) $x_p(t)$ for two sampling rates. The dashed envelope represents $x_c(t)$ (c) the output sequence for the two different sampling rates.

8. The continuous-time Fourier transform of $x_c(t)$, in terms of the sample value of $x_c(t)$ by applying the Fourier transform is,

$$x_p(t) = \sum_{n=-\infty}^{\infty} x_c(nT) \delta(t - nT), \quad \dots (5.14.1)$$

and since the transform of $\delta(t - nT)$ is $e^{-j\omega nT}$, it follows that

$$X_p(j\omega) = \sum_{n=-\infty}^{\infty} x_c(nT) e^{-j\omega nT} \quad \dots (5.14.2)$$

9. Now consider the discrete-time Fourier transform of $x_d[n]$, that is,

$$X_d(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x_d[n] e^{-j\Omega n} \quad \dots (5.14.3)$$

or $X_d(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x_c(nT) e^{-j\Omega n} \quad \dots (5.14.3)$

10. Comparing eq. (5.14.2) and (5.14.3), we see that $X_d(e^{j\Omega})$ and $X_p(j\omega)$ are related through

$$X_d(e^{j\Omega}) = X_p(j\Omega/T) \quad \dots (5.14.4)$$

Also, $X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\omega - k\omega_s)) \quad \dots (5.14.5)$

Consequently,

$$X_d(e^{j\Omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - 2\pi k) / T) \quad \dots (5.14.6)$$

11. The relationship among $X_c(j\omega)$, $X_p(j\omega)$, and $X_d(e^{j\Omega})$ is illustrated in Fig. 5.14.2 for two different sampling rates. Here, $X_d(e^{j\Omega})$ is a frequency-scaled version of $X_p(j\omega)$ and, in particular, is periodic in Ω with period 2π .

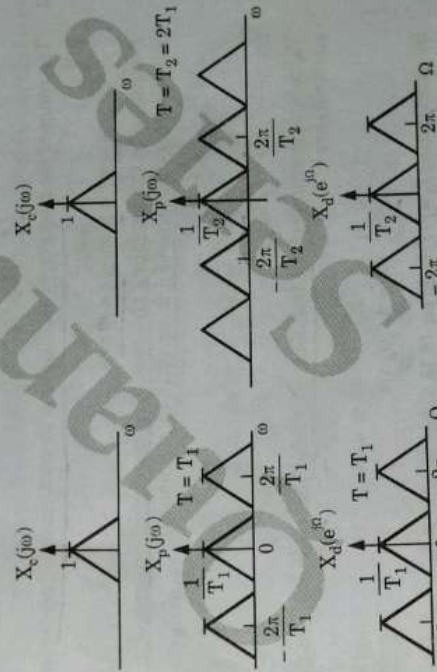


Fig. 5.14.2. Relationship between $X_c(j\omega)$, $X_p(j\omega)$, and $X_d(e^{j\Omega})$ for two different sampling rates.

12. The relationship $\Omega = \omega T$ exist between continuous and discrete-time variables, which we can get by comparing eq. (5.14.5) and (5.14.6).

Que 5.15. Explain the sampling of sinusoidal signals.

Answer

1. Consider a continuous-time sinusoidal signal of the form

$$x(t) = A \cos(2\pi ft) \quad \dots(5.15.1)$$

which, when sampled periodically at a rate $f_s = 1/T_s$ samples/s, yields

$$x(t)|_{t=nT_s} = x(nT_s) = x(n) = A \cos(2\pi f n T_s)$$

$$x(n) = A \cos\left(2\pi n \frac{f}{f_s}\right) \quad \dots(5.15.2)$$

$$x(n) = A \cos(2\pi n f_d) \quad \dots(5.15.4)$$

where, $f_d = \frac{f}{f_s}$ = frequency for discrete-time signal

$$\dots(5.15.5)$$

or, equivalently, as

$$2\pi f_d = \frac{2\pi f}{f_s}$$

$$\omega_d = \frac{\omega}{f_s} = \omega T_s \quad \dots(5.15.6)$$

2. The range of the frequency variable f or ω for continuous-time sinusoids are

$$-\infty < f < \infty \text{ or } -\infty < \omega < \infty \quad \dots(5.15.7)$$

3. According to the sampling theorem, the sampling frequency $f_s \geq 2f$. Substituting this value of f_s in eq. (5.15.5), we obtain

$$f_d \leq \frac{1}{2}$$

$$\omega_d \leq \pi$$

Therefore, the range of the frequency variable f_d or ω_d for discrete-time sinusoids are

$$-\frac{1}{2} \leq f_d \leq \frac{1}{2} \text{ or } -\pi \leq \omega_d \leq \pi \quad \dots(5.15.8)$$

4. By substituting $f_d = f/f_s$ from eq. (5.15.5) and $\omega_d = \omega/f_s$ from eq. (5.15.6) into eq. (5.15.8), we find that the frequency of the continuous-time sinusoids when sampled at a rate $f_s = 1/T_s$ must fall in the range

$$-\frac{1}{2T_s} \leq f \leq \frac{1}{2T_s} \quad \dots(5.15.9)$$

$$\text{or, } -\frac{\pi}{T_s} \leq \omega \leq \pi f_s = \frac{\pi}{T_s} \quad \dots(5.15.10)$$

5. Periodic sampling of a continuous-time signal implies a mapping of the infinite frequency range for the variable f (or ω) into a finite frequency range for the variable f_d (or ω_d).

6. Since the highest frequency in a discrete-time signal is $\omega_d = \pi$ or $f_d = 1/2$, it follows that, with a sampling rate f_s , the corresponding highest values of f and ω are

$$f_{\max} = \frac{f_s}{2} = \frac{1}{2T_s} \quad \dots(5.15.11)$$

$$\omega_{\max} = \pi f_s = \frac{\pi}{T_s}$$

Que 5.16. Consider the two continuous-time sinusoidal signals $x_1(t) = \cos(20\pi t)$ and $x_2(t) = \cos(100\pi t)$ which are sampled at a rate $f_s = 40$ Hz. Find the corresponding discrete time signals.

AKTU 2017-18, Marks 07

Answer

1. Given,

$$x_1(t) = \cos(20\pi t)$$

$$x_2(t) = \cos(100\pi t)$$

2. The two sinusoidal signals can be written in standard form for $\cos(2\pi ft)$ as follows:

$$x_1(t) = \cos 2\pi(10)t$$

$$x_2(t) = \cos 2\pi(50)t$$

and Sampling rate, $f_s = 40$ Hz

3. The corresponding discrete time signal can be given as:

$$x_1(n) = \cos 2\pi\left(\frac{10}{40}\right)n$$

$$x_1(n) = \cos \frac{\pi}{2}n$$

$$x_2(n) = \cos 2\pi\left(\frac{50}{40}\right)n$$

$$x_2(n) = \cos \frac{5\pi}{2}n$$

and

Que 5.17. The signals $x_1(t) = 10 \cos(100\pi t)$ and $x_2(t) = 10 \cos(50\pi t)$ are both sampled with $f_s = 75$ Hz. Show that the two sequences of samples obtained are identical.

Answer

1. If the signal is sampled at $f_s = 75$ Hz, the discrete-time signal is

$$x_1(t)|_{t=nT_s} = x_1(nT_s) = x_1(n) = 10 \cos(100\pi n T_s) \\ = 10 \cos\left(\frac{100\pi}{75}n\right)$$

$$\begin{aligned}
 x_1(n) &= 10 \cos\left(\frac{4\pi}{3}n\right) \\
 &= 10 \cos\left(2n\pi - \frac{2\pi}{3}n\right) \\
 x_1'(n) &= 10 \cos\left(\frac{2\pi}{3}n\right)
 \end{aligned}$$

2. Similarly, we have

$$\begin{aligned}
 x_2(t)|_{t=nT_s} &= x_2(nT_s) = x_2(n) = 10 \cos(50\pi nT_s) \\
 &= 10 \cos\left(\frac{50\pi}{75}n\right)
 \end{aligned}$$

$$x_2(n) = 10 \cos\left(\frac{2\pi}{3}n\right) = x_1(n)$$

VERY IMPORTANT QUESTIONS

Following questions are very important. These questions may be asked in your SESSIONALS as well as UNIVERSITY EXAMINATION.

Q. 1. State and prove sampling theorem. Also give some applications of sampling theorem.

Ans. Refer Q. 5.2.

Q. 2. Determine the Nyquist sampling rate and Nyquist sampling interval for

$$x(t) = \text{sinc}(100\pi t) + 3 \text{sinc}^2(60\pi t)$$

Ans. Refer Q. 5.4.

Q. 3. Explain reconstruction of signal from its samples using interpolation.

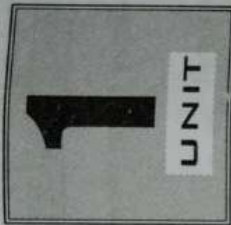
Ans. Refer Q. 5.9.

Q. 4. What is Shannon's sampling theorem? Also discuss aliasing by taking an example.

Ans. Refer Q. 5.13.

Q. 5. Consider the two continuous-time sinusoidal signals $x_1(t) = \cos(20\pi t)$ and $x_2(t) = \cos(100\pi t)$ which are sampled at a rate $f_s = 40$ Hz. Find the corresponding discrete time signals.

Ans. Refer Q. 5.16.



Introduction to Signals and Systems (2 Marks Questions)

1.1. Define a signal.

AKTU 2018-19, Marks 02

AKTU 2021-22, Marks 02

Ans. A signal is a single-valued function of one or more variables that conveys some information. For example, an audio waveform.

1.2. Distinguish between energy and power signal.

AKTU 2018-19, Marks 02

Ans.

S. No.	Energy signal	Power signal
1.	Energy E of signal $x(t)$ is defined as $E = \int_{-\infty}^{\infty} x(t) ^2 dt$	Average power P of continuous signal $x(t)$ is defined as $P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) ^2 dt$
2.	$x(t)$ is said to be an energy signal if and only if $0 < E < \infty$, and $P = 0$.	$x(t)$ is said to be power signal if and only if $0 < P < \infty$, and $E = \infty$.
3.	Non-periodic signals are energy signal.	Periodic signals are power signal.

1.3. Draw the signal $u(n) - u(n-3)$.

AKTU 2021-22, Marks 02

Ans.

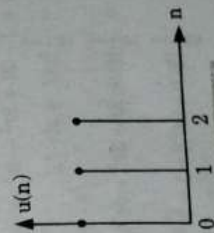


Fig. 1.3.1.

1.4. Determine whether it is energy or power signal $x(t) = tu(t)$.

AKTU 2022-23, Marks 02

Ans. Given $x(t) = tu(t)$

The given signal is a ramp signal and it is neither energy nor a power signal.

1.5. Determine the even and odd part of the signal.

$$x(t) = \cos\left(\omega t + \frac{\pi}{3}\right)$$

Ans. Given, $x(t) = \cos\left(\omega t + \frac{\pi}{3}\right)$

$$x_e(t) = \frac{x(t) + x(-t)}{2} = \frac{\cos\left(\omega t + \frac{\pi}{3}\right) + \cos\left(-\omega t + \frac{\pi}{3}\right)}{2}$$

$$= \frac{\cos\left(\omega t + \frac{\pi}{3}\right) + \cos\left(\omega t - \frac{\pi}{3}\right)}{2}$$

$$= \frac{\cos \omega t \cdot \frac{1}{2} - \sin \omega t \cdot \frac{\sqrt{3}}{2} + \cos \omega t \cdot \frac{1}{2} + \sin \omega t \cdot \frac{\sqrt{3}}{2}}{2}$$

$$= \frac{\cos \omega t}{2}$$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

$$= \frac{\cos\left(\omega t + \frac{\pi}{3}\right) - \cos\left(\omega t - \frac{\pi}{3}\right)}{2} = \frac{2 \cdot \frac{\sqrt{3}}{2} \sin \omega t}{2}$$

$$= -\frac{\sqrt{3}}{2} \sin \omega t$$

1.7. Find even and odd component of the following signal

$$x(t) = 1 + 2t + 3t^2 + 4t^3$$

Ans.

AKTU 2022-23, Marks 02

$$x(t) = 1 + 2t + 3t^2 + 4t^3$$

$$x(-t) = 1 - 2t + 3t^2 - 4t^3$$

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)]$$

$$= \frac{1}{2} [(1 + 2t + 3t^2 + 4t^3) + (1 - 2t + 3t^2 - 4t^3)]$$

$$= \frac{1}{2} [2(1 + 3t^2)] = 1 + 3t^2$$

$$x_o(t) = \frac{1}{2} [x(t) - x(-t)]$$

$$= \frac{1}{2} [(1 + 2t + 3t^2 + 4t^3) - (1 - 2t + 3t^2 - 4t^3)]$$

$$= \frac{1}{2} [2(2t + 4t^3)] = 2t + 4t^3$$

1.8. Find the fundamental period of the given signal.

$$x(n) = \sin\left(\frac{6\pi n}{7} + 1\right)$$

Ans.

The given signal,

$$x(n) = \sin\left(\frac{6\pi n}{7} + 1\right)$$

$$\omega_0 = \frac{6\pi}{7}$$

$$\text{So, period } N = \frac{2\pi}{\omega_0} = \frac{2\pi}{\frac{6\pi}{7}} \quad [\text{where 'm' is a positive integer}]$$

$$m = 0, 1, 2, 3, \dots$$

$$N = \frac{7m}{3}$$

$$\text{for } m = 3, N = 7$$

hence, the fundamental period of $x(n)$ is 7.

1.9. Determine the fundamental period of the following if the signal is periodic $x(t) = \cos(\pi t) + \cos(2\pi t)$.

AKTU 2021-22, Marks 02

Ans.

1. Given, $x(t) = \cos(\pi t) + \cos(2\pi t)$

2. Assume $x_1(t) = \cos \pi t$, and

$$x_2(t) = \cos 2\pi t$$

$$x(t) = \cos \pi t$$

$$\omega_1 = \pi$$

$$= 2\pi f_1 = \pi$$

$$f_1 = \frac{1}{2}$$

$$T_1 = 2$$

$$x_2(t) = \cos 2\pi t$$

4. For

$$2\pi f_1 = 2\pi$$

 $\Rightarrow$

$$f_2 = 1$$

$$\omega_2 = 2\pi$$

$$T_2 = 1$$

$$\frac{T_1}{T_2} = \frac{2}{1} = 2$$

5. It is a rational number. So $x(t)$ is a periodic function.

6. The fundamental period is

$$\text{LCM}(T_1, T_2) = 2$$

1.10. Check the periodicity of the signals given below. Determine the fundamental time period if signal is periodic:

$$x(t) = \sin(10t - 1) - \sin(4t - 1)$$

AKTU 2017-18, Marks 02

Ans.

1. Given, $x(t) = \sin(10t - 1) - \sin(4t - 1)$

Let $x(t) = x_1(t) - x_2(t)$

2. Now, $x_1(t) = \sin(10t - 1)$

Comparing with

$$x(t) = \sin(\omega_1 t + \theta_1) \Rightarrow \omega_1 = 10 \Rightarrow 2\pi f_1 = 10$$

or

$$f_1 = \frac{10}{2\pi} = \frac{5}{\pi}$$

So,

$$T_1 = \frac{1}{f_1} = \frac{\pi}{5} \text{ sec}$$

3. Now, $x_2(t) = \sin(4t - 1)$

Comparing with

$$x_2(t) = \sin(\omega_2 t + \theta_2) \Rightarrow \omega_2 = 4 \Rightarrow 2\pi f_2 = 4$$

$$f_2 = \frac{4}{2\pi} = \frac{2}{\pi}$$

or

$$T_2 = \frac{1}{f_2} = \frac{\pi}{2} \text{ sec}$$

4. The ratio of two period = $\frac{T_1}{T_2} = \frac{\pi/5}{\pi/2} = \frac{2}{5}$ (rational number).

$$\text{So, fundamental period} = \text{L.C.M}\left(\frac{\pi}{5}, \frac{\pi}{2}\right) = \pi$$

1.11. Draw the signal $x(t) = u(t) - u(t - 2)$.

Ans. Given,

$$x(t) = u(t) - u(t - 2)$$

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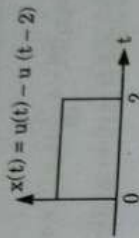


Fig. 1.11.1.

1.12. Verify whether the given system described by the equation is linear and time-invariant: $x(t) = t^2$.

Ans. The given system is,

$$x(t) = t^2$$

$$x_1(t) = (t - t_0)^2$$

[Delayed input]

$$x_2(t - t_0) = (t - t_0)^2$$

[Delayed output]

Now

$$x_1(t) = x_2(t - t_0) = (t - t_0)^2$$

Hence, the given system is time-invariant.

Now

$$x(t) = t^2$$

$$x_1(t) = at^2$$

[scaling in input]

$$x_2(at) = (at)^2 = a^2 t^2$$

[scaling in output]

Since,

$$x_1(t) \neq x_2(at)$$

Hence, the given system is non-linear.

1.13. Check whether the system is linear or non-linear.

$$2 \frac{d^2 y(t)}{dt^2} + 4 \frac{dy(t)}{dt} + 3y(t) = x(t + 1)$$

$$\text{Ans. Given, } 2 \frac{d^2 y(t)}{dt^2} + 4 \frac{dy(t)}{dt} + 3y(t) = x(t + 1)$$

1. Applying weight sum of the response due to input signal is

$$2 \frac{d^2}{dt^2} [a_1 y_1(t)] + 4 \frac{d}{dt} [a_1 y_1(t)] + 3 [a_1 y_1(t)] = a_1 x_1(t + 1) \quad \dots (1.13.1)$$

$$2 \frac{d^2}{dt^2} [a_2 y_2(t)] + 4 \frac{d}{dt} [a_2 y_2(t)] + 3 [a_2 y_2(t)] = a_2 x_2(t + 1) \quad \dots (1.13.2)$$

2. Adding the eq. (1.13.1) and (1.13.2), we get

$$2 \frac{d^2}{dt^2} [a_1 y_1(t) + a_2 y_2(t)] + 4 \frac{d}{dt} [a_1 y_1(t) + a_2 y_2(t)] + 3 [a_1 y_1(t) + a_2 y_2(t)] = a_1 x_1(t + 1) + a_2 x_2(t + 1) \quad \dots (1.13.3)$$

3. When input is given in weight, then

$$2 \frac{d^2}{dt^2} [a_1 y_1(t) + a_2 y_2(t)] + 4 \frac{d}{dt} [a_1 y_1(t) + a_2 y_2(t)] + 3 [a_1 y_1(t) + a_2 y_2(t)] = a_1 x_1(t+1) + a_2 x_2(t+1) \quad \dots (1.13.4)$$

Since eq. (1.13.3) = eq. (1.13.4), hence it is linear system.

1.14. Consider a discrete-time system with input $x[n]$ output

$$y[n] = n[x[n]]^2$$

Is this system time-variant or time-invariant ?

Ans. Given, $y[n] = n[x[n]]^2$

For the delayed input

$$y[n] = n[x[n - n_0]]^2$$

For delayed output

$$y[n - n_0] = (n - n_0)[x[n - n_0]]^2$$

Since

$$y[n] \neq y[n - n_0]$$

Hence the system is time varying.

1.15. Check whether the given system are time-variant and

$$\text{causal } y(t) = tx(t).$$

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Ans. For time-variant :

$$\text{Given, } y(t) = tx(t)$$

Case I : Delaying input by T [i.e., replace $(t - T)$ in place of t in $x(t)$]

$$y(t, T) = [x(t - T)]$$

Case II : Delaying output by T [i.e., replace $(t - T)$ for every t in $y(t)$]

$$\text{or } y(t - T) = [(t - T)x(t - T)]$$

$$\text{Since, } y(t - T) \neq y(t, T)$$

Thus, system is time-variant.

For causality :

$$\text{For } t = 0, \quad y(0) = 0 \times x(0)$$

$$\text{For } t = 1, \quad y(1) = x(1)$$

$$\text{For } t = -1, y(-1) = -1 \times x(-1)$$

Since, system depends on present and past value only; therefore it is a causal system.



2

UNIT

Linear Shift-Invariant (LSI) System (2 Marks Questions)

2.1. What do you mean by causality of system ?

Ans. A system is said to be a 'causal system' or a 'non-anticipatory system' if its output at any instant of time depends only on the input at that instant and the previous instants but not on the input at future instant.

2.2. What are the necessary conditions for an LTI system to be stable ?

Ans. For an LTI system to be stable :

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty ; \quad \text{in continuous-time (CT) domain.}$$

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty ; \quad \text{in discrete-time (DT) domain.}$$

2.3. State the expression of convolution integral.

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Ans. The expression of convolution integral is as follows

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \quad \dots (2.3.1)$$

where $x(t)$ is input signal and $h(t)$ = impulse response

By change of variable, $\lambda = (t - \tau)$, we may write eq. (2.3.1) as

$$y(t) = \int_{-\infty}^{\infty} x(t - \tau) h(\tau) d\tau$$

hence,

$$\begin{aligned} y(t) &= x(t) \otimes h(t) \\ &= \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = \int_{-\infty}^{\infty} x(t - \tau) h(\tau) d\tau \end{aligned}$$

2.4. An LTI system is described by the differential equation

$$\frac{dy(t)}{dt} + 4y(t) = x(t), \text{ Determine its impulse response } h(t) \text{ and then } H(f).$$

Ans. Given,

$$\frac{dy(t)}{dt} + 4y(t) = x(t)$$

Taking Laplace transform

$$sY(s) + 4Y(s) = X(s)$$

$$H(f) = \frac{Y(s)}{X(s)} = \frac{1}{s+4}$$

$$x(t) = \delta(t), X(s) = 1$$

$$\text{For impulse input } H(f) = Y(s) = \frac{1}{s+4}$$

The impulse response is,

$$h(t) = y(t) = e^{-4t}u(t)$$

2.5. Obtain the convolution of :

$$x(t) = u(t) \text{ and } h(t) = 1 \text{ for } -1 \leq t \leq 1$$

$$\text{Given, } x(t) = u(t) \text{ and } h(t) = 1 \text{ for } -1 \leq t \leq 1$$

we can also write

$$h(t) = u(t+1) - u(t-1)$$

we know that

$$x(t) \otimes h(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} [u(\tau+1) - u(\tau-1)] u(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} u(\tau+1) u(t-\tau) d\tau - \int_{-\infty}^{\infty} u(\tau-1) u(t-\tau) d\tau$$



Fig. 2.5.1.

In the 1st part $u(\tau+1) = 1$ for $\tau > -1$ and $u(t-\tau) = 1$ for $\tau < t$, and in 2nd part $u(\tau-1) = 1$ for $\tau > 1$ and $u(t-\tau) = 1$ for $2 < t$.

$$\begin{aligned} x(t) \otimes h(t) &= \int_{-1}^t 1 d\tau - \int_{t-1}^t 1 d\tau \\ &= (t+1)u(t+1) - (t-1)u(t-1) \end{aligned}$$

2.6. Explain significance of convolution in a communication system.

Ans. Convolution in communication systems is crucial for signal processing. It enhances data transmission by filtering and extracting relevant information from signals. In applications like channel coding and equalization, convolution ensures efficient data representation, reducing errors and improving overall communication reliability.

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2.7. Write two properties of convolution.

Ans.

1. **Commutative property :** The commutative property of convolution states that

$$x_1(t) \otimes x_2(t) = x_2(t) \otimes x_1(t)$$

2. **Distributive property :** The distributive property of convolution states that

$$x_1(t) \otimes [x_2(t) + x_3(t)] = [x_1(t) \otimes x_2(t)] + [x_1(t) \otimes x_3(t)]$$

2.8. What is the difference between cross correlation and auto correlation ?

Ans.

S.No.	Cross correlation	Auto correlation
1.	Measures the similarity between two different signals.	Measures the similarity between a signal and delayed version of itself.
2.	Cross correlation is not necessarily symmetric, as it depends on the order of the signals.	Auto correlation is always symmetric.

2.9. Mention relation between impulse response with frequency response.

Ans.

1. Let $h(t)$ be the impulse response of the system.

2. The $H(s)$ be the transfer function of $h(t)$ given by

$$H(s) = \int_{-\infty}^{\infty} h(t) e^{-st} dt \quad \dots(2.9.1)$$

3. The frequency response is given by substituting $s \rightarrow j\omega$ in eq. (2.9.1) as,

$$H(s)|_{j\omega} = H(j\omega) = \text{Frequency response}$$

2.10. Write the differential equation of the circuit shown in Fig. 2.10.1.

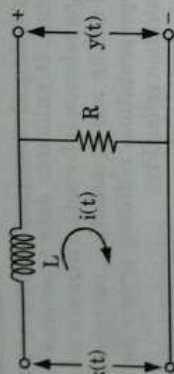


Fig. 2.10.1.

Ans. Writing KVL equation, we have

$$L \frac{di(t)}{dt} + R i(t) = x(t)$$

and

$$R i(t) = y(t)$$

$$\therefore \frac{L}{R} \frac{dy(t)}{dt} + y(t) = x(t)$$



Transforms (2 Marks Questions)

3.1. Write Dirichlet's conditions for the existence of Fourier series.

Ans. The conditions under which a periodic signal can be represented by a Fourier series are known as Dirichlet's conditions. They are as follows :

In each period,

- The function $x(t)$ must be a single valued function.
- The function $x(t)$ has only a finite number of maxima and minima.
- The function $x(t)$ has a finite number of discontinuities.
- The function $x(t)$ is absolutely integrable over one period, that is $\int_0^T |x(t)| dt < \infty$.

3.2. Prove the time differentiation property of Fourier transform.

Ans. We know, $x(t) = F^{-1} [X(f)] = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$

Taking the time derivative

$$\begin{aligned} \frac{d}{dt} x(t) &= \frac{d}{dt} \left[\int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df \right] \\ &= \int_{-\infty}^{\infty} \{ j2\pi f X(f) \} e^{j2\pi ft} df \end{aligned}$$

Taking Fourier transform

$$F \left[\frac{d}{dt} x(t) \right] = j2\pi f X(f)$$

The n th time derivation can be calculated as

$$F \left[\frac{d^n}{dt^n} x(t) \right] = (j2\pi f)^n X(f)$$

3.3. Determine the sufficient condition for the existence of CTFT.

AKTU 2021-22, Marks 02

Ans. The conditions for the existence of CTFT are:

- The function $x(t)$ has a finite number of maxima and minima in every finite interval of time.
- The function $x(t)$ has a finite number of discontinuities in every finite interval of time.

3.4. Differentiate between CTFT and DTFT.

AKTU 2021-22, Marks 02

Ans.

S.No.	CTFT	DTFT
1.	The CTFT, $X(j\omega)$ is not periodic in ω .	The DTFT, $X(\omega)$ is periodic in ω .
2.	Its inverse involves integration over all frequencies from $-\infty$ to ∞ .	Inverse DTFT involves integration over a frequency interval of only 2π .

3.5. Find the Fourier transform of

$$x(t) = \sin(\omega t) \cos(\omega t).$$

Ans. Given, $x(t) = \sin(\omega t) \cos(\omega t)$

$$= \frac{1}{2} \sin(2\omega t)$$

$$x(t) = \frac{1}{2} \times \frac{1}{2j} (e^{j2\omega t} - e^{-j2\omega t})$$

$$x(t) = \frac{-j}{4} (e^{j2\omega t} - e^{-j2\omega t})$$

Taking Fourier transform

$$F[x(t)] = \frac{-j}{4} [F(e^{j2\omega t}) - F(e^{-j2\omega t})]$$

$$X(\omega') = \frac{-j}{4} [2\pi\delta(\omega' - 2\omega) - 2\pi\delta(\omega' + 2\omega)]$$

3.6. Find the discrete-time Fourier transform of the signal.

$$x[n] = a^n u[-n-1]$$

Given, $x[n] = a^n u[-n-1]$

Discrete-time Fourier transform of $x[n]$

$$X(\omega) = \sum_{n=-\infty}^{\infty} a^n u[-n-1] e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{n=-1} (ae^{j\omega})^{-n} = \sum_{n=1}^{\infty} (ae^{j\omega})^n$$

$$X(\omega) = \frac{ae^{j\omega}}{1 - ae^{j\omega}}$$

3.7. What is the condition of existence (convergence) of the Laplace transform?

Ans. The condition for existence of Laplace transform for some finite value of σ is given as

$$\int_{-\infty}^{\infty} |x(t)e^{-\sigma t}| dt < \infty$$

3.8. Define initial value theorem for Laplace transform.

Ans. This theorem enables us to determine the initial value of a function $x(t)$, directly from its Laplace transform $X(s)$. It states that

$$\lim_{t \rightarrow 0} x(t) = \lim_{s \rightarrow \infty} sX(s), \text{ if } x(t) \text{ is continuous at } t = 0.$$

3.9. Write time shifting property of Laplace transform.

AKTU 2022-23, Marks 02

Ans. If the $x(t)$ signal is delayed in time by t_0 unit then its Laplace transform is multiplied by e^{-st_0} factor.

$$\text{If } x(t) \xrightarrow{LT} X(s)$$

$$x(t - t_0) \xrightarrow{} e^{-st_0} X(s)$$

where t_0 is constant.

3.10. Determine the initial value for the given Laplace transform

$$X(s) = \frac{4}{s^2 + 3s + 8}.$$

AKTU 2022-23, Marks 02

$$\text{Ans. Given } X(s) = \frac{4}{s^2 + 3s + 8}$$

$$x(0^+) = \lim_{s \rightarrow \infty} sX(s) = \lim_{s \rightarrow \infty} s \left(\frac{4}{s^2 + 3s + 8} \right)$$

$$\lim_{s \rightarrow \infty} \frac{4s}{1 + 3s + 8s^2} = 0 \left(s = \frac{1}{s} \right)$$

3.11. Given, $F(s) = \frac{s+8}{s^2+6s+13}$, find $f(0)$ using the initial value

theorem.

Ans. $sF(s) = \frac{s^2 + 8s}{s^2 + 6s + 13} = 1 + \frac{2s - 13}{s^2 + 6s + 13}$

From the initial value theorem, we know that

$$f(0) = \lim_{s \rightarrow \infty} sF(s)$$

$$f(0) = \lim_{s \rightarrow \infty} \left[1 + \frac{2s - 13}{s^2 + 6s + 13} \right] = 1.$$

3.12. Find out the Laplace transform of the signal with its ROC.

$$x(t) = e^{-t}u(t) + e^{-4t}u(t)$$

OR

Determine the Laplace transform and find out ROC for

$$x(t) = e^{-t}u(t) + e^{-4t}u(t)$$

AKTU 2017-18, Marks 02

Ans. Given, $x(t) = e^{-t}u(t) + e^{-4t}u(t)$

Taking Laplace transform of both side

$$X(s) = \frac{1}{s+1} + \frac{1}{s+4}$$

Then common ROC is $s > -4$

3.13. Write the s-domain transfer function of a first order system.

Ans. Transfer function $H(s) = \frac{1}{1 + s\tau}$

where τ = Time constant of the system.

3.14. Find the DTFT of the causal sequence $x(n) = a^n u(n)$; $|a| < 1$.

AKTU 2017-18, Marks 02

Ans. The DTFT is given as

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} a^n \cdot e^{-j\omega n} = \sum_{n=0}^{\infty} (a \cdot e^{-j\omega})^n$$

$$X(\omega) = 1 + ae^{-j\omega} + a^2 e^{-2j\omega} + a^3 e^{-3j\omega} + \dots$$

$$X(\omega) = \frac{1}{1 - ae^{-j\omega}} \quad \text{and} \quad |ae^{-j\omega}| < 1$$

3.15. Using final value theorem, find final value of signal corresponding to Laplace transform

$$X(s) = \frac{s+1}{s(s+2)(s+8)}$$

AKTU 2017-18, Marks 02

Ans. Given, $X(s) = \frac{(s+1)}{s(s+2)(s+8)}$

According to final value theorem,

$$\begin{aligned} x(\infty) &= \lim_{s \rightarrow 0} [s \cdot X(s)] = \lim_{s \rightarrow 0} \frac{s(s+1)}{s(s+2)(s+8)} \\ &= \lim_{s \rightarrow 0} \frac{(s+1)}{(s+2)(s+8)} = \frac{1}{2 \times 8} = \frac{1}{16} \end{aligned}$$

3.16. What are advantages of Laplace transform ?

AKTU 2018-19, Marks 02

Ans.

1. Signals which are not convergent in Fourier transform are convergent in Laplace transform.
2. Convolution in time domain can be obtained by multiplication in s-domain.
3. LTI systems can be analysed easily using Laplace transforms.

3.17. What are the limitations of Fourier transform ?

AKTU 2018-19, Marks 02

Ans.

1. Fourier transform gives only spectral details of the signal without considering temporal properties. Hence it is not suitable for analyzing signals with time varying spectra (non-stationary signals).
2. It has fixed time and frequency resolution i.e., 100 % frequency information but 0 % time information.

3.18. Define power spectral density.

AKTU 2018-19, Marks 02

Ans. The distribution of average power of the signal in the frequency-domain is called power spectral density (PSD).



4

UNIT

Z-Transform (2 Marks Questions)

4.1. What is the condition for existence of z-transform ?

AKTU 2022-23, Marks 02

Ans. $\sum_{n=-\infty}^{\infty} |x(n)z^{-n}| < \infty$, this is the condition for existence of z-transform.

4.2. Define initial value theorem for z-transform.

Ans. For a causal discrete-time signal, the initial value may be determined by,

$$x[0] = \lim_{n \rightarrow 0} x[n] = \lim_{z \rightarrow \infty} X(z)$$

4.3. What is the final value theorem for z-transform ?

Ans. The final-value of a discrete time signal can be determined as,

$$x[\infty] = \lim_{n \rightarrow \infty} x[n] = \lim_{z \rightarrow 1} [(1-z)^{-1} X(z)]$$

4.4. What is the significance of ROC ?

AKTU 2022-23, Marks 02

Ans. The region of convergence is the area in the pole / zero plot of the transfer function in which the function exists. For purposes of useful filter design, we prefer to work with rational functions, which can be described by two polynomials one each for determining the poles and the zeroes respectively.

4.5. Give any two properties of ROC of z-transform.

1. The ROC of a right sided signal is of the form $|z| > \alpha$.
2. The ROC of a left sided signal is of the form $|z| < \beta$.

4.6. State convolution property of z-transform.

OR

State the convolution property for continuous and discrete time domain signal in z-transform.

AKTU 2017-18, Marks 02

Ans. This property states that the convolution of two discrete-time signal can be expressed as simple multiplication of their z-transforms.

$$x[n] = x_1[n] \otimes x_2[n] \xrightarrow{ZT} X(z) = X_1(z) \cdot X_2(z)$$

4.7. Draw ROC of two sided sequence in z transform.

AKTU 2022-23, Marks 02

Ans.

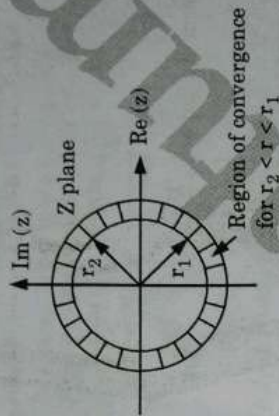


Fig. 4.7.1.

4.8. Find the z-transform of $u(n)$.

AKTU 2021-22, Marks 02

Ans. we know $x[n] = u(n)$

$$u[n] = \begin{cases} 1 & ; \text{ for } n \geq 0 \\ 0 & ; \text{ for } n < 0 \end{cases}$$

$$\begin{aligned} x(z) &= z[x(n)] = z[u(n)] = \sum_{n=-\infty}^{\infty} u(n) z^{-n} \\ &= \sum_{n=0}^{\infty} |z^{-n} = 1 + z^{-1} + z^{-2} + z^{-3} + \dots \\ &= \frac{1}{1 - z^{-1}} = \frac{z}{z - 1} \end{aligned}$$

4.9. Find z transform of the signal $x(n) = (1/2)^n \cdot u(n)$ and its ROC.

AKTU 2021-22, Marks 02

Ans. $x(n) = \left(\frac{1}{2}\right)^n u[n]$

$$\text{we know } Z[a^n u(n)] = \frac{z}{z - a} ; |z| > a$$

$$z \left[\left(\frac{1}{2} \right)^n u[n] \right] = \frac{z}{z - \frac{1}{2}}; |z| > \frac{1}{2}$$

ROC: $|z| > 2$

4.10. Find z-transform of the signal

$$x[n] = \left(\frac{1}{2}\right)^n u[n] \otimes \left(\frac{1}{4}\right)^n u[n]$$

Ans.

Given,

$$x[n] = \underbrace{\left(\frac{1}{2}\right)^n u[n]}_{x_1[n]} \otimes \underbrace{\left(\frac{1}{4}\right)^n u[n]}_{x_2[n]}$$

$$z\text{-transform of } x_1[n] = \frac{1}{1 - \frac{z^{-1}}{2}}$$

$$z\text{-transform of } x_2[n] = \frac{1}{1 - \frac{z^{-1}}{4}}$$

From the convolution property

$$X(z) = \frac{1}{1 - \frac{z^{-1}}{2}} \times \frac{1}{1 - \frac{z^{-1}}{4}} \times \frac{4z}{2z - 1} \times \frac{1}{4z - 1} \xrightarrow{\pi} X_1(z) X_2(z) = X(z)$$

$$X(z) = \frac{8z^9}{(2z-1)(4z-1)}$$

4.11. What is the relationship between z-transform and Fourier transform?

Ans. The z-transform of signal $x(n]$ is given by,

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \dots(4.11.1)$$

Substitute $z = re^{j\omega}$ in eq. (4.11.1)

$$X(re^{i0}) = \sum_{n=-\infty}^{\infty} x[n](re^{i0})^{-n} \quad \dots(4.11.2)$$

$$= \sum_{n=-\infty}^{\infty} \{x[n]r^{-n}\}e^{-j\omega n} = F\{x[n]r^{-n}\}$$

Thus, $X(re^{j\omega})$ is the Fourier transform of the sequence $x[n]$ which is multiplied by a real exponential r^{-n} which may be growing or decaying which increasing n depending on whether r is greater or less than unity.

If $r = 1$, then $|z| = 1$ and eq. (4.11.2) becomes

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = F[x[n]]$$

The z -transform reduces to Fourier transform in the complex z -plane on the contour of a circle with unit radius.

4.12. What is output $y(n)$ when z^n is applied at input of discrete-time linear time-invariant system with impulse response $h(n)$?

Ans. For a discrete-time linear time-invariant system with impulse response $h[n]$, the response $y[n]$ of the system to a complex exponential input of the form z^n is

$$v(n) = H(z)z^n$$

where,

$$H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n}$$

4.13. Find the z-transform and ROC of $x[n] = 2^n u[-n - 1]$

Ans. Its z -transform is given by

$$x[n] \xleftrightarrow{z} \sum_{+ \infty} 2^n u[-n-1] z^{-n}$$

$$X(z) = \sum_{n=-\infty}^{\infty} z^n z^{-n} = \sum_{n=-\infty}^{\infty} z^{-n} z^{-n} = -1 + \sum_{n=0}^{\infty} (z^{-1} z)^n$$

If $|2^{-1}z| < 1$, or equivalently, $|z| < |2|$,

$$X(z) = \frac{1}{1 - \frac{z}{2}} - 1 = \frac{2}{2 - z} - 1 = \frac{z}{2 - z} \quad |z| < |2|$$

4.1.4. Given the z -transform pair $x[n] \xleftrightarrow{z} \frac{z}{(z^2 + 4)}$ with ROC

$|z| < 2$. Find $Y(z)$ when $y[n] = 2^n x[n]$

$$v[n] = 2^n x[n]$$

We have from scaling property

$$q^{\eta_V}[n] \hookrightarrow X(a^{-1}z)$$

Also we have,

$$x[n] \xrightarrow{z} \frac{z}{z^3 + 4}, \text{ thus } Y(z) = \frac{\frac{z}{z^3 + 4}}{\left(z - \frac{2}{z}\right) + 4} = \frac{z}{z^2} = \frac{2z}{z^3 + 16}$$



5

UNIT

Sampling of Time Signals (2 Marks Questions)

5.1. What is sampling ?

Ans. Sampling is the process of converting continuous-time signal $x(t)$ into a discrete-time signal $x[n]$.

5.2. State sampling theorem.

Ans. Sampling theorem states that a band-limited signal of finite energy, which has no frequency components higher than ω_m rad/s (or f_m Hz), may be completely recovered from a knowledge of its samples if the sampling frequency $\omega_s \geq 2\omega_m$ samples/rad/s (or $f_s \geq 2f_m$ samples/s). Thus, the sampling frequency $\omega_s \geq 2\omega_m$ or $f_s \geq 2f_m$.

5.3. State Nyquist theorem.

AKTU 2021-22, Marks 02

Ans. The Nyquist theorem specifies that a sinusoidal function in time or distance can be regenerated with no loss of information as long as it is sampled at a frequency greater than or equal to twice per cycle.

5.4. What is meant by anti-aliasing filter ?

AKTU 2022-23, Marks 02

Ans. An anti-aliasing filter is just a low pass filter with cut-off frequency (i.e., the -3dB frequency) set to Nyquist frequency. This filter cuts out any higher order frequency content in the input signal as any frequencies higher than the Nyquist frequency would be aliased.

5.5. What is Aliasing effect ?

Ans. A signal sampled at less than Nyquist rate (i.e., $f_s < 2f_m$) is said to be under sampled and in such case, $X(\omega)$, the spectrum of $x(t)$ is no longer replicated in $X_s(\omega)$. This effect in which the individual terms in equation

$$X_s(\omega) = \frac{1}{T} \sum_{n=-\infty}^{\infty} X(\omega - n\omega_0)$$

overlap is referred to as aliasing.

5.6. What do you mean by Nyquist rate and Nyquist interval ?

Ans. The minimum sampling frequency $f_s = 2f_m$ is called the Nyquist rate of sampling.

The maximum sampling interval $T_s = \frac{1}{2f_m}$ is called the Nyquist interval.

5.7. What are the two problems occur during signal reconstruction ?

Ans.

1. If $f_s = 2f_m$, so there will not be any guard band so, to reconstruct $x(t)$ from $x_s(t)$, one has to use ideal filter with a cut-off frequency equal to f_m . It is impossible to realize ideal filter.
2. Aliasing effect.

5.8. Find sampling frequency of signal :

$$2 \cos(2\pi \times 10^3 t) \cos(4\pi \times 10^3 t)$$

AKTU 2022-23, Marks 02

Ans. Given : $2 \cos(2\pi \times 10^3 t) \cos(4\pi \times 10^3 t)$

$$2 \cos(2000 \pi t) \cos(4000 \pi t)$$

$$= \cos(6000 \pi t) \cos(2000 \pi t)$$

$$\omega_1 = 6000 \pi \text{ rad/sec}$$

$$\omega_2 = 2000 \pi \text{ rad/sec}$$

$$f_1 = \frac{\omega_1}{2\pi} = 3000 \text{ Hz}, \quad f_2 = \frac{\omega_2}{2\pi} = 1000 \text{ Hz}$$

$$f_{\max} = \text{Max}(f_1, f_2)$$

$$= \text{Max}(3000, 1000)$$

$$= 3000 \text{ Hz}$$

$$= 2 \times f_{\max}$$

$$= 2 \times 3000$$

$$= 6 \text{ kHz}$$

Nyquist rate

Nyquist interval

$$T_s = \frac{1}{6000}$$

$$= 0.16 \text{ msec}$$

5.9. What is guard band ?

Ans. When $\omega_s > 2\omega_m$, the spectral replicates have a large separation between them. This separation region is called guard band.

5.10. Name three different types of sampling techniques.

Ans. The three types are :

1. Ideal or instantaneous or impulse sampling.
2. Natural sampling
3. Flat-top sampling

B. Tech.
(SEM. III) ODD SEMESTER THEORY
EXAMINATION, 2017-18
SIGNAL AND SYSTEM

Time : 3 Hours

Max. Marks : 70

Note : Attempt all sections. If require any missing data; then choose suitably.

Section-A

1. Attempt all questions in brief. (2 × 7 = 14)
 a. Check the periodicity of the signals given below. Determine the fundamental time period if signal is periodic :

$$x(t) = \sin(10t - 1) - \sin(4t - 1)$$

Ans. Refer Q. 1.10, Page SQ-4J, 2 Marks Questions, Unit-1.

- b. Find the DTFT of the causal sequence $x(n) = a^n u(n)$; $|a| < 1$.
Ans. Refer Q. 3.14, Page SQ-14J, 2 Marks Questions, Unit-3.

- c. Check whether the given system are time-variant and causal $y(t) = tx(t)$.

Ans. Refer Q. 1.15, Page SQ-6J, 2 Marks Questions, Unit-1.

- d. State the convolution property for continuous and discrete time domain signal in z-transform.

Ans. Refer Q. 4.6, Page SQ-16J, 2 Marks Questions, Unit-4.

- e. Determine the Laplace transform and find out ROC for

$$x(t) = e^{-t}u(t) + e^{-4t}u(t)$$

Ans. Refer Q. 3.12, Page SQ-14J, 2 Marks Questions, Unit-3.

- f. Draw the signal $x(t) = u(t) - u(t - 2)$.

Ans. Refer Q. 1.11, Page SQ-4J, 2 Marks Questions, Unit-1.

- g. Using final value theorem, find final value of signal corresponding to Laplace transform

$$X(s) = \frac{s+1}{s(s+2)(s+8)}$$

Ans. Refer Q. 3.15, Page SQ-15J, 2 Marks Questions, Unit-3.

- 5.11. Define aperture effect.

Ans. By using flat-top samples, an amplitude distortion is introduced in the reconstructed signal $x(t)$. The distortion caused by the non-uniform spectral gain $H(\omega)$ on higher frequency of $x(t)$, is called aperture effect.

- 5.12. What do you mean by interpolation ?

OR

What is interpolation in sampling ?

AKTU 2018-19, Marks 02

Ans. The process of reconstruction of continuous-time signal $x(t)$ from its sampled version is called interpolation.

- 5.13. What is relation between continuous-time and discrete-time frequency variable ?

Ans. The relation is,

$$\Omega = \omega T$$

Ω = Discrete-time frequency variable

ω = Continuous-time frequency variable

T = Sampling period.

where,



Section-B

2. Attempt any three parts of the following : (7 × 3 = 21)

a. Find and sketch the autocorrelation function $R_{xx}(\tau)$ for $x(t) = e^{-at} u(t)$, $a > 0$.

Ans. This question is out of syllabus from session (2023-24).

b. Find the convolution for given sequence

$$x[n] = \begin{cases} 1 & \text{for } n = -2, 0, 1 \\ 2 & \text{for } n = -1 \\ 0 & \text{else} \end{cases}$$

$$\text{and } h[n] = \delta[n] - \delta[n-1] + \delta[n-2] - \delta[n-3]$$

Ans. Refer Q. 2.9, Page 2-8J, Unit-2.

c. Find the Fourier transform of the signals given below :

$$\text{i. } x(t) = \begin{cases} A, & |t| < T_0 \\ 0, & |t| > T_0 \end{cases}$$

$$\text{ii. } x(t) = e^{-at} u(t)$$

Draw the magnitude and phase response of the transformed signal.

Ans. Refer Q. 3.10, Page 3-11J, Unit-3.

d. If $X(s) = \frac{5s-7}{(s-1)(s+2)}$ with $-2 < R(s) < -1$. Find $x(t)$.

Ans. Refer Q. 3.36, Page 3-32J, Unit-3.

e. State and prove sampling theorem and discuss the effect of under sampling.

Ans. Refer Q. 5.3, Page 5-5J, Unit-5.

Section-C

3. Attempt any one part of the following : (7 × 1 = 7)

a. i. What is signal ? Give brief classification of signals.

Ans.

i. Refer Q. 1.3, Page 1-3J, Unit-1.

ii. Prove that power of energy signal is zero over infinite time.

Ans.

ii. Refer Q. 1.6, Page 1-7J, Unit-1.

b. Plot $x(t) = u(t) - r(t-1) + 2r(t-2) - r(t-3) + u(t-4) - 2u(t-5)$. Find the even and odd parts of this signal.

Ans. Refer Q. 1.9, Page 1-8J, Unit-1.

4. Attempt any one part of the following :

a. Determine whether the following continuous time system $y(t) = x(t) \cos(100\pi t)$ is

i. Static or dynamic

ii. Linear or non-linear

iii. Shift variant or shift invariant

iv. Causal or non-causal

v. Stable or unstable.

Ans. Refer Q. 1.20, Page 1-16J, Unit-1.

b. Find energy and power of the signal

$$\text{i. } x(t) = \cos(t)$$

$$\text{ii. } x(t) = Ae^{-\alpha t} u(t), \alpha > 0$$

Ans. Refer Q. 1.4, Page 1-5J, Unit-1.

5. Attempt any one part of the following :

a. An LTI system with impulse response $h_1(n)$

$$h_1(n) = \left(\frac{1}{3}\right)^n u(n)$$

is connected in parallel with another causal LTI system with impulse response $h_2(n)$. The resulting parallel interconnection has the frequency response.

$$H(e^{j\omega}) = \frac{-12 + 5e^{-j\omega}}{12 - 7e^{-j\omega} + e^{-2j\omega}}$$

Find the impulse response $h_2(n)$.

Ans. Refer Q. 3.21, Page 3-20J, Unit-3.

b. Find the Fourier transform of the signal $x(t) = e^{-at} u(t)$ and plot its magnitude and phase spectrum.

Ans. Refer Q. 3.10, Page 3-11J, Unit-3.

6. Attempt any one part of the following :

a. i. Show that if $x_3(t) = ax_1(t) + bx_2(t)$ then $X_3(s) = aX_1(s) + bX_2(s)$.

ii. If Laplace transform of $x(t)$ is $(s+2)/(s^2+4s+5)$. Determine Laplace transform of $y(t) = x(2t-1)u(2t-1)$.

(7 × 1 = 7)

Ans. Refer Q. 3.29, Page 3-26J, Unit-3.

ii. Refer Q. 3.34, Page 3-31J, Unit-3.

b. A causal LTI system is described by difference equation.

$$y(n) = y(n-1) + y'(n-2) + x(n-1)$$

Find the system function $H(z)$ for this system. Plot the poles zeros of $H(z)$ and indicate the region of convergence.

B. Tech.
(SEM. III) ODD SEMESTER THEORY
EXAMINATION, 2018-19
SIGNAL AND SYSTEM

Time : 3 Hours

Max. Marks : 70

Note: Attempt all sections. If require any missing data; then choose suitably.

Section-A

1. Attempt all questions in brief. (2 × 7 = 14)

a. Distinguish between energy and power signal.

Ans. Refer Q. 1.2, Page SQ-1J, 2 Marks Questions, Unit-1.

b. Define power spectral density.

Ans. Refer Q. 3.18, Page SQ-15J, 2 Marks Questions, Unit-3.

c. Explain significance of convolution in a communication system.

Ans. Refer Q. 2.6, Page SQ-9J, 2 Marks Questions, Unit-2.

d. What are advantages of Laplace transform ?

Ans. Refer Q. 3.16, Page SQ-15J, 2 Marks Questions, Unit-3.

e. What are the limitations of Fourier transform ?

Ans. Refer Q. 3.17, Page SQ-15J, 2 Marks Questions, Unit-3.

f. Define a signal.

Ans. Refer Q. 1.1, Page SQ-1J, 2 Marks Questions, Unit-1.

g. What is interpolation in sampling ?

Ans. Refer Q. 5.12, Page SQ-22J, 2 Marks Questions, Unit-5.

Section-B

2. Attempt any three of the following : (7 × 3 = 21)

a. Classify signals according to signal characteristics.

Ans. Refer Q. 1.3, Page 1-3J, Unit-1.

b. Explain the principle of linearity of DT system.

Ans. Refer Q. 1.13, Page 1-12J, Unit-1.

c. Explain the following properties of Fourier transform :
 time scaling, conjugate functions.

Ans. Refer Q. 4.19, Page 4-18J, Unit-4.

7. Attempt any one part of the following : (7 × 1 = 7)

a. Consider the two continuous-time sinusoidal signals

$x_1(t) = \cos(20\pi t)$ and $x_2(t) = \cos(100\pi t)$ which are sampled at a rate $f_s = 40$ Hz. Find the corresponding discrete time signals.

Ans. Refer Q. 5.16, Page 5-21J, Unit-5.

b. Explain system bandwidth and rise time for low pass filter and prove that $t_r = 0.35/B$.

Ans. This question is out of syllabus from session (2023-24).



Ans. Refer Q. 3.6, Page 3-7J, Unit-3.

- d. State and prove initial and final value theorem of Laplace transform.

Ans. Refer Q. 3.30, Page 3-28J, Unit-3.

- e. State and prove sampling theorem.

Ans. Refer Q. 5.2, Page 5-3J, Unit-5.

Section-C

3. Attempt any one part of the following : (7 × 1 = 7)

- a. What is Shannon's sampling theorem ? Also discuss aliasing by taking an example.

Ans. Refer Q. 5.13, Page 5-16J, Unit-5.

- b. Explain the impulse train sampling of discrete time signals.

Ans. Refer Q. 5.6, Page 5-7J, Unit-5.

4. Attempt any one part of the following :

- a. State whether the following signals $x(t)$ is periodic or not, giving reasons. If it is periodic, find the corresponding period.

$$x(t) = 2 \cos 100 \pi t + 5 \sin 50 t.$$

Ans. Refer Q. 1.10, Page 1-9J, Unit-1.

- b. For a LTI system with unit impulse response $h(t) = e^{-2t}u(t)$. Determine the output to the input $x(t) = e^{-t}u(t)$.

Ans. Refer Q. 3.37, Page 3-33J, Unit-3.

5. Attempt any one part of the following :

- a. Find the energy spectral density of $f(t) = e^{-st}u(t)$. (7 × 1 = 7)

Ans. Refer Q. 3.42, Page 3-37J, Unit-3.

- b. Find impulse response of system described by the equation $2y'(t) + 3y(t) = x(t)$.

Ans. Refer Q. 3.38, Page 3-33J, Unit-3.

6. Attempt any one part of the following :

- a. State and prove frequency shifting theorem of DTFT. (7 × 1 = 7)

Ans. Refer Q. 3.22, Page 3-21J, Unit-3.

- b. Explain Fourier transform of single sided exponential pulse.

Ans. Refer Q. 3.10, Page 3-11J, Unit-3.

7. Attempt any one part of the following :

- a. Find Laplace transform of following signal and draw ROC $x(t) = \cos(3t + \pi/4)u(t)$. (7 × 1 = 7)

Ans. Refer Q. 3.39, Page 3-34J, Unit-3.

- b. Determine z-transform of $x(n) = \sin \omega_0 n u(n)$.

Ans. Refer Q. 4.2, Page 4-2J, Unit-4.



B. Tech.

(SEM. IV) EVEN SEMESTER THEORY
EXAMINATION, 2021-22
SIGNAL SYSTEM

Time : 3 Hours

Max. Marks : 100

Note : 1. Attempt all sections. If require any missing data; then choose suitably.

Section-A

1. Attempt all questions in brief. (2 × 10 = 20)

a. Define a signal with example.

Ans. Refer Q. 1.1, Page SQ-1J, 2 Marks Questions, Unit-1.

b. Draw the signal $u(n) - u(n-3)$.

Ans. Refer Q. 1.3, Page SQ-1J, 2 Marks Questions, Unit-1.

c. Check whether the given system is causal and time variant $y(t) = t \cdot x(t)$.

Ans. Refer Q. 1.15, Page SQ-6J, 2 Marks Questions, Unit-1.

d. State Nyquist theorem.

Ans. Refer Q. 5.3, Page SQ-20J, 2 Marks Questions, Unit-5.

e. Determine the sufficient condition for the existence of CTFT.

Ans. Refer Q. 3.3, Page SQ-12J, 2 Marks Questions, Unit-3.

f. Find z-transform of the signal $x(n) = (1/2)^n \cdot u(n)$ and its ROC.

Ans. Refer Q. 4.9, Page SQ-17J, 2 Marks Questions, Unit-4.

g. Determine the fundamental period of the following if the signal is periodic $x(t) = \cos(\pi t) + \cos(2\pi t)$.

Ans. Refer Q. 1.9, Page SQ-3J, 2 Marks Questions, Unit-1.

h. State the expression of convolution integral.

Ans. Refer Q. 2.3, Page SQ-7J, 2 Marks Questions, Unit-2.

i. Compare CTFT and DTFT.

Ans. Refer Q. 3.4, Page SQ-12J, 2 Marks Questions, Unit-3.

j. Find the z-transform of $u(n)$.

Ans. Refer Q. 4.8, Page SQ-17J, 2 Marks Questions, Unit-4.

Section-B

2. Attempt any three of the following :

(10 × 3 = 30)

- a. i. State and prove the frequency shifting theorem of CTFT.
ii. Explain the principle of linearity property corresponding to CTFT.

Ans.

- i. Refer Q. 3.24, Page 3-22J, Unit-3.
ii. Refer Q. 3.6, Page 3-7J, Unit-3.

- b. Consider $x(t) = \cos(2\pi f t)$. Determine it is a power signal or energy signal.

Ans.

Refer Q. 1.5, Page 1-6J, Unit-1.

- c. Determine the even and odd components of the following signals :

- i. $x(t) = \cos(t) + \sin(t) + \cos(t) \cdot \sin(t)$
ii. $x(n) = \{1, 1, 1, 1, 1\}$

↑

Ans. Refer Q. 1.12, Page 1-11J, Unit-1.

- d. Find the Fourier transform of the signal given below :
 $x(t) = e^{-at} u(t)$ and sketch the magnitude and phase spectrum.

Ans.

Refer Q. 3.10, Page 3-11J, Unit-3.

- e. Using Fourier transform find the convolution of
 $x_1(t) = e^{-2t} u(t)$, $x_2(t) = e^{-3t} u(t)$

Ans.

Refer Q. 3.13, Page 3-15J, Unit-3.

Section-C

3. Attempt any one part of the following : (10 × 1 = 10)

- a. Find the inverse Laplace of the following :

$$X(S) = \frac{2}{(s+4)(s-1)} \text{ if the region of convergence is}$$

- i. $-4 < \operatorname{Re}(s) < 1$
ii. $\operatorname{Re}(s) > 1$

Ans.

Refer Q. 3.40, Page 3-35J, Unit-3.

- b. Using Laplace transform obtain the impulse response of the given second-order system

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = x(t)$$

Ans.

Refer Q. 3.43, Page 3-37J, Unit-3.

4. Attempt any one part of the following :

(10 × 1 = 10)

- a. Determine the z-transform of $x(n) = \cos(\omega_p n) u(n)$ and sketch the ROC.

Ans.

Refer Q. 4.9, Page 4-10J, Unit-4.

- b. Determine the inverse z -transform of the following function

$$H(z) = \frac{0.2z}{(z+0.4)(z-0.2)} \quad \text{ROC: } |z| > 0.4$$

Ans. Refer Q. 4.12, Page 4-12J, Unit-4.

5. Attempt any one part of the following: (10 × 1 = 10)

- a. Explain and prove Parseval's Theorem.

Ans. Refer Q. 3.27, Page 3-24J, Unit-3.

- b. Analyze the Discrete Time Fourier Transform of the following

$$x(n) = 0.5^n u(n) + 2^{-n} u(-n-1)$$

Ans. Refer Q. 3.23, Page 3-21J, Unit-3.

6. Attempt any one part of the following: (10 × 1 = 10)

- a. Implement the convolution integral on the signals

$$x(t) = e^{-2t} u(t) \text{ and } h(t) = u(t).$$

Ans. Refer Q. 2.6, Page 2-5J, Unit-2.

- b. Implement the convolution sum on the signals

$$x(n) = a^n u(n) \text{ and } h(n) = u(n).$$

Ans. Given:

$$x(n) = a^n u(n), \quad h(n) = u(n)$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

$$y(n) = \sum_{k=-\infty}^{\infty} a^k u(k) u(n-k)$$

$$x(n)h(n-k) = \begin{cases} a^k, & 0 < k < n \\ 0, & \text{elsewhere} \end{cases}$$

$$\therefore y(n) = \sum_{k=0}^n a^k \cdot 1 = \frac{1-a^{n+1}}{1-a}$$

7. Attempt any one part of the following: (10 × 1 = 10)
a. State and prove the sampling theorem and discuss the effect of under sampling.

Ans. Refer Q. 5.3, Page 5-5J, Unit-5.

- b. Explain reconstruction of signal from its samples using interpolation.

Ans. Refer Q. 5.9, Page 5-11J, Unit-5.



B. Tech.
(SEM. IV) EVEN SEMESTER THEORY
EXAMINATION, 2022-23
SIGNAL SYSTEM

Time : 3 Hours

Max. Marks : 100

Note: 1. Attempt all sections. If require any missing data; then choose suitably.

Section-A

1. Attempt all questions in brief. (2 × 10 = 20)

- a. Determine whether it is energy or power signal $x(t) = tu(t)$.

Ans. Refer Q. 1.4, Page SQ-2J, 2 Marks Questions, Unit-1.

- b. Find even and odd component of the following signal $x(t) = 1 + 2t + 3t^2 + 4t^3$.

Ans. Refer Q. 1.7, Page SQ-2J, 2 Marks Questions, Unit-1.

- c. What is the difference between cross correlation and auto correlation?

Ans. Refer Q. 2.8, Page SQ-9J, 2 Marks Questions, Unit-2.

- d. Determine the initial value for the given laplace transform

$$X(s) = \frac{4}{s^2 + 3s + 8}.$$

Ans. Refer Q. 3.10, Page SQ-13J, 2 Marks Questions, Unit-3.

- e. What is the significance of ROC?

Ans. Refer Q. 4.4, Page SQ-16J, 2 Marks Questions, Unit-4.

- f. Write time shifting property of laplace transform.

Ans. Refer Q. 3.9, Page SQ-13J, 2 Marks Questions, Unit-3.

- g. What is the condition of z transform to exist.

Ans. Refer Q. 4.1, Page SQ-16J, 2 Marks Questions, Unit-4.

- h. Draw ROC of two sided sequence in z transform.

Ans. Refer Q. 4.7, Page SQ-17J, 2 Marks Questions, Unit-4.

- i. What is meant by anti-aliasing filter?

Ans. Refer Q. 5.4, Page SQ-20J, 2 Marks Questions, Unit-5.

- j. Find sampling frequency of signal :

$$2 \cos(2\pi \times 10^3 t) \cos(4\pi \times 10^3 t)$$

Ans. Refer Q. 5.8, Page SQ-21J, 2 Marks Questions, Unit-5.

Section-B

2. Attempt any three of the following : $(10 \times 3 = 30)$

- a. Check whether the given system is static, linear, causal and time invariant system or not $y(t) = at^2 x(t) + bt x(t-4)$

Ans. Refer Q. 1.22, Page 1-18J, Unit-1.

- b. Consider a causal LTI system with frequency response $H(\omega)$

$$= \frac{1}{4 + j\omega}. \text{ For a particular input } x(t) \text{ the system is observed to produce the output}$$

$$y(t) = e^{-2t}u(t) - e^{-4t}u(t). \text{ Find the input } x(t).$$

Ans. Refer Q. 3.25, Page 3-23J, Unit-3.

- c. Find the fourier transform of a signal $x(t) = \frac{1}{a^2 + t^2}$.

Ans. Refer Q. 3.16, Page 3-16J, Unit-3.

- d. Find the Z-transform and ROC of $X(z)$ for

$$x(n) = 3\left(\frac{5}{7}\right)^n u(n) + 2\left(-\frac{1}{3}\right)^n u(n)$$

Ans. Refer Q. 4.8, Page 4-8J, Unit-4.

- e. State and prove sampling theorem

Ans. Refer Q. 5.2, Page 5-3J, Unit-5.

Section-C

3. Attempt any one part of the following :

- a. Sketch the following signal

i. $2u(t+2) - 2u(t-3)$

ii. $r(t) - r(t-1) - r(t-3) + r(t-4)$

Ans. Refer Q. 1.11, Page 1-9J, Unit-1.

- b. Check whether the following digital systems are BIBO stable or not

i. $y(n) = ax(n) + b$

ii. $y(n) = e^{-x(n)}$

Ans. Refer Q. 1.23, Page 1-19J, Unit-1.

4. Attempt any one part of the following :

- a. Find the convolution of the following signal by graphical method $(10 \times 1 = 10)$

$$x(t) = e^{-3t}u(t); h(t) = u(t+3)$$

Ans. Refer Q. 2.7, Page 2-6J, Unit-2.

- b. Using fourier transform, find the convolution of the signals $x_1(t) = e^{-2t}u(t)$ and $x_2(t) = e^{-3t}u(t)$

Ans. Refer Q. 3.13, Page 3-15J, Unit-3.

5. Attempt any one part of the following :

$$(10 \times 1 = 10)$$

- a. Determine the exponential fourier series for the waveform shown

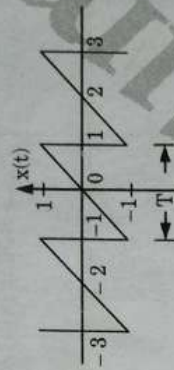


Fig. 1.

Ans. Refer Q. 3.4, Page 3-5J, Unit-3.

- b. Find the fourier transform of the signal $x(t) = e^{-a|t|} \operatorname{sgn}(t)$

Ans. Refer Q. 3.12, Page 3-14J, Unit-3.

6. Attempt any one part of the following :

$$(10 \times 1 = 10)$$

- a. Find inverse Z-Transform of

$$X(z) = \frac{2z-7}{z^2-5z+6}; \text{ ROC: } |z| < 2$$

Ans. Refer Q. 4.11, Page 4-11J, Unit-4.

- b. A causal LTI system is described by the difference equation $y(n) = y(n-1) + y(n-2) + x(n) + 2x(n-1)$. Find the system function and frequency response of the system. Plot the poles and zeroes and indicate ROC. Also determine the stability and impulse response of the system.

Ans. Given :

$$y(n) = y(n-1) + y(n-2) + x(n) + 2x(n-1)$$

Taking z transform on both sides, we get

$$Y(z) = z^{-1}Y(z) + z^{-2}Y(z) + X(z) + 2z^{-1}X(z)$$

$$Y(z)(1 - z^{-1} - z^{-2}) = X(z)(1 + 2z^{-1})$$

The system function is

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1 + 2z^{-1}}{1 - z^{-1} - z^{-2}} = \frac{z(z+2)}{z^2 - z - 1}$$

$$= \frac{z(z+2)}{(z-1.62)(z+0.62)}$$

The frequency response of system is

$$H(e^{j\omega}) = \frac{z(z+2)}{(z-1.62)(z+0.62)} \Big|_{z=e^{j\omega}}$$

$$= \frac{e^{j\omega}(e^{j\omega}+2)}{(e^{j\omega}-1.62)(e^{j\omega}+0.62)}$$

$H(z)$ has zeros at $z=0$ and $z=-2$

$H(z)$ has poles at $z=1.62$ and $z=-0.62$

To find impulse response $h(n)$

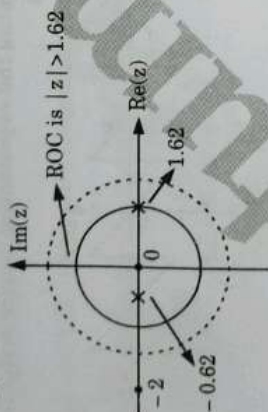


Fig. 1.

$$\frac{H(z)}{z} = \frac{z+2}{(z-1.62)(z+0.62)}$$

$$= \frac{A}{z-1.62} + \frac{B}{z+0.62}$$

$$= \frac{1.62}{z-1.62} - \frac{0.62}{z+0.62}$$

$$H(z) = 1.62 \left(\frac{z}{z-1.62} \right) - 0.62 \left(\frac{z}{z+0.62} \right)$$

Taking inverse z transform, the impulse response is

$$h(n) = 1.62 (1.62)^n u(n) - 0.62 (-0.62)^n u(n)$$

7. Attempt any one part of the following:

a. Solve $y[k+2] + 5y[k+1] + 6y[k] = 3f[k+1] + 5f[k]$ (10 × 1 = 10)

If the initial conditions are $y[-1] = 11/6$, $y[-2] = 37/36$ and the input $f[k] = 2^{-k} V[k]$ and indicate zero input and zero state component in the output $y[k]$.

Ans. Refer Q. 4.18, Page 4-18J, Unit-4.

b. Determine the Nyquist sampling rate and Nyquist sampling interval for

$$x(t) = \text{sinc}(100\pi t) + 3 \text{sinc}^2(60\pi t)$$

Ans. Refer Q. 5.4, Page 5-5J, Unit-5.



B. Tech.
(SEM. IV) EVEN SEMESTER THEORY
EXAMINATION, 2023-24
SIGNAL SYSTEM

Time : 3 Hours

Max. Marks : 70

Note : 1. Attempt all sections. If require any missing data, then choose suitably.

SECTION A

1. Attempt all questions in brief. (2 × 7 = 14)

a. Determine the fundamental period of the signal $x(t) = \sin(4t-1)$.

Ans. Given, $x(t) = \sin(4t-1)$

We have, $x(t) = \sin(\omega t - \theta)$

On comparing eq. (1) and eq. (2) then we get

$$\omega = 4, \quad \theta = 1$$

$$\Rightarrow 2\pi f = 4$$

$$f = \frac{4}{2\pi} = \frac{2}{\pi}$$

$$\text{Time period, } T = \frac{1}{f} = \frac{\pi}{2} \text{ sec}$$

b. Write the general formula for calculating convolution in continuous time domain.

Ans. Refer Q. 3.8, Page 3-9J, Unit-3.

c. Find the Fourier transform of continuous time signal $x(t) = \delta(t+1)$.

Ans.

1. Given, $x(t) = \delta(t+1)$

$$\delta(t) \xrightarrow{FT} 1$$

2. Using time-shifting property

$$\delta(t - (-1)) \xrightarrow{FT} e^{-j\omega(-1)} = 1$$

$$\delta(t+1) \xrightarrow{FT} e^{j\omega}$$

d. Calculate the z -transform of discrete time signal $x(n) = \delta(n+1)$.

ii. Stable system :

1. The stability of a system is defined, if the bounded input produces bounded output (BIBO).
2. For stable system, the output signal must satisfy the following condition :

$$|y(t)| \leq M_y < \infty \text{ for all values of } t.$$

where, $M_y \rightarrow$ finite positive number.

3. The input signal must also satisfy the following condition :

$$|x(t)| \leq M_x < \infty \text{ for all values of } t.$$

where, $M_x \rightarrow$ finite positive number.

- iii. **Causal system :** A system is said to be causal (non-anticipative) if the output of the system at any time t depends only on the present and past values of the input but not on future inputs. Causal systems are real time system.

- iv. **Dynamic system :** In contrast, a system is said to be dynamic or memory system if the response depends upon past or future inputs.

- b. A causal LTI system is described by difference equation.

$$y(n) = y(n-1) + y(n-2) + x(n-1)$$

Find the system function $H(z) = Y(z)/X(z)$ for this system. Plot the pole-zero plot of $H(z)$ and indicate the region of convergence.

Ans. Given, $y(n) = y(n-1) + y(n-2) + x(n-1)$

Taking z-transform,

$$y(z) = z^{-1}y(z) + z^{-2}y(z) + z^{-1}x(z)$$

$$y(z) = [1 - z^{-1} + z^{-2}] = z^{-1}x(z)$$

$$H(z) = \frac{y(z)}{x(z)} = \frac{z^{-1}}{1 - z^{-1} + z^{-2}}$$

Multiply z^2 in numerator and denominator, we get

$$H(z) = \frac{y(z)}{x(z)} = \frac{z}{z^2 - z + 1}$$

$$= \frac{z}{\left(z - \left(\frac{1}{2} + \frac{j\sqrt{5}}{2}\right)\right)\left(z - \left(\frac{1}{2} - \frac{j\sqrt{5}}{2}\right)\right)}$$

$$\text{Poles } z = \frac{1}{2} + \frac{j\sqrt{5}}{2} \text{ and } z = \frac{1}{2} - \frac{j\sqrt{5}}{2}$$

Zero

$$\text{ROC : } |z| < \left(\frac{1}{2} + \frac{j\sqrt{5}}{2}\right) \text{ and } |z| < \left(\frac{1}{2} - \frac{j\sqrt{5}}{2}\right)$$

Ans. Given, $x(n) = \delta(n+1)$

$$\delta(n) \xrightarrow{zT} 1$$

Using time shifting property

$$\delta(n - (-1)) \xrightarrow{zT} z^{-(-1)} = z$$

$$\delta(n+1) \xrightarrow{zT} z$$

- e. Sketch the sampled signal of $x(t) = \cos(2\pi t)$.

Ans. $x(t) = \cos(2\pi t)$
 $\cos(2\pi t)$

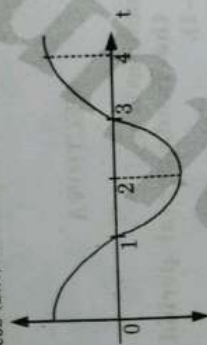


Fig. 1.

- f. Sketch the continuous time signal $x(t) = u(-t+1)$.

Ans. Given, $x(t) = u(-t+1)$



Fig. 2.

- g. What is Parseval's theorem ?

Ans. Refer Q. 3.27, Page 3-24J, Unit-3.

SECTION B

2. Attempt any three of the following : (7 × 3 = 21)

- a. Define the following terms related to system with mathematical expressions :

i. Linear system

ii. Stability

iii. Causality and

iv. Dynamic system.

Ans.

- i. **Linear system :** A system which obeys the principle of superposition and principle of homogeneity is called a linear system.

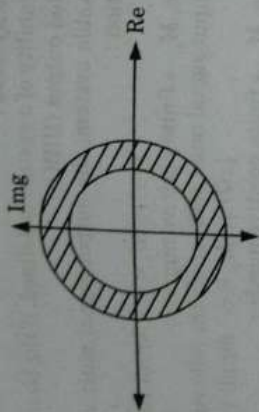


Fig. 3.

- c. Sketch the signal $y(t) = e^{-a|t|}$ and find the Fourier transform of this same signal with magnitude and phase curve.

Ans.

1. Given, $y(t) = e^{-a|t|}$

2. Taking Fourier transform,

$$Y(\omega) = \int_{-\infty}^{\infty} e^{-a|t|} e^{-j\omega t} dt$$

3. This integral can be split into two parts :

$$Y(\omega) = \int_{-\infty}^0 e^{(a+j\omega)t} dt + \int_0^{\infty} e^{-(a-j\omega)t} dt$$

4. For first part,

$$\text{For } t < 0, \int_{-\infty}^0 e^{(a+j\omega)t} dt = \frac{1}{a+j\omega}$$

For second part,

$$\text{For } t \geq 0, \int_0^{\infty} e^{-(a-j\omega)t} dt = \frac{1}{a-j\omega}$$

5. Combining both parts,

$$Y(\omega) = \frac{1}{a+j\omega} + \frac{1}{a-j\omega}$$

6. This simplifies to,

$$Y(\omega) = \frac{2a}{a^2 + \omega^2}$$

7. Magnitude,

$$|Y(\omega)| = \left| \frac{2a}{a^2 + \omega^2} \right|$$

8. The phase of $Y(\omega)$ is zero since the function is symmetric and real.
Phase of $Y(\omega) = 0$.

- d. What is ROC ? Discuss any three properties of z-transform with mathematical expressions and example.

Ans.

A. ROC : Refer Q. 4.7, Page 4-7J, Unit-4.

B. Properties of z-transform : Refer Q. 4.3, Page 4-3J, Unit-3.

- c. Describe natural sampling process with graphs and mathematical equations.

Ans. Refer Q. 5.7, Page 5-8J, Unit-5.

SECTION C

3. Attempt any one part of the following :
a. Define time invariant system. Also check whether the given system is time variant or invariant system : $(7 \times 1 = 7)$

i. $y_1(t) = tx(t)$,

ii. $y_2(n) = 2x(n) + 3$,

iii. $y_3(t) = 2x(-t)$.

Ans.

A. Time-invariant system :

Time-invariance is the property of a system which makes the behaviour of the system independent of time. This means that the behaviour of the system does not depend on the time at which the input is applied.

If $x(t) \rightarrow y(t)$

then $x(t-T) \rightarrow y(t-T)$

B. Numerical :

i. Given, $y(t) = tx(t)$

Delaying input by T

$$Y_1(t-T) = (t-T)x(t-T)$$

$$y(t) \neq y_1(t-T)$$

Thus, system is time-variant

ii. Given, $y_2(n) = 2x(n) + 3$

Delaying input by n_0

$$y_2(n-n_0) = 2x(n-n_0) + 3$$

$$y_2(n) = y_2(n-n_0)$$

Thus, system is time-invariant

iii. Given, $y_3(n) = 2x(-t)$

Delaying by t_0

$$y_3(t-t_0) = 2x(-(t-t_0))$$

$$y_3(t-t_0) = 2x(-t+t_0)$$

$$y_3(t) \neq y(t-t_0)$$

Thus, system is time-variant.

- b. Define the following with mathematical expressions and examples :

- i. Energy and power signals,

ii. Even and odd signals.

Ans. Refer Q. 1.3, Page 1-3J, Unit-1.

4. Attempt any one part of the following :

(7 × 1 = 7)

a. What is discrete time convolution ? Find the convolution $y(n) = x(n) * h(n)$ of the discrete time signals $x(n) = h(n) = u(n)$.

Ans.

A. Discrete time convolution : Discrete time convolution is a mathematical operation that combines two discrete-time signals to create a third signal.

B. Numerical :

1. We use definition of convolution for discrete-time signals :

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

2. Given, $x(n) = u(n)$

$$\text{So, } y(n) = \sum_{k=-\infty}^{\infty} u(k) u(n-k)$$

3. Since $u(k) = 0$ for $k < 0$ and $u(n-k) = 0$ for $k > n$, the limits of the summation become 0 to n .

$$y(n) = \sum_{k=0}^n 1$$

4. After solving then we get

$$y(n) = n + 1 \text{ for } n \geq 0$$

$$y(n) = 0 \text{ for } n < 0$$

b. What is LTI system ? An LTI system is described by

$$\text{differential equation } \frac{dy(t)}{dt} + 5y(t) = 3x(t).$$

Calculate :

i. Transfer function, $H(s) = Y(s)/X(s)$.

ii. Output, $y(t)$ when $x(t) = u(t)$.

Ans.

A. LTI system : A linear time-invariant (LTI) system is a system that has a linear relationship between its input and output, and its response is not dependent on time.

B. Numerical :

$$1. \text{ Given, } \frac{dy(t)}{dt} + 5y(t) = 3x(t)$$

2. Taking Laplace transform

$$sY(s) + 5Y(s) = 3X(s)$$

$$Y(s)(s+5) = 3X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{3}{s+5}$$

4. Taking inverse Laplace transform,

$$h(t) = \frac{y(t)}{x(t)} = 3e^{-5t}$$

$$y(t) = 3e^{-5t}u(t)$$

[∵ Given $x(t) = u(t)$]

5. Attempt any one part of the following :

(10 × 1 = 10)

a. Write any three properties of Fourier transform. Find the Fourier transform of the given signal using properties $y(t) = te^{-4t}u(t)$.

Ans.

A. Properties of Fourier transform : Refer Q. 3.6, Page 3-7J, Unit-3.

B. Numerical :

Given, $y(t) = te^{-4t}u(t)$

Fourier transform of $e^{-4t}u(t) = \frac{1}{j\omega + 4}$

Using the property:

$$t \cdot x(t) \xrightarrow{FT} j \frac{d}{d\omega} X(\omega)$$

So, Fourier transform of $t \cdot x(t)$

$$= j \frac{d}{d\omega} \left(\frac{1}{j\omega + 4} \right)$$

$$= \frac{j(-j)}{(4 + j\omega)^2} = \frac{1}{(4 + j\omega)^2}$$

b. Determine the inverse Laplace transform of the following functions using partial fraction method :

$$\text{i. } X(s) = \frac{4(s+3)}{s(s+1)(s+2)}$$

$$\text{ii. } Y(s) = \frac{4}{(s+1)(s+2)^2}$$

Ans.

$$\text{i. Given, } X(s) = \frac{4(s+3)}{s(s+1)(s+2)}$$

$$X(s) = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}$$

On solving partial fraction then we get
A = 6, B = -8, C = 2

So, $X(s) = \frac{6}{s} + \frac{8}{(s+1)} + \frac{2}{(s+2)}$

Taking inverse laplace transform, we get
 $x(t) = 6 - 8e^{-t} + 2e^{-2t}$

ii. Given, $Y(s) = \frac{4}{(s+1)(s+2)^2}$

$$Y(s) = \frac{A}{(s+1)} + \frac{B}{(s+2)} + \frac{C}{(s+2)^2}$$

On solving partial fraction,

$$A = 4, B = -6, C = -4$$

$$Y(s) = \frac{4}{(s+1)} - \frac{6}{(s+2)} - \frac{4}{(s+2)^2}$$

Taking inverse laplace transform,

$$y(t) = 4e^{-t} - 6e^{-2t} - 4te^{-2t}$$

6. Attempt any one part of the following :

a. Determine the inverse z-transform $y(n)$ of the following
 $(7 \times 1 = 7)$

functions $Y(z) = \frac{z^2}{(z^2 + 3z + 2)}$, given ROC $|z| > 2$.

Ans.

$$Y(z) = \frac{z^2}{(z^2 + 3z + 2)}$$

$$Y(z) = \frac{z^2}{z^2 + 2z + z + 2}$$

$$= \frac{z^2}{z(z+2) + 1(z+2)}$$

$$= \frac{z^2}{(z+1)(z+2)}$$

$$Y(z) = \frac{A}{(z+1)} + \frac{B}{(z+2)}$$

Taking partial fraction,

$$A = 1, B = -4$$

$$Y(z) = \frac{1}{z+1} + \frac{(-4)}{z+2}$$

Taking z-transform,

$$y(n) = (-1)^n u[n] - 4(-2)^n u[n]$$

b. Determine the z-transform of $x(n) = (1/2)^n u(n)$ and depict the ROC and the location of poles and zeros in the z-plane.

Ans.

1. Given, $x(n) = \left(\frac{1}{2}\right)^n u(n)$

2. We know, $X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$

$$X(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n}$$

3. This is a geometric series with the common ratio $r = \frac{1}{2} z^{-1}$. The

sum of the geometric series is given by :

$$X(z) = \frac{1}{1 - \frac{1}{2} z^{-1}} \text{ for } \left| \frac{1}{2} z^{-1} \right| < 1$$

4. Simplify this, we get

$$X(z) = \frac{z}{z - \frac{1}{2}} \text{ for } |z| > \frac{1}{2}$$

ROC: For the z-transform to exist, the magnitude of z must satisfy

$$|z| > \frac{1}{2}$$

Poles: $z = \frac{1}{2}$

Zero: $z = 0$

Depiction in the z-plane :

i. The zero is at $z = 0$

ii. The poles is at $z = \frac{1}{2}$

iii. The ROC is the region outside the circle with radius $\frac{1}{2}$ in the

z-plane.

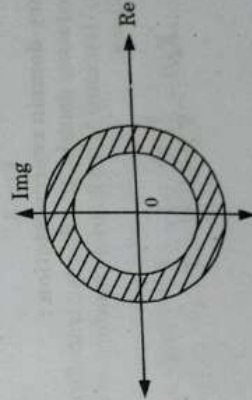


Fig. 4.

7. Attempt any one part of the following: (7 × 1 = 7)

a. Explain ideal sampling with time domain and frequency domain graphs and mathematical equations.

Ans.

1. Ideal sampling is a theoretical concept in signal processing where a continuous-time signal is sampled by multiplying it with an infinite train of Dirac delta functions, resulting in a series of instantaneous samples.
2. This process is fundamental in converting analog signals to discrete ones for digital processing.

A. Time-domain representation :

1. Consider a continuous-time signal $x(t)$.
2. Ideal sampling involves multiplying $x(t)$ by a Dirac comb function $\Delta_T(t)$, defined as :

$$\Delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

Here, T is the sampling period, and $\delta(t)$ represents the Dirac delta function.

3. The sampled signal $x_s(t)$ is :

$$x_s(t) = x(t) \cdot \Delta_T(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

4. This operation yields :

$$x_s(t) = \sum_{n=-\infty}^{\infty} x(nT) \cdot \delta(t - nT)$$

In this expression, $x(nT)$ are the sampled values of $x(t)$ at intervals of T .

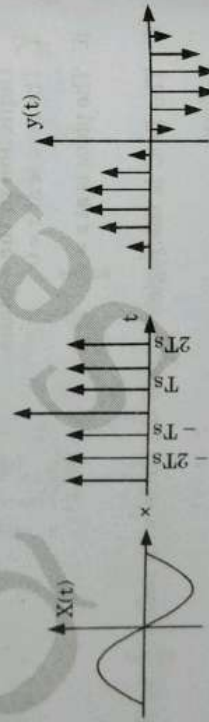


Fig. 5.

B. Frequency-domain representation :

1. In the frequency domain, the Fourier transform of the sampled signal $X_s(f)$ is related to the Fourier transform of the original signal $X(f)$ as follows :

$$X_s(f) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X\left(f - \frac{k}{T}\right)$$

2. This equation indicates that the spectrum of the sampled signal $X_s(f)$ consists of the original spectrum $X(f)$ replicated at intervals of $\frac{1}{T}$ (the sampling frequency).
3. If the sampling frequency is not sufficiently high, these replicas can overlap, leading to aliasing.

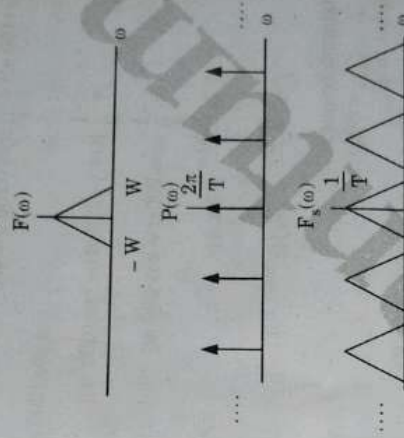


Fig. 6.

- b. What is nyquist rate and nyquist interval ? Calculate the nyquist rate and nyquist interval for the following continuous-time sinusoidal signals :

- i. $x_1(t) = \cos(20\pi t) + \cos(40\pi t)$
- ii. $x_2(t) = \cos(200\pi t) \cos(400\pi t)$.

Ans.

A. Nyquist rate and nyquist interval : Refer Q. 5.1, Page 5-2d, Unit-5.

B. Numerical :

- i. Given, $x_1(t) = \cos(20\pi t) + \cos(40\pi t)$

$$f_1 = \frac{20\pi}{2\pi} = 10 \text{ Hz}$$

$$f_2 = \frac{40\pi}{2\pi} = 20 \text{ Hz}$$

The maximum frequency, $f_{\max} = 20 \text{ Hz}$

The nyquist rate, $f_{\text{Nyquist}} = 2 \times f_{\max} = 2 \times 20 \text{ Hz} = 40 \text{ Hz}$

The nyquist interval = $\frac{1}{f_{\text{Nyquist}}} = \frac{1}{40 \text{ Hz}}$

7. Attempt any one part of the following :

(7 × 1 = 7)

a. Explain ideal sampling with time domain and frequency domain graphs and mathematical equations.

Ans.

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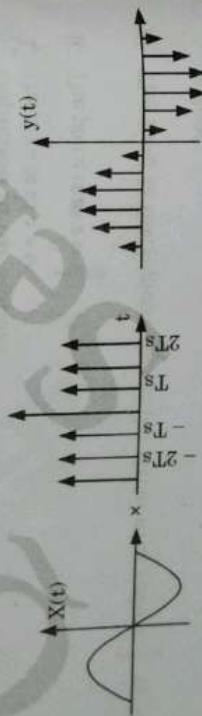


Fig. 5.

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3. If the sampling frequency is not sufficiently high, these replicas can overlap, leading to aliasing.

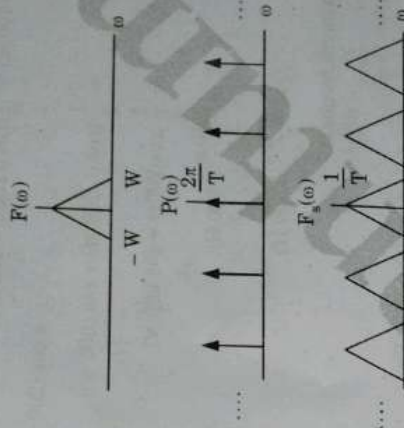


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- i. $x_1(t) = \cos(20\pi t) + \cos(40\pi t)$
- ii. $x_2(t) = \cos(200\pi t) \cos(400\pi t)$.

Ans.

A. Nyquist rate and nyquist interval : Refer Q. 5.1, Page 5-2J, Unit-5.

B. Numerical :

- i. Given, $x_1(t) = \cos(20\pi t) + \cos(40\pi t)$

$$f_1 = \frac{20\pi}{2\pi} = 10 \text{ Hz}$$

$$f_2 = \frac{40\pi}{2\pi} = 20 \text{ Hz}$$

The maximum frequency, $f_{\max} = 20 \text{ Hz}$

The nyquist rate, $f_{\text{Nyquist}} = 2 \times f_{\max} = 2 \times 20 \text{ Hz} = 40 \text{ Hz}$

The nyquist interval = $\frac{1}{f_{\text{Nyquist}}} = \frac{1}{40 \text{ Hz}}$

- ii. Given, $x_p(t) = \cos(200\pi t) \cos(400\pi t)$
Using trigonometric identity,

$$\cos(A) \cos(B) = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

$$\cos(200\pi t) \cos(400\pi t) = \frac{1}{2} [\cos((200\pi - 400\pi)t) + \cos(200\pi + 400\pi t)]$$

$$= \frac{1}{2} [\cos(-200\pi t) + \cos(600\pi t)]$$

$$= \frac{1}{2} [\cos 200\pi t + \cos 600\pi t]$$

$$f_1 = \frac{200}{2\pi} = 100 \text{ Hz}$$

$$f_2 = \frac{600}{2\pi} = 300 \text{ Hz}$$

The maximum frequency,

$$f_{\max} = 300 \text{ Hz}$$

The Nyquist rate, $f_{\text{Nyquist}} = 2 \times f_{\max} = 2 \times 300 = 600 \text{ Hz}$

The Nyquist interval,

$$T_{\text{Nyquist}} = \frac{1}{f_{\text{Nyquist}}} = \frac{1}{600} = 0.00166 \text{ seconds} \\ = 1.66 \text{ ms}$$



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